

Decoding the Click

Turning Self-Tracking Data into Insights for Mental Healthcare

Master Thesis



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February, 2026

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Cover photo: Vibeke Hempler, 2012

Published by: DTU, Brovej, Building 321, 2800 Kgs. Lyngby Denmark

Abstract

The introduction and development of self-tracking technologies, such as the *One Button Tracker*, offer clinicians more data on patients' mental health experiences. However, it creates a meaningful problem for them in the form of the *interpretation gap*, as they must process raw behavioral logs under high cognitive load.

This thesis, *Decoding the Click*, investigates the development of a web platform that translates longitudinal self-tracking data into usable clinical narratives, integrating *Large Language Models* into the process.

Using an agile methodology across six iterative sprints, the study uses an OBT-inspired dataset that simulates dependency behaviors to validate the system. Early testing revealed that standalone LLMs suffer from *numeric drift* and *hallucination* issues when performing complex arithmetic tasks. Accordingly, this research establishes the need for a specialized architecture that separates deterministic data processing for quantitative metrics from probabilistic reasoning for qualitative narratives.

The findings demonstrate that, by shifting from traditional *Prompt Engineering* to *Context Engineering* through methods such as Dynamic Context Assembly, raw signals can be more reliably converted into structured insights and visual summaries. Ultimately, this work explores a framework for integrating AI into mental healthcare as a boundary object, enhancing the therapeutic process while strictly respecting the patient-psychologist relationship.

Acknowledgements

This thesis marks the end of a two and a half year long period during which I had the opportunity to delve into the fascinating world of artificial intelligence.

I would like to express my deepest gratitude to my supervisor, Jakob Eg Larsen, for his guidance, feedback, and support throughout this process. I am especially grateful for the opportunity to build on his research and for the opportunity to be part of the One Button Tracker research and development. His deep technical and clinical understanding of the instrument was fundamental in the development of the platform.

I would like to thank Patricia W. Marcelino for her involvement in this thesis, both in her capacity as a psychologist and as a participant in the user study. Her expert feedback provided valuable insights into creating a more clinically viable platform.

Special thanks to the participants who contributed to the user surveys. Their open insights helped establish the foundation for the decisions and the architecture of this thesis.

I am deeply grateful to the volunteers who participated in the iterative think-aloud sessions throughout the development sprints. Their patience with early prototypes and critical feedback was essential in refining the platform.

To all my friends, near and far: thank you for your presence and support.

And a big thanks to my family for their amazing love and support. Thank you for always believing in me.

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1 Introduction

1.1 Motivation

With rapid advancements in analysis and healthcare, data quickly became a core support tool for continuous patient monitoring, early disease detection, and the delivery of personalized medicine [1, 2]. As technological progress continues, particularly in the fields of artificial intelligence and data science, a growing set of tools [3] is emerging to facilitate their integration. Simultaneously, mental health has gained increasing recognition as a key component of overall health, leading to an expansion of innovative solutions [4].

Nonetheless, these solutions frequently encounter limitations in terms of support, usable data, and user simplicity. Conventional psychotherapeutic methodologies frequently depend on retrospective self-reporting, a process which is vulnerable to the influence of *recall bias* [5, 6]. This is due to patients encountering difficulties in accurately recalling the context and intensity of symptoms between sessions [5].

To address this deficiency, researchers and clinicians often turn to *Ecological Momentary Assessment (EMA)* [7], a methodology designed to capture patients' experiences, emotions, and behaviors in real-time within their natural environments. However, despite the objective of mobile EMA applications to overcome retrospective recall bias, they often impose a significant cognitive burden on users. This results in *tracking fatigue* and reduced adherence [6]. Furthermore, passive sensors such as heart rate monitors are, in reality, unable to capture mental states and thoughts, for example, cravings or intrusive thoughts, which are essential for clinical treatment and are hardly quantifiable [8].

The *One Button Tracker (OBT)*, developed and researched at the Technical University of Denmark, has emerged to fulfill a specific but crucial role in this field [8]. This technology focuses on making the registration of subjective phenomena easier by requiring minimal effort and recording in the moment [6]. This addresses the limitations of memory and device burden. However, the generation of high-frequency longitudinal data brings forth a new challenge, which is known as *the interpretation gap* [4]. The analysis of raw behavioral logs by psychologists is often considered ineffective due to the density of the data during brief sessions. Fundamentally, in the context of active self-tracking, data analysis is not a unilateral diagnosis but, in fact, a combined process between clinician and patient where both give meaning to this data [9]. In previous OBT studies, data often functioned as a *third entity* in the consultation room, serving as a key tool that helped the patient reflect on their experiences [9, 4]. This gap is not only a challenge for clinical diagnosis but also a problem for the therapeutic process. It is extremely important to give meaning and context to data, which is a two-way process. While the system provides when something happened, the patient and psychologist provide the why [4]. Therefore, the platform aims to bridge this gap not by replacing patient input but by organizing obtained data into more accessible contexts and narratives.

The integration of the OBT with Prompt and Context Engineering and *Large Language Models (LLMs)* is a key research point to the project's objective of translating One Button Tracker data into structured clinical insights [10]. Ultimately, this approach aims to promote a more effective therapeutic process by unburdening OBT data translation and improving the connection between psychologists and patients. However, to achieve the desired narratives, the interactions between models and data must be studied.

1.2 Problem

Unlike traditional medicine, in which blood tests and scans provide verifiable and objective data, mental healthcare has long struggled with the collection of simple, reliable patient data outside of a clinical setting. Conventional psychotherapeutic methodologies are based on retrospective self-reporting, a process that is vulnerable to recall bias [5]. Patients often encounter difficulties in accurately recalling the context, intensity, or frequency of symptoms once they are no longer experiencing distress, resulting in data that deteriorates in quality over time [6].

Modern technological solutions, including manual logging applications and EMA tools, frequently face resistance to support and user-friendliness, even when they aim to address this deficit. These solutions often impose a considerable cognitive burden on users, requiring simple actions such as unlocking phones, navigating interfaces, and responding to surveys [5]. This can result in tracking fatigue, characterized by users' reluctance to engage with these tracking mechanisms due to the perceived burden it imposes on their privacy and autonomy [6]. This reluctance can lead to lower adherence to tracking mechanisms and inconsistent data collection, potentially compromising the reliability and validity of the collected data.

The integration of a device such as the One Button Tracker has been demonstrated to effectively address the adherence problem by enabling low-effort, direct, and easy registration of subjective phenomena [8, 6, 11], allowing for more longitudinal studies. However, as mentioned before, this advancement raises an interpretation gap issue. The OBT generates high-resolution temporal logs, represented as timestamps and click durations. However, this data, although promising, is difficult to interpret.

Psychologists, who are not always trained as data scientists, often lack the expertise and time to effectively analyze and interpret this raw information during brief consultations. When presented with raw data or abstract visualizations, clinicians must invest considerable cognitive effort to decipher patterns, correlations, or causal relationships between triggers and symptoms [4]. In the absence of a mechanism to translate these raw digital signals into natural language narratives, there is a risk that the data will become an artifact that distracts from, rather than enhances, the therapeutic process [9]. The challenge is not only to display data, but also to create narratives, that is, to convert structured log data into concise and usable summaries that align with clinical reasoning.

This project examines the viability of an LLM-dependent platform to address this issue. Nonetheless, due to the mismatch between the stateless nature of large language models and the meaningful longitudinal nature of mental health data [10], a new issue arises. Standard LLM interactions are episodic and lack persistent memory, whereas mental healthcare relies on the evolution of a patient's state over weeks or months [10]. A standard prompt is inadequate in this regard, as it does not account for the patient's historical baseline, the specific clinical protocols defined in previous sessions, and the trajectory of treatment. Moreover, in view of the sensitivity of mental health data, the processing of such information requires strict data governance, needing localized inference solutions to avoid privacy risks associated with cloud-based processing [3, 12, 13].

Therefore, the main challenge lies in the development of a methodology that can dynamically assemble and process OBT data [14, 10]. The objective is to translate new observational data into a format that is both comprehensible and advantageous to the therapeutic process, thus ensuring that the technology serves as a bridge between the patient's daily life and the clinician's expert judgment [3, 4].

1.3 Research Question and Aims

The central theme of this thesis addresses the interpretation gap, the significant cognitive load and analytical burden faced by psychologists when attempting to integrate and analyze high-frequency, raw OBT data into therapeutic practice [4, 8]. The primary research question, therefore, explores the potential of a platform using Large Language Models (LLMs) to translate raw self-tracking data into actionable clinical insights, while ensuring data sovereignty and clinical safety [14, 3]. The objective of this approach is not only to assess the platform's viability from a user experience perspective but also to evaluate the quality of the LLM's output against deterministic baselines [10].

In order to answer this question, the research branches into three main objectives. As mentioned before, the core of the project is to address the existing issue by developing a web-based platform that integrates the One Button Tracker with an LLM-driven analysis pipeline [9, 3]. The objective is to minimize the cognitive load by converting raw OBT data into structured narrative insights and visual summaries, transforming clicks into clinical context [4].

Secondly, the thesis explores the optimal architectural balance for clinical AI by implementing a hybrid system. Rather than rely solely on the probabilistic nature of LLMs, which carries a risk of *hallucination* and *lost-in-the-middle* memory failures [10, 14], this research integrates deterministic data processing for quantitative metrics with prompt and context Engineering techniques for qualitative reasoning. This approach ensures that the system functions as a safe support tool that enhances, rather than replaces, professional judgment [14].

Lastly, even if only partially within the project's scope, it assesses the feasibility of *Local Inference* strategies. The use of open-source models, notably *Llama 3* via the *Ollama* framework, is a key step on the research [12]. The objective is to study a cost-effective, privacy-preserving alternative to cloud-based AI that complies with health data regulations by retaining sensitive patient data within a secure, local infrastructure [15, 16].

1.4 Thesis Contribution

This work builds upon previous studies of the OBT, focusing on the transition from a standalone data-acquisition device to a more integrated clinical support system [17]. The OBT has been shown to effectively minimize user burden and recall bias by enabling instant registration of experiences [8, 6]. Its utility has historically been constrained by the significant cognitive load required for clinicians to manually analyze raw and high-frequency data [4]. This thesis proposes a methodology to address this issue. This methodology is based on translating OBT observational data into not only coherent and relevant metrics but also structured clinical narratives. Within the platform, the psychologist should be able to analyse key patient metrics, such as worst days or most-used *rules* within a *protocol*, as well as insights into the patient's data provided by the LLM.

This contribution establishes a link between patient-generated data and its application in therapeutic contexts, thus facilitating a more effective, data-informed therapeutic process without imposing additional administrative demands on practitioners [9, 18]. In addition, a contribution is made to the fields of Health Informatics and Applied Artificial Intelligence by exploring a prompt and context engineering framework specifically designed for analyzing OBT data [14].

This study addresses the critical issue of interpretation in self-tracking for mental health. It does so by exploring context engineering, which considers context not as a static textual input, but as a dynamic, structured assembly of information [14, 10]. Specifically, the project introduces a hybrid architecture that separates quantitative precision from qualitative reasoning. It uses deterministic data processing to achieve accurate statistical metrics, while considering the probabilistic capabilities of LLMs for narrative generation [3]. This approach effectively reduces the risks of model hallucination, ensuring that AI-generated insights remain clinically grounded and safe [10]. In light of the aforementioned considerations, the data variables, European restrictions and regulations, and ethical psychological dilemmas are also analyzed and investigated to determine the viability of the proposed solution [12, 3].

1.5 Thesis Structure

The structure of this thesis, which is organized into six chapters, moves from the theoretical foundations of the research to the technical implementation and final evaluation of the proposed solution.

In the second chapter, the study's theoretical framework is established. The study examines the current state of data usage in psychological practice, highlighting the limitations of traditional self-reporting and the emerging role of *Artificial Intelligence in Digital Health*. This chapter identifies a research gap in the so-called interpretation gap in self-tracking data. It also introduces prompt and context engineering as a key to addressing the complexities of large language model integration in clinical settings.

The third chapter details the research design used to address the main research questions. This chapter dives deep into the approach and the mixed-methods framework that was used for validation. Given the sensitive nature of the domain, this chapter also outlines the generation and use of synthetic clinical datasets to ensure compliance with privacy regulations. Furthermore, in this chapter, the ethical considerations and protocol established are detailed for expert validation with clinical professionals.

The fourth chapter thoroughly documents the iterative engineering process of platform development. This chapter details the technical evolution of the system from a basic prototype to a more robust hybrid architecture, following an agile methodology spanning across six development sprints. The study provides a technical analysis of the decision to implement Local Inference using the Ollama framework and the Llama 3 model to ensure data protection and avoid a custom-built LLM. Additionally, it investigates the focus on implementing deterministic data processing pipelines alongside probabilistic LLM reasoning to avoid the risk of hallucinations.

The fifth chapter of this study is dedicated to analyzing and detailing the findings derived from the testing phases. This includes an examination of the system's efficacy in translating raw One Button Tracker logs into relevant clinical narratives, the outcomes of user experience evaluations, and the results of manual quality assessments. These assessments compare LLM outputs against deterministic ground-truth metrics.

The sixth chapter summarizes the findings in relation to the initial research aims. The results are interpreted through the scope of prompt and context engineering, with the system's ability to bridge the interpretation gap while maintaining clinical safety being evaluated. In this chapter, the possible future approaches for the platform are analyzed. However, it also discusses the implications of a more user-focused design and acknowledges the limitations of the study.

In the last chapter, the thesis's core is summarized, with particular emphasis on the main contributions to the fields of Health Informatics and Applied AI. The study concludes with remarks on the viability of the OBT as an integrated clinical support instrument and proposes prospective research approaches in the domains of app integration, artificial intelligence, and mental health technologies.

1.6 AI Usage

This research used artificial intelligence technologies to optimize investigative and development workflows. Perplexity and NotebookLM served as auxiliary tools for literature review and semantic search, while DeepL was used to refine the linguistic clarity of specific sections. Furthermore, the Codex model aided in identifying and resolving code syntax and debugging issues during the prototyping phases.

Nonetheless, the responsibility for the research design, critical analysis, code implementation, and the written arguments presented in this document rests solely with the author, as these tools were used exclusively to assist the author in research and development.

2 Background

2.1 Introduction

When discussing the background, it is important to establish the theoretical and technological foundations upon which the Decoding the Click platform is built. This chapter provides a review of the relevant literature and studies that served as the basis for defining the areas of research on acquiring data in mental health and on emerging methodologies and techniques for optimizing large language models for sensitive clinical environments.

Beginning with an analysis regarding the importance of data in psychology, particularly the shift from retrospective self-reporting to Ecological Momentary Assessment practices, this study examines the OBT as a tool for capturing high-fidelity subjective data. In this analysis, the limitation identified by the OBT's development, the interpretation gap, is explained and detailed how it applies to this project. As previously mentioned, this refers to the cognitive burden clinicians can experience when processing high-frequency raw data.

The following section shifts the discussion to AI in digital health, a topic that extends beyond the conventional machine learning framework to include LLMs and prompt- and context-engineering methodologies. This section theoretically and mathematically distinguishes between conventional prompt engineering and dynamic context assembly, establishing the basis for using LLMs to convert longitudinal behavioral logs into clinical narratives.

Finally, the section on ethical considerations addresses deployment constraints in health-care. It provides a thorough review of the importance of data sovereignty and the technical trade-offs between cloud-based models and local inference systems for protecting sensitive patient data. The chapter concludes by summarizing these subjects and identifying the specific research gap this thesis addresses, which is the absence of a framework for translating data into narrative-driven therapeutic insights.

2.2 The Role of Data in Psychology

2.2.1 The Limitations of Retrospective Reporting

Historically, psychological assessment and data collection have relied heavily on retrospective self-reporting, where patients revisit experiences during clinical sessions days or weeks after they occur [5, 8, 6]. Research indicates that this approach is vulnerable to what is termed *recall bias*, since patients often struggle to accurately reconstruct the frequency, intensity, or context of symptoms once removed from the immediate moment of distress [5]. This limitation becomes extremely important in conditions such as *Post-Traumatic Stress Disorder (PTSD)*, where memory deficits and avoidance actions can further compromise the reliability of retrospection, resulting in a discrepancy between the patient's real lived experience and the clinician's understanding [8].

2.2.2 The Shift to Ecological Momentary Assessment

To address these gaps, the field has increasingly investigated and adopted Ecological Momentary Assessment. This methodology prioritizes data collection in the patient's natural environment and at the precise moment of the experience. The use of EMA in the capture of real-time data enhances ecological validity [7, 19], thus providing a more accurate perspective on symptom fluctuations that are otherwise often overlooked by retrospective methods [5]. However, traditional EMA tools, which often involve completing repetitive questionnaires via smartphone applications or electronic diaries, can impose a considerable cognitive burden on users, potentially leading to *tracking fatigue* and reduced adherence over time [6].

2.2.3 Quantitative Data as a Intermediary for Qualitative Experience

Within the paradigm of *Personal Science*, the act of collecting data functions not only as a monitoring instrument, but also as a door for introspection and autonomous action [6]. In fact, *passive sensing* often fails to capture the nature of mental states, such as cravings, intrusive thoughts, or changes in mood. Nonetheless, this automatic collection of objective physiological and behavioral data through wearable sensors provides quantitative physiological data [8].

The integration of tools such as the One Button Tracker promotes a shift towards more active self-tracking of these phenomena. In reality, the process involves the conversion of a qualitative experience, for example, the patient feeling anxiety, into a quantitative data point, such as a timestamped click. This transformation actually enables patients to generate a structured data stream that is firmly rooted within their reality or experiences [18]. This *Patient-Generated Health Data (PGHD)* serves as a boundary object in therapy, facilitating shared decision-making and transforming therapeutic dialogue from abstract recollections into more concrete, data-informed analysis of behavioral patterns [9, 4].

2.3 AI and Machine Learning in Digital Health

2.3.1 From Signal Processing to Pattern Recognition

The application of Artificial Intelligence in health monitoring has historically focused on passive sensing, such as the use of algorithms to detect physiological anomalies in continuous data streams. Convolutional Neural Networks and Recurrent Neural Networks are two traditional machine learning techniques that have been successfully used to classify human activities or detect specific medical events, such as arrhythmia, from raw sensor data [1, 3, 2]. In these *passive* paradigms, the purpose of AI is mainly that of a classifier, processing objective metrics, for example heart rate variability or step counts, in order to alert clinicians to physiological changes [1].

As noted in previous works[5, 8], passive sensors often exhibit a delay relative to the psychological experience. For instance, a heart rate spike occurs after the onset of anxiety, and by the time a passive sensor registers the physiological change, the individual has often already passed the point where cognitive coping strategies are no longer effective[8]. However, active self-tracking captures a more *in-the-moment* subjective precursor [8], enabling the individual to be more aware and better able to use coping strategies.

However, a critical limitation of these traditional machine learning approaches is their inability to contextualize subjective experience [3]. Conventional machine learning models struggle to contextualize these sparse signals when lacking extensive labeled datasets, leading to a gap between the data collected and the narrative insights important for therapeutic interventions [3].

2.3.2 The Shift to Generative AI and Context Engineering

The appearance and rising popularity of Large Language Models uncovers the opportunity for a paradigm shift from traditional classification to more human-friendly narratives [14, 20]. In contrast to stateless classifiers, LLMs are usually capable of processing *multi-modal* data, combining numerical data with textual context to generate human-readable health insights [3]. Recent frameworks, including PhysioLLM and Health-LLM, illustrate that when physiological data is augmented with context, LLMs can surpass conventional baselines in terms of providing personalised health predictions and enhancing user experience [21, 22].

Nonetheless, this capability is supported by the concepts of prompt and context engineering, two methodologies explored on LLM optimization [10, 14]. In context engineering, the usage of dynamic context assemblies, as opposed to static inputs, in the treatment of clinical history and behavioral logs allows systems to be constructed with techniques such as *Retrieval-Augmented Generation* to *ground* the model in the patient's history [14, 23]. This enables the LLM to overcome the limitations of generic advice and to offer more context-aware interpretations of longitudinal data, thereby identifying patterns that might otherwise go unnoticed by standard algorithms, such as difficult weeks or coping mechanisms [3, 14].

2.3.3 The Interpretation Gap

The integration of large language models addresses the specific barrier identified in Personal Science, which previous studies noted as the burden of interpretation [6]. Research indicates that patients and their clinicians often struggle to translate raw data into actionable knowledge without support [9], even if they are willing to track data. Active self-tracking instruments, such as the OBT, generate high-value subjective data. However, this data requires interpretation to be useful in a clinical context [4].

Recent studies suggest that LLMs can serve as this bridge by interpreting raw sensor data without the extensive, manually labeled datasets required by traditional machine learning [3]. The combination of deterministic data processing for accurate counting and statistics with probabilistic LLM reasoning for narrative generation has led to the emergence of hybrid architectures as viable solutions to support clinical decision-making while alleviating therapists' cognitive load [14, 3].

This gap still exists among psychologists who are not only more comfortable with technology but may also who might be trained in data science. Some psychology study programs are undergoing restructuring and incorporating data science as a minor component. Given the growing integration of data into psychology, this shift in study programs is extremely important for future therapeutic processes, as it may evolve with the emergence of new solutions.

2.4 Prompt and Context Engineering

During the investigation and development of this thesis, the platform's architecture underwent multiple changes and methodological applications to optimize LLM output. When exploring these optimization methods, first prompt and then context engineering emerged as focal points that would later prove necessary for achieving the desired solution.

2.4.1 Prompt Engineering

Prompt engineering is an emerging field focused on optimizing interactions with generative AI models [24]. The objective of this optimization is to maximize the potential and utility of these models. This way, a prompt serves as the textual interface, communicating a user's desires through instructions, questions, input data, and examples. However, prompt engineering goes beyond the mere construction of these inputs. The practice consists of an iterative process of developing prompts by constantly modifying or changing techniques to achieve a specific output. To engage in this practice, it is first necessary to understand the model's capabilities and limitations, as well as the context in which it operates.

The transition from a mere *tips and tricks* approach to a more rigorous engineering discipline is possible. The field of prompt engineering mirrors traditional software engineering practices, such as version control and regression testing. This practical process is cyclical and generally comprises three repeated steps, starting with *Inference*, in which the prompt is run against a dataset. Afterwards, to iterate on the prompt, the model's output must be assessed against an established criterion, either quantitatively or qualitatively. In the end, the prompt template should be refined based on the evaluation to improve results. As mentioned before, this is an iterative process, so these three steps are repeated until the prompt achieves the desired outcome. In our platform's case, the outcome would be information such as success rates and intensity windows from the patients' data.

2.4.2 Importance of Prompt Engineering

The fundamental objective of prompt engineering is to navigate and avoid the inherent constraints of Large Language Models. The limitations of the model are usually clear in its basic state, characterized by an absence of persistent memory, its probabilistic nature, which manifests as inconsistency across runs, and its tendency towards hallucination, resulting in content fabrication [10]. The purpose of engineering techniques is to impose structure on stochastic systems.

2.4.3 Context Engineering - A Branch of Prompt Engineering

Context engineering is an emerging formal discipline that extends beyond traditional prompt engineering to address the increasing complexity of modern AI systems, particularly those based on LLMs. While prompt engineering primarily focuses on designing static textual prompts to guide model output, context engineering adopts a broader, more consistent perspective [14]. It concerns the structured optimization of the entire information payload supplied to an LLM during inference.

Rather than treating context as a single, unchangeable string of text, context engineering conceptualizes it as a more dynamic and structured composition of varied informational components. These components may include task instructions, external knowledge sources, tool definitions, system-level constraints, and persistent or episodic memory. These elements are selectively sourced, filtered, and orchestrated in order to maximize model performance, robustness, and task alignment [14].

Formal Definition and Objectives

Theoretically, context engineering transforms the notion of context from a static prompt into a dynamic function. Let the context C be defined as:

$$C = \mathcal{A}(c_1, c_2, \dots, c_n), \quad (2.1)$$

where $\mathcal{A}$ indicates a context assembly function, and each c_i represents an individual context component. These components may include static instructions as well as dynamic elements such as user interaction history, retrieved external knowledge, or the current system state [14].

The main objective of context engineering can be viewed as an optimization problem where the goal is to identify the optimal set of context-generating and assembly functions that maximize the quality of the LLM's output, subject to practical constraints such as the model's finite context window and computational limitations. These concepts emphasize the shift from simpler prompt design to more principled, system-level optimization [14].

Foundational Components of Context Engineering

Formally, context engineering is structured into three fundamental phases that together form a comprehensive system. Initially, *Context Retrieval and Generation* focuses on sourcing and constructing relevant information. It includes traditional prompt engineering for instruction formulation, external knowledge retrieval via mechanisms such as Retrieval-Augmented Generation (RAG), and *dynamic context assembly*, which combines multiple information sources into an input representation [14, 10]. Following that, *Context Processing* involves transforming and refining the gathered information. Within these techniques, long-sequence processing is used to handle extended input, self-refinement mechanisms that iteratively improve model output, and structured information integration, including the use of knowledge graphs or multimodal data representations. In the end, *Context Management* addresses the efficient organization and maintenance of information over time. This includes the design of memory hierarchies that distinguish between short-term and long-term context, context compression strategies that maximize informational density within limited context windows, and optimization methods that tackle the inherently stateless nature of LLMs [14, 10].

System-Level Implementations

The principles of context engineering are incorporated in a variety of advanced AI system architectures that enable LLMs to interact effectively with external environments [14].

Retrieval-Augmented Generation, or RAG, systems dynamically inject external knowledge into the inference process and have evolved toward increasingly modular and autonomous architectures [14, 10]. Memory systems aim to emulate aspects of human cognition by enabling persistent information retention across interactions. Tool-integrated reasoning frameworks [25] allow LLMs to call external functions or tools, such as calculators or web browsers, to improve their reasoning capabilities. Finally, multi-agent systems consist of coordinated collections of agents that communicate and collaborate to solve complex tasks, leveraging shared and distributed context representations [26, 27].

Together, these implementations demonstrate how context engineering serves as a unifying framework for building scalable, adaptive, and intelligent LLM-based systems [14].

2.4.4 Prompt Engineering and Context Engineering

Although prompt engineering and context engineering share the common objective of improving the performance and reliability of LLMs, they differ fundamentally in their conceptual scope, formalization, and system-level implications. The progression from prompt engineering to context engineering reflects a broader shift from traditional text crafting towards principled information balance and architectural optimization [14].

Static Text versus Dynamic System Architecture

Traditionally, prompt engineering treats context (C) as a simple and static string of text, typically expressed as:

$$C = \text{prompt}. \quad (2.2)$$

Within this paradigm, optimization is achieved through iterative refinement of textual elements, such as instructions, questions, demonstrations, and examples, as explained earlier. The main challenge lies in exploring the space of possible text formulations, either manually or automatically, to identify phrasings that best evoke the desired model outcome for a given task [14, 10].

In contrast, context engineering views context as a dynamic, structured system. As previously defined, context is expressed as:

$$C = \mathcal{A}(c_1, c_2, \dots, c_n), \quad (2.3)$$

where $\mathcal{A}$ denotes an assembly function, and each c_i represents a distinct informational component, including dynamic elements such as retrieved external knowledge, user interaction history, system state, or tool specifications [14]. Rather than optimizing a static text string, context engineering addresses a system-level optimization problem, seeking the optimal configuration of retrieval, selection, and assembly functions that together maximize output quality [14].

Scope of Optimization

The optimization objective in prompt engineering is typically framed as maximizing the probability of a desired output Y given a fixed prompt:

$$\max P_{\theta}(Y \mid \text{prompt}), \quad (2.4)$$

where P_{θ} is the model parameterized by θ [14]. Techniques such as *Chain-of-Thought* prompting or *Step-Back* prompting [28] can be employed to encourage reasoning within a single inference window, without altering the broader information pipeline [10].

Context engineering expands this objective to include the optimization of the entire inference-time information flow. The goal is to maximize expected performance across a diverse set of tasks by optimizing multiple subsystems [14]. Similar to the iterative nature of prompt engineering, context engineering also starts with an inference or retrieval stage, where knowledge is sourced to maximize relevance or mutual information with the target output. Afterwards, the processing phase involves handling ultra-long sequences, structured representations, or multimodal inputs that exceed the capacity of standard prompts. Lastly, to operate within strict context windows and computational constraints, the system needs to be managed and executed through context compression and memory hierarchies.

Statefulness and Memory

Prompt engineering is mainly stateless, treating each prompt as a new or independent interaction. Although conversational history can be integrated within prompts, each interaction is effectively treated as an isolated instance, requiring all relevant context to be explicitly provided within the current input window [10].

In contrast, context engineering aims to achieve a stateful environment. It explicitly models persistent memory components (c_{mem}) and dynamic system state (c_{state}) as core elements of the context assembly process [14]. This enables the integration of memory architectures inspired by human cognition, such as episodic and semantic memory, allowing LLM-based systems to maintain continuity and coherence across extended interactions [14].

Scalability and Error Analysis

As task complexity increases, prompt engineering typically relies on manual inspection, experimental trial-and-error, and exploratory refinement [10]. Its scalability is constrained by context-window limitations and the increasing fragility of long, densely packed prompts, which often suffer from issues such as the *lost-in-the-middle* [14].

However, context engineering addresses scalability through modular system design. By breaking context generation into distinct functional stages, such as retrieval, filtering, formatting, and compression, these components can be independently evaluated, optimized, and debugged [14]. This modularity enables systematic error analysis, for example by assessing retrieval quality separately from generation performance, and supports the construction of robust, large-scale LLM-based applications [14].

2.5 Ethical Considerations of AI in Health

2.5.1 Privacy and Data Constraints

While investigating the platform's possible implementation, it became important to examine potential constraints on data privacy, particularly regarding data collection and translation in the mental health domain. This approach, which considered using LLMs to interpret data for the purpose of psychological support within the European Union, reveals a deeper web of legal and privacy constraints and risks [29, 30]. Although key decisions, such as the choice of a local LLM and synthetic datasets, were made, an investigation into the potential privacy and ethical consequences was conducted, with particular emphasis on GDPR and AI-specific regulations. This investigation considered the demographic results from user surveys and experimental sessions conducted during the development of this project. The decision to focus on specific national regulations, such as the Danish, Portuguese, and EU regulations, was deemed appropriate given the relative distribution of nationalities within the pool of survey users.

EU-Wide Basis: The General Data Protection Regulation (GDPR)

The GDPR is the main legal framework governing the processing of personal data across the EU. It applies to any entity that collects, stores, or analyzes personal data, including health-related data, which is classified as *special category data*. Processing such data requires a lawful basis, for example, explicit consent or legal/clinical necessity, strict documentation, and specific security measures [29].

Denmark: Local Implementation

In Denmark, GDPR is complemented by the Danish Data Protection Act [31], which adds country-specific clauses. For instance, the Danish government extends privacy protections to the data of deceased persons for up to 10 years and specifies requirements for data controllers, such as health professionals and researchers, to maintain strict records of processing activities and engage in frequent self-auditing. Where these special categories of data, especially mental health records, are used beyond medical or psychological support, like in research or analytics, Danish law often requires preapproval by the data protection authority and always relies on patient consent or clear anonymization.

Portugal: National Variations

Portugal's implementation of GDPR also recognizes health data as highly sensitive, with additional local rules [32]. Firstly, the obligation to separate storage of health and genetic data from other personal data is one of the key factors for mandatory audit trails on health data access and patient notification upon any access or transfer. Lastly, there are limitations on the reuse of health data for research unless vital for public health or with explicit written consent.

Special Requirements for Psychologists and Mental Health Apps

In both Denmark and Portugal, psychologists are bound by confidentiality requirements that frequently exceed the standards stipulated by general privacy legislation. Disclosure of sensitive data without informed consent, except where required for safety or by law, is prohibited.

The use of LLMs for the processing, summarization, or analysis of mental health data is considered *high-risk* according to the GDPR and the EU's evolving AI Act [33]. A Data Protection Impact Assessment is required, particularly in cases involving profiling or automatic decision-making. Systems must be designed to minimize data to the essentials, ensure transparent processing, and restrict foreign data transfers unless special safeguards are in place. Furthermore, the practice must also ensure that, if the LLM vendor is external or cloud-based, data remains within the EEA or is protected to an equivalent standard.

In matters of consent, it is necessary that it be explicit in all instances involving the processing of health data. Additionally, it is essential that users and patients be informed of the processing, storage, and use of their data, and that they are made aware of the parties responsible for such processing. It is necessary to emphasize the significance of autonomy and the freedom to make one's own choices. This includes ensuring that users and patients are granted the right to withdraw. In cases where consent cannot be obtained, processing of health data is permitted only under established exceptions, such as for treatment or public health.

Record Keeping and Accountability

Since both countries follow EU's GDPR, both require comprehensive, written logs of data flows, processing justifications, and security arrangements. Data must be minimized, meaning that only data relevant and necessary for psychological care or research may be retained. Furthermore, organizations must appoint a Data Protection Officer if processing is regular or systematic.

New Risks and Safeguards with LLMs

Given the platform's features and their reliance on an LLM, this investigation was also conducted to identify new risks and safeguards, especially with the surge of new LLMs on the market. AI systems processing health data must be transparent about how information is summarized, the extent of data exposure, and the source of all outputs. Not only this, but profiling and inference are permitted only under very strict transparency and fairness rules. Lastly, more and more there are audit requirements, including audit data, technical and organizational safeguards, and frequent reviews for bias or security flaws.

2.5.2 Bias in AI Systems

The integration of LLMs into clinical practice raises specific risks related to algorithmic and cognitive biases. In contrast to deterministic software, LLMs are vulnerable to training data bias, where the model maintains stereotypes or cultural assumptions present in its training process [3, 10]. In the field of mental health, where the manifestation of symptoms is profoundly influenced by context, the use of a generic model without specific cultural grounding carries a risk of misinterpretation [3, 34].

Furthermore, LLMs exhibit significant positional bias, which is often referred to as the *lost-in-the-middle* phenomenon. Research indicates that models struggle to access information positioned in the middle of long contexts, often preferring to attend to data at the beginning or end of the input [14, 35]. With regard to longitudinal self-tracking, this presents a critical validity risk where a relapse occurring in the middle of a treatment period could be overlooked by the system if the context payload is not properly engineered [14].

Finally, the phenomenon of *sycophancy*, where models tend to align their responses with the user's stated opinions even when they contradict objective data [10]. This threatens the objectivity of clinical support tools. Should a clinician enter a hypothesis about patient improvement, an unconstrained LLM may skew its analysis of data to confirm this bias, rather than providing an impartial data assessment. This way, in order to safeguard against such biases, it is important to consider not only *Vanilla Prompting* [10, 36], but also a robust architectural separation between deterministic data processing and probabilistic reasoning [14].

2.5.3 Therapeutic Process with integrated AI

The integration of AI also carries risks specially when not controlled and grounded. In mental health, patients often are dealing with self-harm, PTSD, and other disorders, in which an AI narrative focused on a patient's data needs to be extremely neutral and, most importantly, careful with the expressed information.

When it comes to decision-making, it always concerns the direct participants within the therapeutic process. It is important to reiterate that, whether the tool is AI-dependent or not, the therapeutic process is conducted primarily between the therapist and the patient. The integration of these tools should not overthrow professional judgment. Instead, it should assist in comprehending scenarios, experiences, or feelings, thereby improving the therapeutic process.

2.6 LLama 3 as Standard Baseline

The selection of the core Large Language Model was a critical design decision, as it directly influences the platform's feasibility, reliability, and clinical relevance. In this project, the Llama3 model was ultimately selected as the baseline LLM due to a combination of practical, methodological, and research-driven considerations.

One of the main motivations for choosing Llama3 is its lightweight nature and ability to run locally. This characteristic significantly reduces dependency on external, cloud-based services, which often impose constraints on cost, latency, data privacy, and long-term availability while still avoiding the need to develop and train a custom model. Given the sensitive and potentially clinical nature of the data handled by the platform, local execution aligns more closely with privacy-aware, clinically responsible design principles [37]. Moreover, local deployment enables greater control over the system pipeline, facilitating reproducibility and experimentation without relying on third-party APIs or online infrastructure.

Another key factor influencing this decision is that Llama3 is a pre-trained model. As mentioned before, the objective of this thesis is not to design or train a Large Language Model, but rather to investigate the viability and behavior of LLMs when integrated into a clinically oriented platform. The project's focus is not on developing a model from the ground up, as it would divert from the core research aim of evaluating interaction quality, prompt design, and clinical applicability. By leveraging a pre-built model, development effort can be directed toward UI development, system integration, prompt engineering, and evaluation, which are more closely aligned with the thesis's research questions.

Despite these advantages, Llama3 also has certain limitations. As a relatively compact model compared to large, cloud-hosted alternatives, it may exhibit reduced performance in complex reasoning tasks, sophisticated language understanding, or domain-specific clinical interpretation. These limitations necessitate careful prompt design and output evaluation, and they motivate the deeper analysis of model behavior conducted in this thesis. Nevertheless, this model was selected as the standard to allow the investigation to address the research questions.

2.7 Identified Research Gap

A critical gap between data gathering and clinical utility is revealed by a review of the recent studies. While the One Button Tracker and Ecological Momentary Assessment have effectively addressed the *acquisition difficulties* by minimizing patient burden and recall bias, they have unfortunately revealed an *interpretation gap* [8, 6]. The existing literature suggests that the high-frequency, longitudinal data generated by these tools frequently lacks immediate semantic clarity, thereby becoming a significant cognitive load on clinicians who must manually review patterns during consultations or the brief time in between [17]. Visualization tools are extremely useful, nevertheless they can struggle to translate raw behavioral signals into the high-level narrative insights that are important for effective therapeutic decision-making.

Additionally, although Large Language Models present a potential mechanism for this translation, moving from quantitative logs to qualitative narratives, current applications usually depend on cloud-based architectures or passive sensing data [3, 38]. There is a lack of research investigating context engineering specifically for active self-tracking data in mental health, where the system must manage longitudinal memory without falling to biases or issues commonly seen with long context windows [14]. Furthermore, a significant gap remains in implementing these models within a privacy-first, local inference framework [12, 3].

This study seeks to address the specific need for a hybrid, prompt-, or context-engineered architecture that bridges the identified gap between patient-generated data and clinical reasoning. This research investigates the viability of integrating a locally hosted LLM into a data platform dedicated to the OBT. The integration would assist clinicians in reading and interacting with patient data, ensuring compliance with strict data sovereignty regulations and resilience against hallucinatory risks commonly seen in probabilistic models [10].

3 Methodology

3.1 Introduction

To pursue the final solution and execute the necessary tests, evaluations, and assessments, a diverse set of methods was established. This chapter delineates the research design, the testing approach, and the data collected.

3.2 Research Design

The research was planned and conducted using a mixed-methods approach structured around an iterative, agile framework. This approach was chosen in order to not only build a technical tool, the OBT platform, but also to validate its utility within the domain of mental healthcare. This way, the study integrates both qualitative and quantitative methods throughout six developmental sprints to ensure the achieved minimum viable product is not only technically robust but also user-friendly.

3.2.1 Qualitative Methods

The qualitative methods used in this study were applied to examine the interpretation gap and the platform's user experience, using exploratory methods to define the system requirements and assess if the solution is clinically fit.

To assess ethical boundaries, explore the system's architecture requirements, and identify priorities, an interview was conducted with a clinically and company-experienced psychologist. This was an important step that led to analyzing and mapping the clinical needs to system features.

Throughout the development phase, specifically on the third, fourth, and sixth sprints, one-on-one user sessions with Think-Aloud were conducted. Using a synthetic patient scenario based on a real OBT dataset, participants were asked to analyze and articulate their thought process, yielding valuable insights into the clarity of the visualization, the quality of the LLM response, and the overall user experience.

Testing Framework

To ensure a stable and replicable testing framework, a task list covering the major platform features was created. The objective was to construct a viable testing session that could be used on multiple occasions and establish a foundation for the following in-person tests. The experimental phase of the study involved conducting tests with users from diverse backgrounds in individual sessions. The need for local sessions rather than online surveys stems from the fact that all components of the web application, including the LLM, run locally. This decision was made on the basis that the deployment of the web application was considered to be beyond the scope of this thesis.

It is important to note that the sessions were conducted one-to-one. In these sessions, participants were tasked with exploring the app to assess the quality of the features under development. Prior to the session, each participant was provided solely with the necessary context to use and navigate the application, with no additional input to avoid introducing potential biases. The established context was simple. The participants were guided through the process of analyzing a hypothetical scenario, namely Alex R.'s, a fictitious persona with a coffee addiction, as if they were psychological analysts. The subjects were informed of the nature of the data that the OBT was collecting, as well as the rules defined with Alex R.'s protocol for the study of his disorder. After the briefing, participants were given the opportunity to use the web application and evaluate its features, followed by a think-aloud session.

Task List:

- Task 1: Review the key and data insights
- Task 2: Ascertain which week presented the most significant challenges for Alex.
- Task 3: Ascertain the day that proved most challenging for Alex.
- Task 4: Investigate the list of cravings and logs.
- Task 5: Analyse the dashboard and report metrics.
- Task 6: Interact with the LLM for more information.
- Task 7: Provide suggestions for the next sessions with Alex.

In later iterations, an additional task was added to enable proper testing of new features, such as clinician notes.

3.2.2 Quantitative Methods

Complementing the qualitative methods, the quantitative methods focus on technical validation and measurability.

To evaluate the AI's reliability, a testing framework was engineered to compare LLM outputs against a deterministic ground truth, calculating specific accuracy rates to quantify hallucination and numeric drift. These assessments were done manually and tracked the quality of the outputs when applying new engineering techniques.

Nonetheless, at the beginning of this study, an online survey was distributed to collect demographic data, study our target population, and assess technology familiarity scores and adoption constraints. This study would aid in selecting approaches and making choices during development. Another survey, more focused on user feedback, was conducted early in the iterative process.

3.2.3 Iterative Process

As mentioned before, the research followed an agile methodology spanning six sprints, based on and adapted from an informal, individual version of SCRUM [39]. This iterative framework enabled quick prototyping of the One Button Tracker platform, with findings from qualitative user feedback informing quantitative engineering tasks in the next sprint. Additionally, given the individual nature of this thesis, an agile methodology offered flexibility and continuous adaptation throughout the development and documentation process.

3.3 Data Collection and Analysis

Throughout the project's iterative process, a diverse set of data collection methods was used. Regarding the datasets used to develop the platform, they were synthetically generated from authentic OBT datasets collected in previous studies on the device. Additionally, relevant user information was collected. This information pertained mainly to adherence, technology, and mental healthcare, as well as demographic details, collected through user surveys conducted at the study's beginning and during its first iteration. Afterwards, some participants were recruited to provide iterative feedback on qualitative user experience and platform output, and reached out to an expert psychologist to undergo an interview process focused on clinical viability and output.

3.3.1 Dataset

In this study, a synthetic, generated dataset was used. This dataset was designed to simulate the high-frequency, longitudinal data streams characteristic of the One Button Tracker (OBT). To ensure the clinical relevance of the platform while avoiding investigated ethical and privacy risks during the development phase, the data was centered around a fictional patient persona named Alex R., who is undergoing therapy for a more socially tolerated dependency, specifically, coffee addiction.

The dataset was not generated as a static file but rather via a Python script that produces precise structures, realistic timestamps, and adherence to specific OBT logging formats. Based on the OBT format, the generator script defined core settings such as start date and length, and for each day, it sampled the number of logged experiences to generate and the press duration for each click to simulate a raw OBT dataset. While the initial iteration covered a two-week period, the dataset was expanded in later sprints to 84 days and eventually 90 days to rigorously test longitudinal visualization capabilities.

Data Structure and Observation Protocol

As with previous OBT studies, the dataset comprises time-stamped button clicks that simulate real-world patient adherence. These raw logs are interpreted according to a specific observation protocol defined for the fictitious patient, Alex R. Most patients rely on a two-rule protocol defined with their psychologist for their therapeutic process. However, to test the platform in a slightly more challenging but realistic scenario, the main dataset was constructed to contain three rules.

This means that the protocol maps specific physical interactions with the OBT to distinct behavioral events. The first rule, represented by a single click, indicates an intense craving or an immediate urge for coffee. The second rule uses two consecutive clicks to represent a higher-stress or fatigue trigger, identifying the context or antecedent of the craving. Lastly, the additional rule for this scenario was conducted via a single long click, serving as the success criterion and indicating that the patient had successfully used an alternative strategy, such as water or a walk, instead of coffee.

Overall, these three rules mainly focused on creating a realistic yet thorough scenario for the LLM and the platform to address, enabling the study of the viability of both the platform and the LLM individually and together.

In later iterations, not only was the dataset further randomized to simulate different load conditions, creating a distribution with 4–23 events per day, but an additional patient with a two-rule protocol was added to ensure that peak days and risk windows were non-repetitive and realistic, as within most OBT case studies, protocols operated with one or two rules.

A Synthetic Dataset for Coffee Addiction

Based on the prior investigation, the decision to use a fictional coffee addiction scenario rather than real clinical data was derived from three main factors.

As previously mentioned, the platform was implemented locally to avoid operational costs and, more importantly, to prevent exposure of real patient information. This way, using synthetic data also avoided the privacy risks associated with processing real Protected Health Information during the development and testing of the LLM pipeline.

Ethically, coffee consumption was chosen as a socially accepted dependency, as opposed to more serious dependencies or disorders that had previously been addressed in the OBT context. This enabled the evaluation of the system's feasibility and viability and the LLM's insights without suffering the immediate ethical risks, necessary training, and context associated with monitoring severe disorders such as self-harm or alcohol abuse. However, it still allows for future generalization to these conditions.

Very early in the research process, it became clear that the project was lacking optimal data. This meant that the development would have to deal with the lack of access to high-quality, pre-existing therapeutic logs where the platform could be integrated. When the real One Button Tracker data was introduced, the decision was to adapt the mock data to mimic the real data and enable the project to continue to proceed independently while still effectively simulating the therapeutic context.

Use for the Dataset

The dataset was essential to different functions throughout the iterative development process. During the qualitative testing sessions, it provided a consistent narrative context. Participants were briefed on Alex's protocol and asked to role-play as psychological analysts. This enabled them to evaluate the dashboard's features and the clarity of the visualizations without requiring an understanding of complex or obscure medical histories.

Furthermore, maintaining full control over how the dataset was generated while ensuring its integrity compared to the OBT enabled thorough platform testing, as the dataset was randomized to create variable peaks. This allowed comparisons between, for example, high-volume and low-volume days. This was essential to verify that the platform's features could maintain readability and visual consistency under different data loads.

Lastly, the synthetic nature of the data was important to create a deterministic ground truth. As the data was script-generated, the exact number of cravings, triggers, and success rates could be determined. This enabled the construction of the evaluation pipeline that compared the LLM's narrative outputs against these known values, quantifying hallucinations and numeric drift, and verifying whether the model was accurately interpreting the logs rather than inventing statistics.

3.3.2 Survey Design

Data collection began with a structured online questionnaire hosted on Google Forms and distributed via social media, which biased the Portuguese and Danish networks of contacts as they were more accessible. The survey was designed to support a user-centred design approach, ensuring that the development of the One Button Tracker platform was based on the actual habits, privacy concerns, and clinical needs of the target population. The questionnaire contained 21 items and could be completed in approximately 5 to 10 minutes. It was divided into four focused sections.

The survey began by collecting data on age, gender identity, country, and occupation. In light of research on privacy and data constraints, it was important to investigate cultural differences and local politics that could influence adherence to digital health tools. Furthermore, collecting location data enabled an assessment of whether the platform needed to adapt to specific regional regulations or cultural attitudes towards therapy, such as GDPR variants. Occupational data helped determine whether the user base was predominantly tech-savvy or from the general public, which influenced the required complexity of the interface within the clinical context.

Participants were also asked to rate their comfort with new technology, identify their primary smartphone operating system, and report the frequency of their current use of a personal-tracking app. These questions were designed to assess technical feasibility and the risk of tracking fatigue, building upon previous studies. Understanding the split between iOS and Android users was important for deciding whether to build a native mobile app or a responsive web application that could serve mobile and desktop environments. Additionally, measuring tracking frequency helped determine whether the OBT's low-effort value proposition addressed a genuine point for users who never or rarely track data due to the burden of manual entry.

To identify barriers to adoption and privacy concerns, users were asked to select their biggest concerns and rate the importance of simplicity and ease of use, which was critical for validating the system's development decisions. Higher levels of concern about privacy and security directly support the thesis's core technical contribution and the use of local inference rather than cloud-based processing to ensure data sovereignty. Additionally, the focus on simplicity would validate the minimalist hardware design of the One Button Tracker over complex, multi-modal sensors.

The survey also served as an opportunity to publicize the research being conducted, and, more importantly, its mental health context. Questions regarding the user's background investigated their clinical history, the nature of their struggles, what types of data they would be comfortable sharing, and which insights would be most valuable. This section informed the context engineering strategy implemented later in the development process. Understanding that users prioritize identifying triggers and causes meant that the prompt engineering for the LLM could be adapted to focus on causal relationships rather than generic summaries. Furthermore, determining whether users viewed data as a personal tool or as something to share with a psychologist informed the design of viable directions for the final prototype.

In a similar vein to some of the previous questions, users were finally asked about their interest in and potential involvement in the project.

During the early stages of development, another, smaller survey was distributed to gather input on low-fidelity prototypes and to understand the expectations for the output from the large language model.

Upon being presented with phone prototypes, the users were asked about the clarity of the interface's design and labels in communicating the sections and navigation. Nonetheless, to encourage more open-ended feedback, users were asked about the types of information or feedback they would expect to see in this layout. Regarding the dashboard prototypes, users were asked the same questions as for the phone prototypes. These questions ranged from direct, linearly scaled questions that focused on the layout and presentation of information to open-ended questions. Similarly, to better understand the platform's desired features, a new question was posed to testers about the types of charts and LLM integration they would find appealing. This feedback proved to be a key factor in shaping the direction and focus of development in later stages.

In the final section of this survey, an attempt was made to undertake a user quality assessment of the Key Insights and to engage in a limited interaction with a tuned LLM within the context of the OBT. In this study, the hypothetical coffee addiction dataset was used for the first time. In this section, the objective was to provide a more comprehensive illustration of the agreement type and its application in a clinical setting. However, to further investigate how a possible LLM integrity issue could occur, users were inquired once again, this time with the scenario and protocol, about the kinds of questions they would ask the LLM, both as a psychologist and as a patient.

The final assessment was designed to evaluate the model's performance. This approach was adopted to present the initial key insights generated by the trained model. As seen in Figure 4.2, the users were presented with a selection of the early insights that had been generated by the LLM. Lastly, a test Q&A session derived from the scenario and interaction mentioned before with the LLM was presented, Figure 4.3 and A.1, and a final question regarding the completeness, detail, suitability, and usefulness of the LLM for clinical consultation was asked.

3.3.3 Survey Data Analysis

The survey data was analyzed using a descriptive statistical approach to establish a baseline user profile and assess the viability of the proposed technical features. The raw data, exported as a structured CSV file, was processed to extract demographic distributions, technological expertise, and adoption barriers, along with their relative weights for each insight.

For continuous and ordinal variables, such as scale-based comfort questions, standard measures of tendency were calculated. Specifically, the mean and median scores were calculated to quantify the target population's digital literacy. Nevertheless, categorical data, including age, gender, and occupation, were analyzed using frequency distributions to identify the most common demographic segments. For geographic questions, responses were segmented by location to assess cultural or regional biases, revealing a split predominantly between Portuguese and Danish respondents, as mentioned before. As for device usage, smartphone operating system data was aggregated to determine the necessary development frameworks, which supported the decision to build a web-based responsive application rather than a native app. For multiple-choice questions, an aggregation method was used. Responses were parsed to calculate the percentage of support for each specific item relative to the total sample size.

Overall, to interpret the data distribution effectively, the processed metrics were visualized using Python plotting libraries and the visualizations available in Google Forms. These visualizations ranged from histograms used to display distributions to horizontal bar charts constructed to rank and aid analysis of user preferences and priorities, while heatmaps were used to map the geographic concentration of participants.

4 System Development and Architecture

4.1 Introduction

As outlined in the methodology chapter, the development of the OBT platform followed an iterative process spanning across six sprints. The sprints contained tasks in investigation, development, testing, and documentation, with each sprint's progress dependent on the preceding one. The platform's development was directly influenced by feedback received in each testing session. Consequently, these iterations were constructed sequentially in order to optimize the development phase.

The qualitative user feedback received during the testing sessions constituted a direct review and evaluation of the platform at the time, given its stage of development. Although this feedback was derived from a qualitative test session, it is presented alongside its iterative development to facilitate a deeper understanding of the decisions and changes made at each stage.

This chapter provides a comprehensive examination of the system's developmental process, structured into three primary sections: early development, prompt engineering platform, and the final hybrid platform. The objective of this chapter is to provide a detailed description and analysis of the modifications made in response to user feedback collected during the various stages of development.

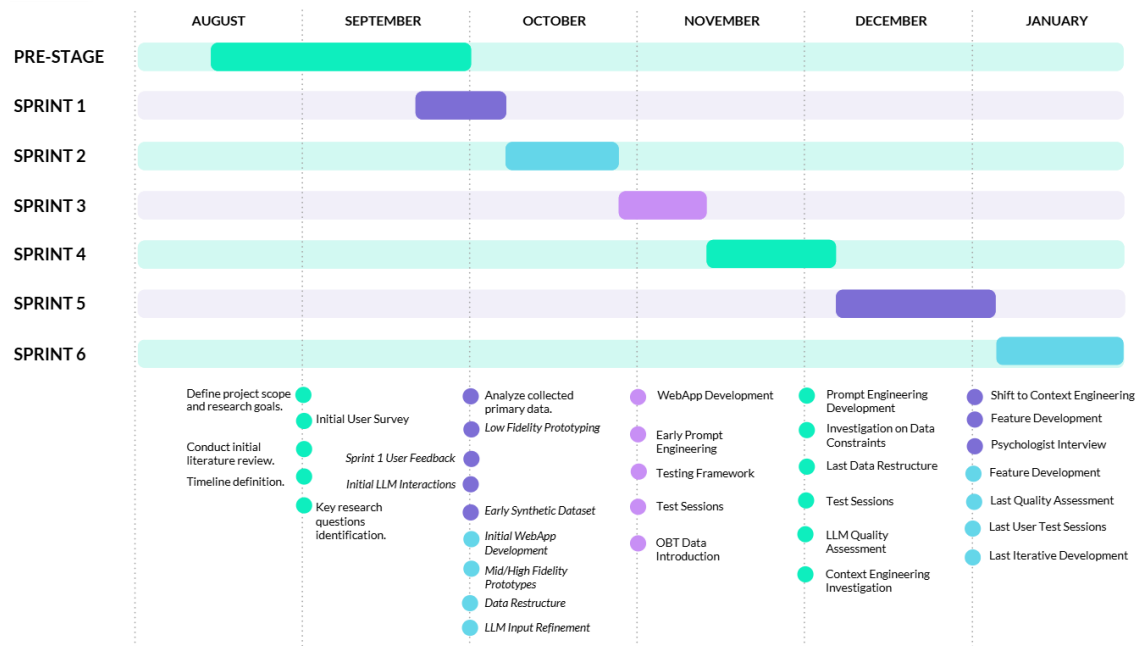


Figure 4.1: The updated iterative process. The timeline provides a realistic reflection of the duration of each sprint and the major tasks and changes conducted during it. This Timeline has been constructed on the foundation of the initial iterative plan, with adaptations to reflect the changes that have occurred during the developmental process. The investigation commenced with an examination of the developmental progress from the initial prototypes to the final development, thus encompassing the entire developmental trajectory.

4.2 Early Development Stages

During the early iterations of development, the focus was to settle on a starting direction and proceed while investigating and studying the possible platform. This way, by building on the user survey conducted at the beginning of the thesis and the literature reviewed, the first iteration laid the foundation for the user experience, LLM usage, and synthetic data handling. This sprint was designed to validate core assumptions about the project's technical feasibility and user interface. By using synthetic data and pre-trained models, it was possible quickly prototype and test the fundamental pipeline while avoiding risks in the project for future, more complex iterations. The emphasis on performance over clinical accuracy at this stage was crucial to ensuring a functional system from which to build.

In an initial exploration, it was considered using *Hugging Face* models with *CPU offloading* for *local inference*. However, this approach was abandoned due to anticipated performance bottlenecks, such as latency and resource constraints, incompatible with quick, iterative testing [3, 13]. The toolchain was decisively shifted to leverage Ollama as the local execution environment [37], hosting the Mistral 7B parameter model [40], and later changed to Llama 3 [41].

This architectural decision not only enables high-speed, local inference execution and decouples the project from cloud resource dependencies, but also creates a stable baseline environment against which all future fine-tuned models can be directly compared. The Python Ollama client was successfully integrated to manage all API communication.

Data generation was a critical step in validating the technical feasibility of the narrative pipeline. The initial plan called for using a tool to generate a synthetic dataset simulating a plausible user-tracking scenario. While the core dataset elements were later generated programmatically via a Python script to ensure precise structure and adherence to OBT logging format, this script successfully produced the three required interdependent components. Initially, the conversation context consisted of a narrative text block that defined the patient's background and therapeutic objective. Most importantly, the script also handled the OBT-defined protocol, represented as a dictionary that defines the meanings of the three rules: Single Click, Two Consecutive Clicks, and One Long Click. Lastly, the synthetic two-week log of time-stamped clicks simulating real-world patient experiences.

Generating this documented, synthetic dataset enabled quick prototyping and testing of the LLM's analytical capabilities, avoiding reliance on real clinical data and reducing risks in future, more complex iterations. This dataset, together with the previously conducted LLM integration, resulted in a simple test interaction, as shown in Figures 4.2 and 4.3.

Start of LLM Output**KEY INSIGHTS:**

1. Alex R.'s coffee cravings are frequent and persistent, with an average of 4-5 cups daily as reported. The OBT logs show that cravings occur multiple times a day across all days, peaking on Mondays and Tuesdays.
2. Stress and fatigue seem to be significant triggers for Alex's coffee consumption, as indicated by the high number of 'High Stress/Fatigue Trigger' clicks. These triggers are particularly prevalent in the morning (7 AM – 10 AM) and late afternoon (3 PM – 6 PM).
3. Coping mechanisms used to address coffee cravings are sporadic, with only occasional 'Coping Success / Alternative Used' clicks. The use of these alternatives could be increased for more effective habit disruption.

These insights suggest that Alex R.'s coffee addiction is a recurring issue throughout the day and is likely triggered by stress or fatigue. Encouraging the use of pre-defined coping mechanisms may help break the automatic habit loop associated with coffee consumption.

End of LLM Output

Figure 4.2: The initial LLM interaction with direct insights into Alex R.'s context and rules. The initial implementation and configuration of using the Mistral7B model was later switched to the Ollama3 model. In this instance, the model demonstrated the ability to interpret basic context, patient description, and protocol definition. During this interaction, the data structure was simplified to a more processed version of OBT data, using rules rather than clicks. This interaction was demonstrated in the course of Sprint 1's User Survey.

Here is the test Q&A:

First Question: Should I, as the psychologist, suggest a 4th or a 5th rule for Alex?

Mistral7B Response:

Based on the data provided, it appears that Alex R. is using the pre-defined OBT protocol effectively to log his coffee cravings and associated triggers (stress/fatigue). However, the use of alternative coping mechanisms seems sporadic.

To encourage more frequent use of these alternatives, I would recommend introducing a new rule:

4. THREE CONSECUTIVE CLICKS: 'Unsuccessful Coping Attempt / Alternative Not Used' Meaning: The patient attempted to use a pre-defined coping mechanism but was unsuccessful in preventing coffee consumption (e.g., deep breathing did not reduce craving). This will help identify patterns and areas for improvement in the development of more effective coping strategies.

Second Question: On our new session, Alex has told me that, on Mondays and Wednesdays, a cup of coffee helps him focusing on football practices in the end of the day. What does the data say about this?

Mistral7B Response:

The OBT logs show that Alex experiences high stress/fatigue triggers throughout the week, with a slight peak on Mondays and Tuesdays. However, there is no specific pattern indicating that he uses coffee to help him focus during football practices at the end of the day (17:00–19:00) on Wednesdays or any other days.

To better understand if this habit exists, it would be helpful to ask Alex about his coping mechanism usage during those specific time slots or consider adding a new OBT rule to log his state during football practices (e.g., 'Coping Attempt / Alternative Used During Football Practice'). This additional data could help pinpoint whether coffee consumption is indeed an effective coping strategy for Alex during these practices.

Figure 4.3: Second LLM Interaction - Example of Psychologist-Model Interaction. As in the previous interaction, this also served as a test object in Sprint 1's User Survey. In this interaction, a simple conversation is simulated to inquire about Alex R's context and to analyze the LLM's responses. The LLM adopted a direct approach, frequently offering suggestions and tips, adopting a stance that was more direct and less intrusive than a neutral stance. One last question was asked and shown in the user questionnaire. This question can be found in A.1.

Based on insights gathered from initial user questionnaires, early user interface prototypes were developed using Figma[42]. The emphasis at this stage was on low-fidelity design to rapidly validate core assumptions about user flow and information architecture, ensuring the design aligns with user needs early in the project before significant development resources were allocated. These prototypes would establish a solid foundation for a standard application, following a structure similar to that of fitness and health tracking applications. The resulting prototypes are shown in Figure 4.4 and Figure 4.5.

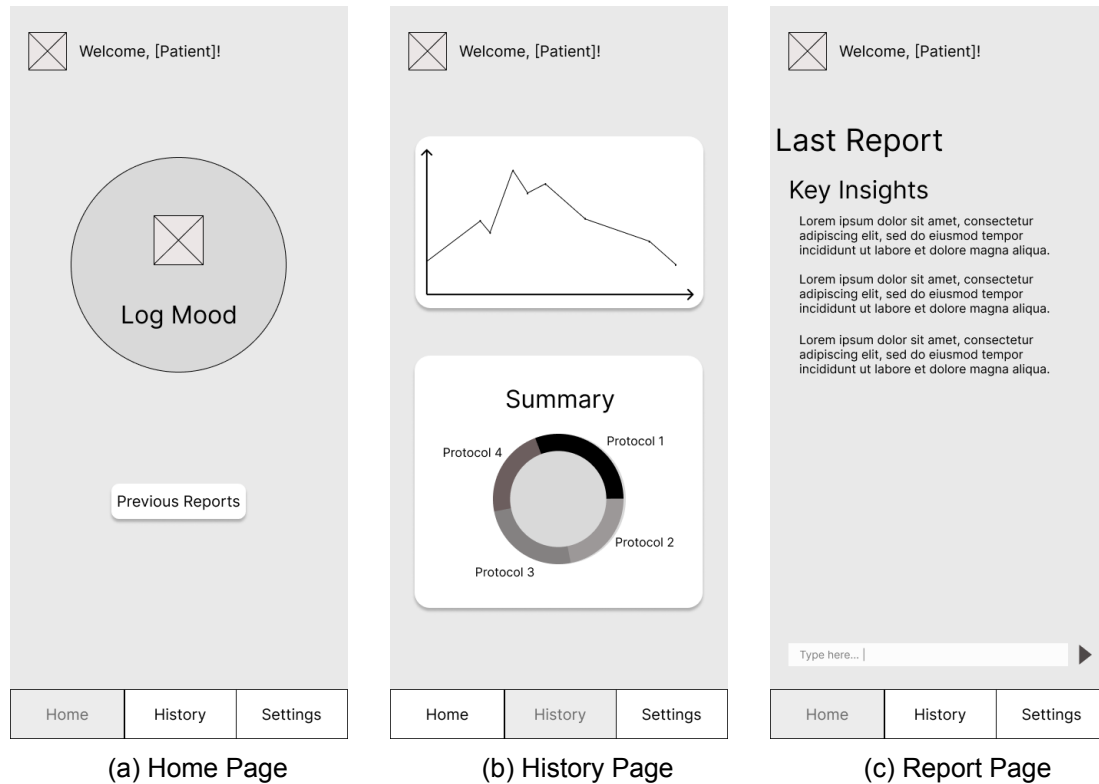
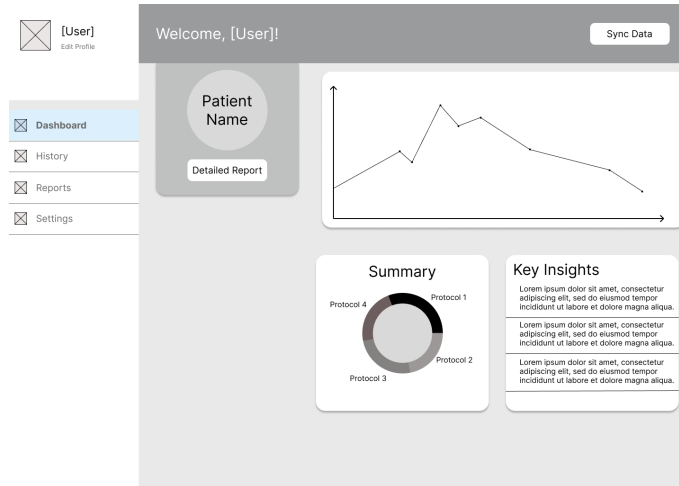
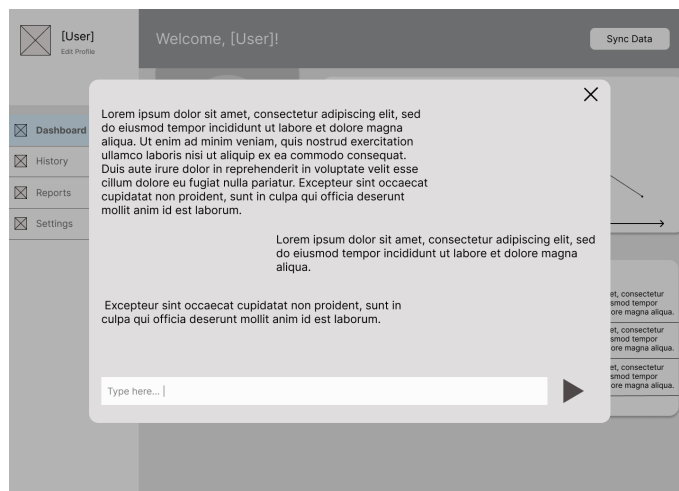


Figure 4.4: Low-Fidelity Interface for Mobile Phones. Built upon the initial user survey, the low-fidelity interfaces include the major features retrieved from the initial study conducted. Among these features, priority was given to an informative dashboard, a report-style page with an option to interact with the LLM. These features would later be changed, and the phone prototypes disregarded to focus on a desktop WebApp view.



(a) Home Page



(b) Chat Page

Figure 4.5: Low-Fidelity Interface for Desktop. Similar to the phone prototypes, the desktop prototypes were also built upon the initial user survey. As such, the same features of reporting, LLM interaction, and an informative dashboard were prioritized. Unlike phone prototypes, these prototypes served as the starting point for further development, as they were iterated on and refined in subsequent iterations.

Following the first iteration and the feedback received, one of the main objectives was to transition from conceptual validation to foundational technical implementation. This involved converting low-fidelity designs into actionable mid- to high-fidelity prototypes, establishing a front-end development framework with React, and defining a rigorous data structure to support a clinical context for the LLM. So, this development cycle focused on stabilizing the core application structure and elevating the user interface from basic wireframes to high-fidelity, component-based prototypes. The work successfully established a robust design foundation and expanded the application's core feature set. By leveraging modern web frameworks React and Tailwind ([43], [44]), the entire client-side interface was self-contained and highly maintainable. At this point, the focus shifted to desktop prototypes, deemed more reasonable given the emphasis on the psychologist's perspective.

All UI elements, including headers, metric cards, and navigation, were formalized into reusable components in accordance with the defined design blueprint. Two critical clinical views were designed and prototyped, moving beyond the initial Dashboard scope, where the History View, Figure 4.6, focused on data density, scannability, and client-side filtering controls to manage a large chronological log of OBT events and the Report View, Figure 4.7, established a clean, print-friendly structure with clear section headers, designed to be used as a final clinical document. The transition to high-fidelity prototypes ensures that the application complies with strict aesthetic guidelines, making the final product ready for integration into a clinical setting while still adhering to modern design principles ([45]) to optimize the user experience. At this stage, a significant portion of the Dashboard was relying on visual placeholders. Nevertheless, it was about successfully defining where the data charts should go, but the actual implementation of complex data visualization libraries was still pending.

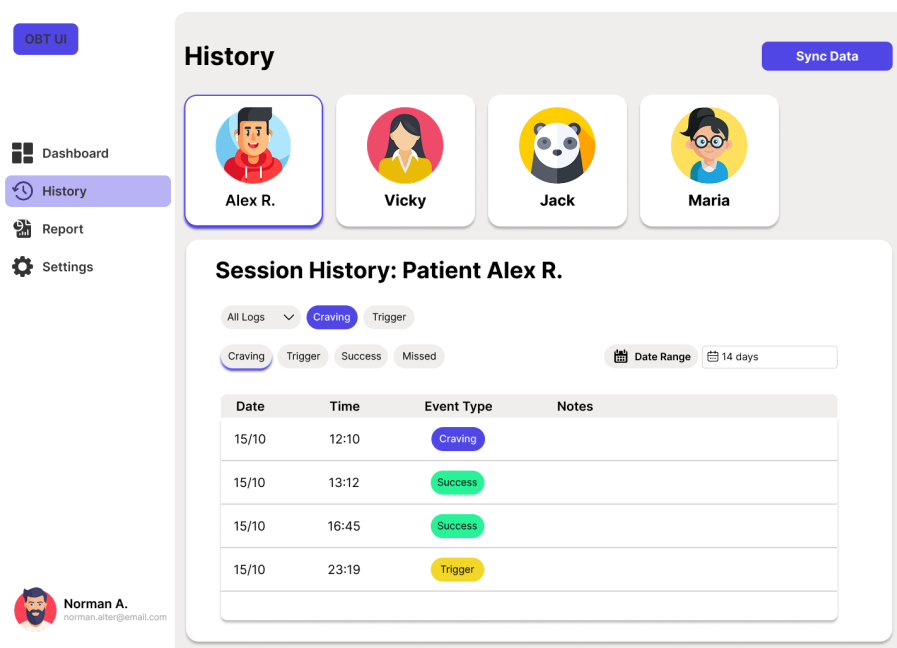


Figure 4.6: High Fidelity Prototypes - History Feature. Definition and prototyping of the History feature within the platform. The history feature was initially defined as a list of logs to still incorporate the raw OBT logs into the platform. This way, even if the summaries and narratives represent the core features, users would still have access to the OBT logs within the platform. With the high-fidelity prototypes, the views were guided towards a more modernized, clinical feel while still following strict aesthetic guidelines.

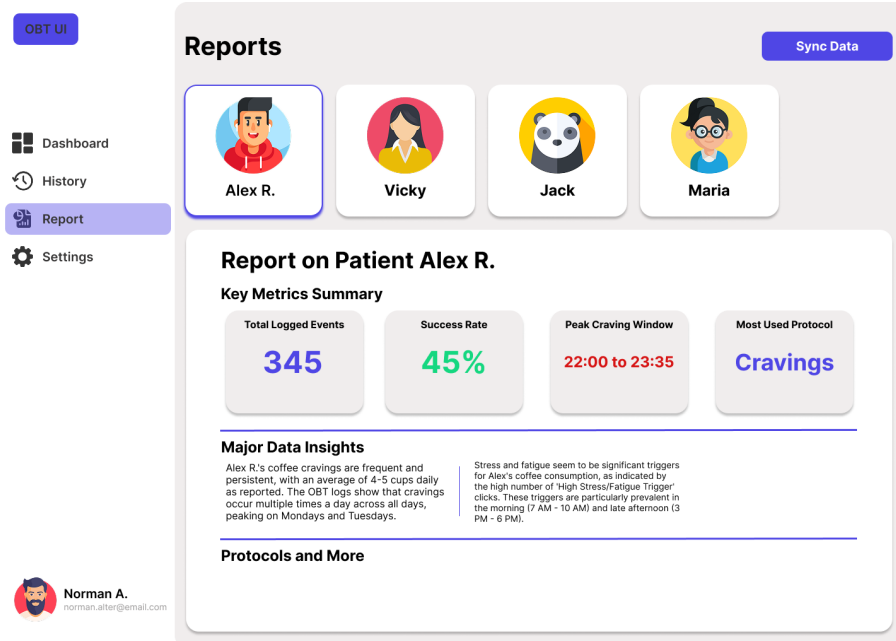


Figure 4.7: High Fidelity Prototypes - Reports Feature. During this iterative stage, the Report feature transitioned to a more concise, text- and print-friendly way of presenting information, while avoiding the graphs and visualizations that would have been incorporated into the dashboard views. This feature was designed as the ideal way to use the platform during therapy sessions to avoid the information overload that the dashboard could cause.



Figure 4.8: Mid/High Fidelity Prototypes - Dashboard. Seen as the main and core feature/view of the platform, containing information from graphs to LLM interaction and narrative insights. The dashboard was designed to serve as the main deep dive into a patient's data and insights, without using the LLM chat feature. The information included within the dashboard at this point was also based on the early user questionnaires.

The final design successfully implements the required color palette and styling for a professional clinical tool, Figure 4.8.

By the end of the early iterations, one key achievement was the construction of a working Minimum Viable Product that would serve as the basis for the rest of the development. This way, a full-stack prototype that follows the dashboard in Figure 4.8, while wiring every surface to LLM-derived insights, was built. The React/Vite front end reproduces the sidebar, patient card, trigger timeline, donut, narrative blocks, and chat console. Each component is received from live JSON delivered by the Python service, so the visuals refresh when new simulations run or follow-up questions are asked. This also meant it was necessary to engineer a Flask backend that could simulate 14 days of One Button Tracker logs, aggregate counts into patient metrics, query the local Llama3 model via Ollama to generate structured dashboard prose, and expose endpoints for the UI to consume. However, this pipeline still contains a deterministic fallback that preserves functionality if the LLM is unavailable.

At this point, common prompt engineering practices were introduced to provide conversational analysis by persisting the LLM context per session. After the initial call, clinicians could ask follow-up questions via the chat panel. The backend then injects the latest metrics and raw log excerpts into each prompt, ensuring that the model's responses remain grounded in the generated data.

These early iterations defined and established the foundation for a deeper dive into prompt engineering for future iterations.

4.2.1 User Feedback

During the early stages of development, a user questionnaire was conducted, focused on early prototypes and LLM interaction. The responses indicated a favorable reception. Nonetheless, to encourage more open-ended feedback, the testers were asked about the nature of the information or feedback they would expect to see in this layout. A close examination of the responses reveals an overarching emphasis on the presentation of data patterns and the manifestation of mood fluctuations. In these responses often was mentioned the need of the graphs being accompanied by a report or a summary of the data being visualized. Lastly, a comment was made about the significance of a neutral background in encouraging users to remain in the application and continue entering additional information. This was a crucial aspect to consider in development, as it exemplifies the importance of user experience. Interestingly, when asked about desired changes, the testers gave only minimal responses, aside from renaming a label and adding more detail and interactivity to the graph.

Regarding the dashboard prototypes shown in Figure 4.5, the testers were asked the same questions as for the phone prototypes. These questions ranged from direct, linearly scaled questions that focused on the layout and presentation of information to open-ended questions.

In line with the phone interface's configuration, testers approved the direction and organization of the information presented. Nevertheless, as previously, the testers were asked about their preferences and potential modifications to the layout. They indicated a need to switch between patients, a feature that was not a primary focus. Additionally, they proposed enhancements to the alerts, number limits, and the overall comprehensiveness of the data and information displayed.

A new question posed to the testers asked about the types of charts they would find appealing. They exhibited a pronounced preference for timeline charts, with a small mention of a possible heatmap feature.

4.3 Prompt Engineering Architecture

The following development cycles focused not only on expanding and exploring LLM-focused features but also on data restructuring and user experience. This approach builds on the structure established in previous iterations, integrating the LLM with features identified as necessary in the initial user surveys or prior OBT studies. At the end of this development stage, the platform was already seeing significant improvements and was organized mainly as a linear pipeline (Figure 4.9).

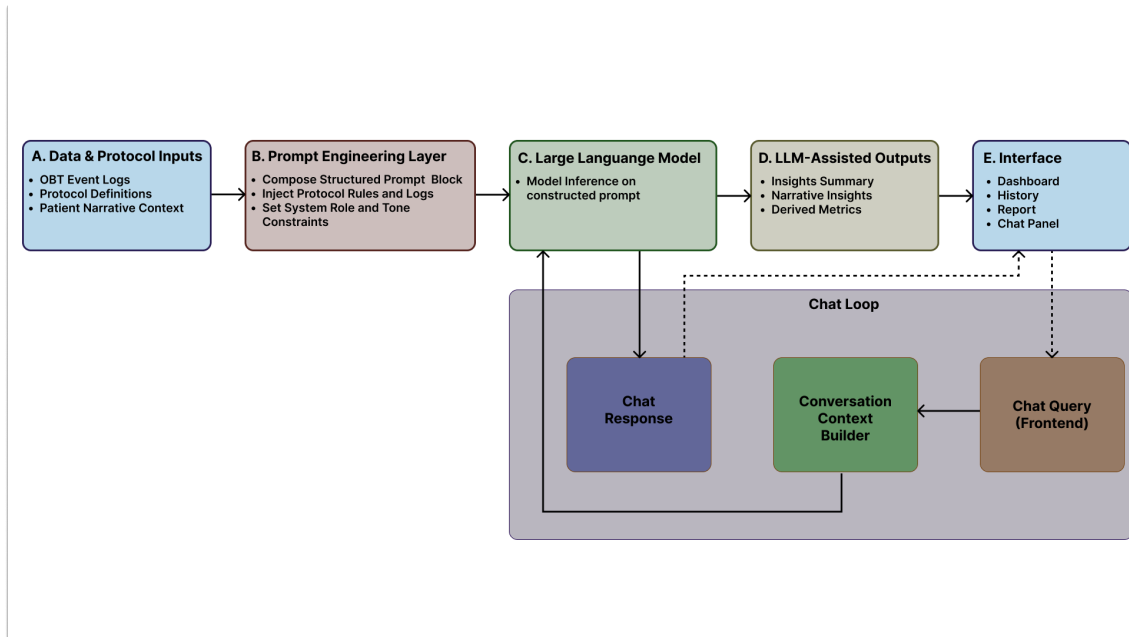


Figure 4.9: Prompt Engineering Architecture during Sprints 3 and 4. The architecture that was in place before context engineering was implemented. It's a linear architecture with a preparation phase, with the data and protocol layer aiding the prompt engineering Layer in constructing the prompt block, roles, and constraints to be injected into the LLM. From the LLM, the platform derives its main information and insights. The chat loop represents the architecture around the chat feature, which requires a constant connection to the LLM.

The front-end feature set was extended to facilitate clinician cognition during evaluative walkthroughs. Within the application's source code, the Weekly Highlights module, Figure 4.10, was re-imagined as a gradient-based temporal track. In this new iteration, each week is represented by a floating capsule that shows the dominant rule category, an LLM-generated insight, and the relative intensity of logged events. Complementary style primitives were introduced to ensure consistent color semantics, geometry, and hover functionalities. At the same time, a high-density OBT heatmap was created and, although initially incorporated into the dashboard, was later migrated from the congested dashboard column into a new *Patterns* workspace, accessible via the navigation sidebar. This dedicated page uses a compact, scroll-friendly layout for daily activities. Collectively, these feature additions translate the previously developed analytics into narrative objects that can be tested during think-aloud evaluations.

Furthermore, the refined UI components provide discrete checkpoints for measuring comprehension. The dedicated Patterns page isolates the high-volume heatmap, so moderated sessions can focus on pattern recognition without layout interference from other cards. Finally, the backend's improved payloads and consistent LLM context make it

possible to script reproducible scenarios in which every usability test can replay the same synthetic study window, ensuring that observed behaviours are attributable to interface functionality rather than fluctuating data quality.

With the introduction and study of real One Button Tracker data, the real data format was built directly on the earlier data pipeline. The created scripts were changed to synthesise 84 days of button presses that include realistic double-click gaps and long-press durations, and to convert them into a clustered, threshold-aware JSON representation in the processed version. The Flask backend consumes this processed information A.1, deriving aggregated metrics, weekly segments, and per-day summaries that feed the new timeline and Patterns view. Moreover, the preprocessing synopsis is injected into both the LLM prompt and the API payloads, ensuring that every visual element on the front-end can be traced back to the structured dataset generated by this restructured flow.

At this stage, the system was implemented as a client-server architecture, with a React single-page application as the presentation layer and a Flask backend for analytics and orchestration. The backend generates synthetic OBT data, computes aggregated clinical metrics, and enriches these metrics using a local LLM, depending on the feature. The frontend renders multiple clinician-oriented views, such as *Dashboard* (A.2), *History* (A.4), *Patterns* (A.3), and *Report* (A.5), from structured API payloads. This separation ensures that the analytical logic and data synthesis remain centralized, while the UI focuses on interpretability and clinical usability.

However, before transitioning to context engineering mechanics, the platform was iterated once again based on user feedback, with a particular focus placed on enhancing visibility and ensuring optimal workflow integration. The *weekly timeline* was restructured into a true multi-week strip with a blue to purple to orange gradient, which was later changed, restored stems/dots, and richer per-week cards that include ranges, dominant rules, success rates, per-rule counts, and clinician notes. Guardrails were implemented to ensure that LLM payloads cannot collapse coverage. The weekly highlights were incorporated into the timeline tiles while maintaining pie summaries. Furthermore, cycle progress and headline metrics were ensured to always reflect the full dataset via stricter fallback validation in comparison to LLM merge validation. The history view had been enhanced with the introduction of filtering, flexible date ranges, intensity scoring, sortable tables, and badges, with the objective of enhancing the navigability of extensive logs. Reports were shown on a weekly basis, together with detailed analysis, insightful commentary, and clear definitions of the patient's protocol. This approach ensures that only relevant information is shown, effectively filtering out excessive technical details. The patterns were reoriented to a weeks-on-rows configuration and weekdays-on-columns configuration, with the heatmap palette aligned to the timeline gradient so that low-volume/okay days were represented by cool colours and high-volume days by warm colours. The LLM consultation was constrained to concise, clinical replies, with a fixed-height chat container and aligned input and footer, and clinician annotations are locally persisted and surfaced in timelines and reports. Notably, deterministic fallbacks and stricter merge logic contribute to enhanced resilience in scenarios where AI output is partial.

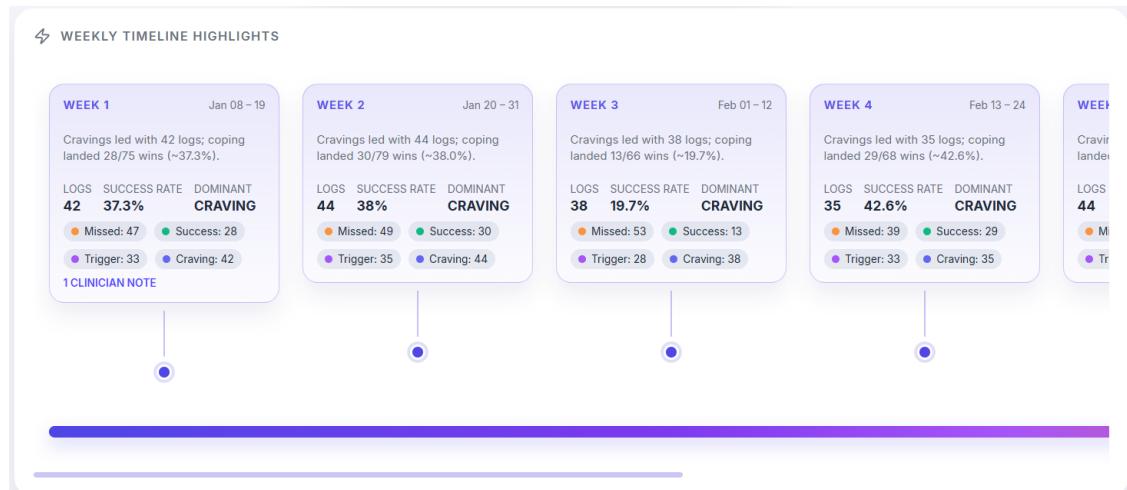


Figure 4.10: Sprint 4 - Weekly Timeline Highlights Container. The new form of representing the weekly timeline. This feature, working as a section of the dashboard, now contains information on logs, success rate, dominant rule, and LLM-derived information. With the report containing note features, the weekly timeline highlights also started to reflect the number of notes per week alongside the other metrics.

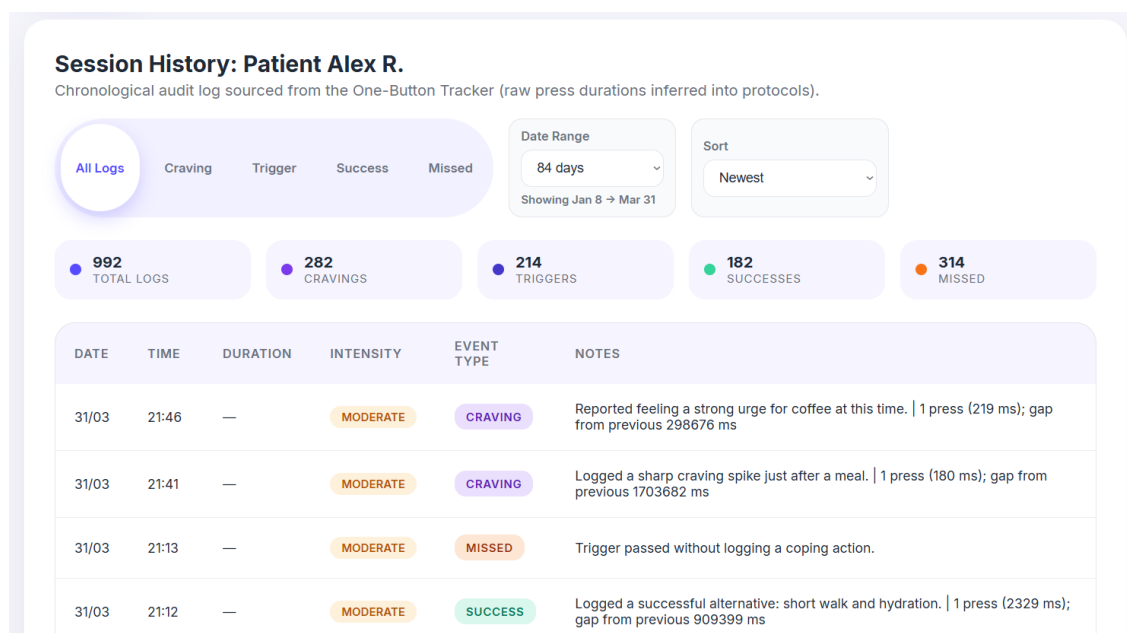


Figure 4.11: Sprint 4 - History Feature. The redesigned history feature displayed a more interactive list of OBT logs, organized by rule type. This way, new sorting types and filters, such as the event type and date ranges, were added in order to aid users in understanding the patient's history.

Nevertheless, following a review of the content on the platform, it was determined that the content was still deemed unsatisfactory. Consequently, minor changes were deemed necessary before starting the respective test phase. This approach enabled the refinement of clinician-facing surfaces, with the objective of enhancing readability and ensuring visual alignment with the platform's deep-blue identity. The dashboard chat was constrained within its card while maintaining the fixed-height consultation flow. The layout

of the patterns, Figure 4.12, underwent modification, with the week-row/day-column configuration retained, but with a shift in the colour scheme of the container to a chromatic scale of ice to slate to deep-platform-blue gradient. The bars, however, continued to use the warm/cool gradient for per-rule counts. The text was set to a dark neutral colour to enhance contrast, and the quality badges remained unchanged. The notes taken by clinicians have been moved from the dashboard to the report's *Clinician Workspace* (Figure 4.13), where they can be edited and paired with hotspots and next-step summaries. In addition, the report had been modified to include structured insights rather than raw log excerpts. A realignment of history controls was undertaken (Figure 4.11) to achieve greater consistency. This realignment required the tightening of date-range styling. Palette would then utilize the platform's deep blue as the darkest heatmap stop to ensure chromatic consistency with the rest of the UI.

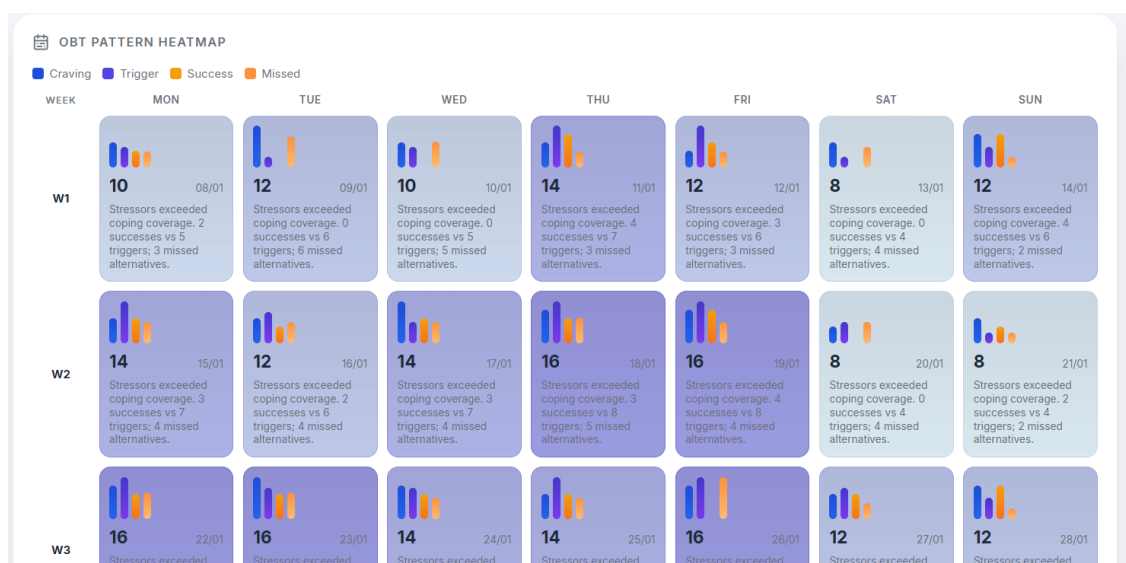


Figure 4.12: Sprint 4 - Heatmap Feature. The heatmap or Patterns feature suffered a diverse set of changes throughout the iterative process. During Sprint 4, changes to coloring, bar plots, the information displayed, and overall organization were deployed. This led to a heatmap feature that would be considered clinically proven if kept reliable and error-free. With this feature, users could have a reliable overview of difficult days for the patient.

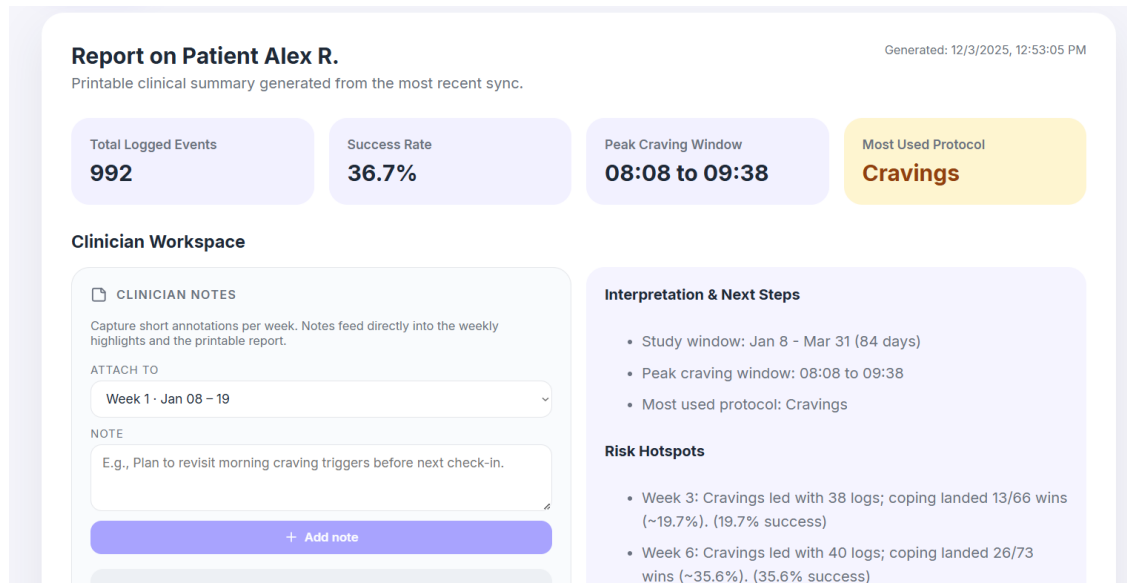


Figure 4.13: Sprint 4 - Report Feature. By the end of Sprint 4, the main new feature of adding notes to a patient profile was added. This way, the report was still maintained as a quick, accessible report on a patient, including information such as success rate, peak craving windows, and most used rules, alongside metrics derived from the LLM. Combining with the new notes feature, the Report was close to its final proposed solution and ready to test with new users and psychologists in the following iterations.

4.3.1 User Feedback

In the midst of prompt engineering implementation, during this development cycle, evaluations on user experience were conducted to test the LLM's output performance.

Following the gathering of a handful of users for the purpose of evaluating the platform, the testing framework delineated in the methodology section was used.

Within the think-aloud sessions, participants indicated that the interface initially felt confusing despite projecting a *medical* visual language and orientation required a short acclimation period. After familiarization, however, they judged the presentation to be highly informative, noting that every tab corresponded clearly to its intended functionality and collectively conveyed the expected feature set. Users appreciated the richness of the History tab but observed that Weekly Highlights and the dashboard timeline rendered only two weeks of data, preventing a complete comparison across the study period. They proposed replacing the repetitive text-based craving insights with a comprehensive pie or iconographic representation that uses color to encode weekly differences, thereby exposing the full dataset in a single view.

Evaluators reported that the color scale in the heat map provided minimal perceptual distance between what was considered *difficult* and *good* days. This way, they recommended adopting a different palette with greater dynamic range. Combining the heat map with the chat interface was considered effective for clarifying the worst days, yet participants asked for explicit documentation of how the visualization defines a *bad* or *successful* day, the question was whether the classification depends on aggregate rule counts or solely on missed or negative events, which they flagged for review in the next development cycle.

Regarding the *History* view, feedback was largely positive, but testers criticised the date-range selector as non-informative and advocated truncating the logs into paginated views. They also requested standard sorting capabilities to make the long list manageable within clinical workflows. Although the key insights and headline metrics were deemed accurate and useful, users detected apparent inconsistencies that suggested some dashboard cards might not be loading the complete dataset. For the reporting workspace, they asked for the ability to deep dive into week-specific reports rather than only reviewing an aggregate. Additionally, they recommended omitting technical pre-processed metrics such as derived thresholds from clinician-facing narratives, since those details hinder interpretability.

The conversational tool, the chat feature, delivered meaningful answers, yet participants characterized the responses as overly long and conversational. They preferred concise, clinically worded guidance. The chat container should remain fixed in height with internal scrolling anchored to the newest message so the page does not grow excessively. Several users also suggested tuning the tone to be more direct and time-efficient. Despite the improvement areas noted above, users found the system highly supportive for preparing future consultations with Alex. They also proposed a new feature that would enable psychologists to append notes or contextual annotations, thereby closing the loop between clinical reasoning and the platform's analytics.

This feedback resulted in direct implementation into the development cycle, including the addition and change of new features like the chat with the LLM and the clinician notes.

In the last iteration of this stage, before the transition to a more hybrid platform, another test evaluation was conducted in order to receive more feedback on user experience and platform output.

When over viewing the dashboard and related tasks, users commented that the timeline bar, indicated with the blue to purple to orange bar indicated a level of intensity as time went by which was misleading and not relevant to the data shown in the boxes above, thus being suggested for this bar to be removed or reworked for more friendly or clinically neutral color, nonetheless, it was uniformly considered is an effective visual representation of the data handling and understanding process even making it easier to assess which weeks are harder for the patient. Another issue occurred when interacting with the chat feature of the LLM was that whenever they would send a message not only they couldn't stop the thinking process, which is a common feature in many LLM solutions in the market, but also when the user would send the message and the LLM would respond the page's view would be pushed down, which was noted as a possible nuisance. Even with some errors, users noted that the combination of features and information presented on the dashboard made it very easy to better understand Alex's situation and respective progress throughout the 7-week period.

In the *Patterns* heatmap, users approved the chosen color palette for the mentioned feature. They still needed some tweaks, mainly on the colors of the bar plots, since the success and missed colors were often mistaken due to their similarity, and on the most intense days, the text was too difficult to read. Furthermore, some users considered the color of the trigger to be highly similar to that of the craving and suggested that different colors would be used to facilitate differentiation. There are some issues with understanding each rule within a protocol and each bar plot in it, where some users needed help distinguishing between rules, their respective representations, and their definitions.

Regarding the *History* view, one of the main concerns raised was the importance of the intensity data information and its notes. Since it's neither part of the data nor part of the OBT's measurement capabilities, these data columns were introduced in the previous sprint. Nevertheless, since it's neither natural to the data structure nor to the synthetic data created from OBT examples, the intensity, for example, is showing as moderate across all logs, making filtering this data column rather useless. Given the column's fictitious nature, it was quickly noted that it wouldn't make sense, as it's not part of the measurements, and the LLM can't take accountability for that information. Regarding the notes, the user commented on the need and purpose of having this information since it wasn't integrated into the information given from the OBT, but rather calculated during the pre-process phase.

Lastly, given the restructuring nature of the changes done during this iteration, the report section of the platform got the most feedback. One of the main comments was the repetition of information present in this view. This has to do with the LLM's response to the data and the need for major insights, which leads to repeating statistics on the data and showing them all across the view, where they should be more diverse or else removed. One of the possible solutions was to rethink the prompt system and create more specialized prompts to avoid redundancy in information, following the approach of *prompt engineering*, thus simplifying the clinical report as well.

Statistics organization was also a hot topic during the experiment, where in some cases, users noted the need to change or reorganize how the platform portrays information. This being said, sections such as the risk hotspots were highlighted for review of the respective information, mainly for the sorting of statistics, being suggested to sort by week instead of by failure rate. Regarding the new clinical notes feature, even though it was received with positive feedback, upon failure testing, it was seen that when adding multiple notes, they would start to line up below one general tab, whereas it should be shown under the specific week or period that the note was created on. Furthermore, it was also noted that this feature lacked an edit option to complement the functional foundation of this section. Lastly, some areas required visual tweaks, such as the clinical summary, the interpretation and next steps areas, and the risk hotspots, where the text was seen with visually odd spaces and alignments.

Overall, besides the feedback mentioned before, the platform received positive returns when it comes to its usage and information portrayed, especially considering the tasks and the output the LLM was giving, thus making it important to focus on these small changes in the following iteration to achieve and optimize the best LLM-driven platform.

4.4 Transition to a Hybrid Architecture

4.4.1 Motivation for a Context Engineering Approach in This Thesis

Given the objectives and constraints of the thesis, a context engineering approach was considered and found to be substantially more appropriate than traditional prompt engineering. The core requirement of the proposed system is the handling of longitudinal mental health data that evolves weekly and in which historical information directly influences future interpretations. Such data is inherently dynamic and stateful, rendering a static, prompt-centric interaction paradigm insufficient.

Prompt engineering assumes that all relevant information can be provided explicitly within a single inference window. However, in the context of clinical or psychological monitoring, each interaction is conditioned by prior sessions, long-term patient characteristics, and evolving behavioral patterns. Treating each session as an isolated text input not only fails to capture this temporal dependency but also introduces risks related to context overflow, noise accumulation, and loss of clinically relevant structure. Context engineering, by contrast, explicitly addresses information logistics: how data is stored, retrieved, compressed, and reasoned over time.

4.4.2 Context Engineering Applied to OBT

OBT platform uses context engineering to make LLM outputs predictable, grounded, and safe across three surfaces, dashboard, report, and chat, since the other features are not LLM dependent. On the server file, clickstream data are transformed into structured aggregates and narratives, which are then assembled into surface-specific prompts. In the dashboard, created a function that sends metrics and a schema hint and requires JSON-only responses, and then another that validates and falls back to deterministic data when needed. Regarding the report, the function created specifically for the report runs separately, with rules and constraints to avoid repeating dashboard insights, joined by another function that enforces structure and safety, and finishes with a verification of the LLM output to assess its validity and determine whether it was accepted or fell back. Lastly, in the chat feature, a function was created that feeds a compact snapshot of metrics and context, keeping responses short, grounded, and aiming for a non-diagnostic feel.

Dynamic context assembly is used inject state narratives, structured snapshots, and placeholders for checked resources. Prompts include explicit reasoning and guardrails, as well as other functions to serve as validators against malformed or fenced JSON. Together, these layers convert raw button presses into stable, clinically safe outputs tuned to each UI surface. In this last development section, a context engineering layer was added to make these changes viable (Figure 4.14).

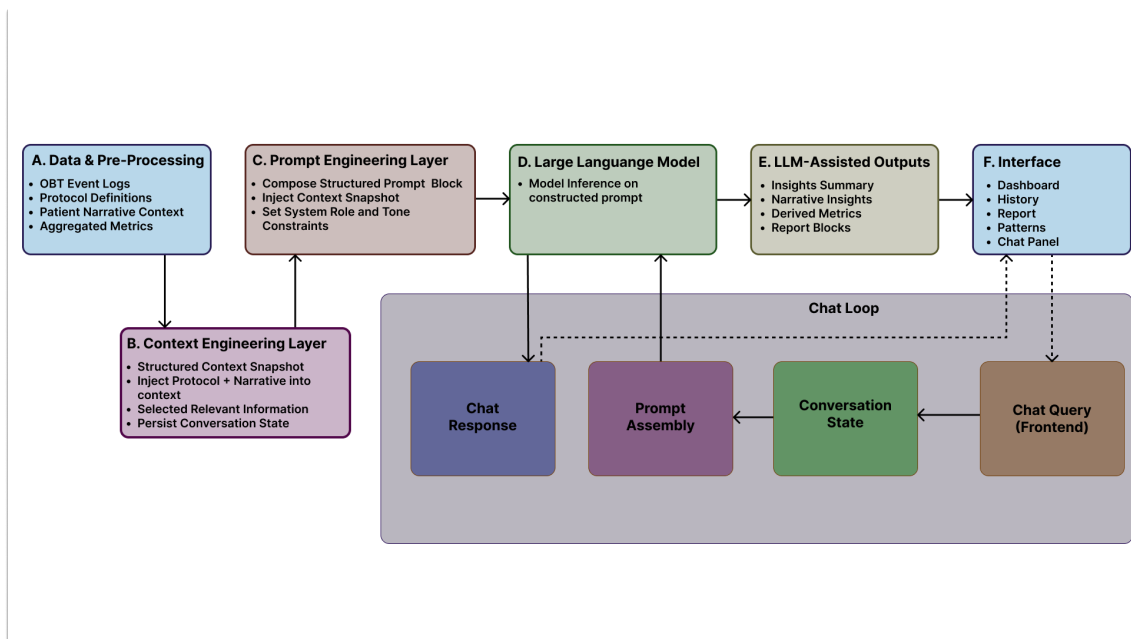


Figure 4.14: Hybrid Architecture of the Platform. Added another complex layer of Context Engineering with methods such as Dynamic Context Assembly with a stronger Data Pre-Processing layer. The shift to context engineering also required changes to the chat loop, since it now requires preserved states to keep the LLM output grounded and deterministic, not only at the beginning but also during user interactions. This is still a Hybrid Architecture, thus the core is still the same as iterations 3 and 4, but optimized and applied with Context Engineering.

4.4.3 User Feedback

Into the last stages of development, another round of test sessions was conducted. This way, the cycle progress visualization was considered that it could be enhanced through refinement of the colour scheme, as the current palette hinders clear distinction between cycles. The enhancement of colour contrast has been shown to improve visual clarity and user interpretation.

The OBT pattern heatmap was also identified as potentially confusing. It has been demonstrated that issues arise in relation to the interaction design when the user hovers over tabs. This is due to the fact that the displayed information does not always fit well within the graph boundaries. This had the potential to obstruct comprehension and negatively affect the user experience. From a scalability perspective, the design may not be optimally suited to longer study durations. For instance, if the study were extended to 20 weeks, the continuous presentation of all data in a single view would likely become excessively long. The introduction of pagination, such as dividing data into multiple pages, could enhance both usability and navigation.

Regarding the clinical summary section, a significant amount of unused space was identified at the top of the page. This layout was sub-optimal in terms of efficiency. To address this issue, it was recommended that measures be taken to use available screen space more effectively and to increase the density of the information presented.

It was noted that the placement of the date range selection and sorting controls in the session history view was not consistent. It was then recommended that the layout be adjusted by moving the sorting functionality closer to the date range controls. This adjustment would result in a more coherent and intuitive interface design.

In this last iteration, these changes were implemented and addressed, ensuring the platform was as close to a viable solution as possible.

4.4.4 The Final Development

In this final development section, the visualization layer was re-engineered to enhance clinical readability and consistency. The Patterns heatmap functions as a true intensity map: tile backgrounds transition from light to deep blue according to per-day log volume, while category bar plots maintain vivid, consistent colours, as in Figure 4.15. The use of a unified colour scheme across the heatmap, timeline, and cycle progress eliminated vagueness across different views.

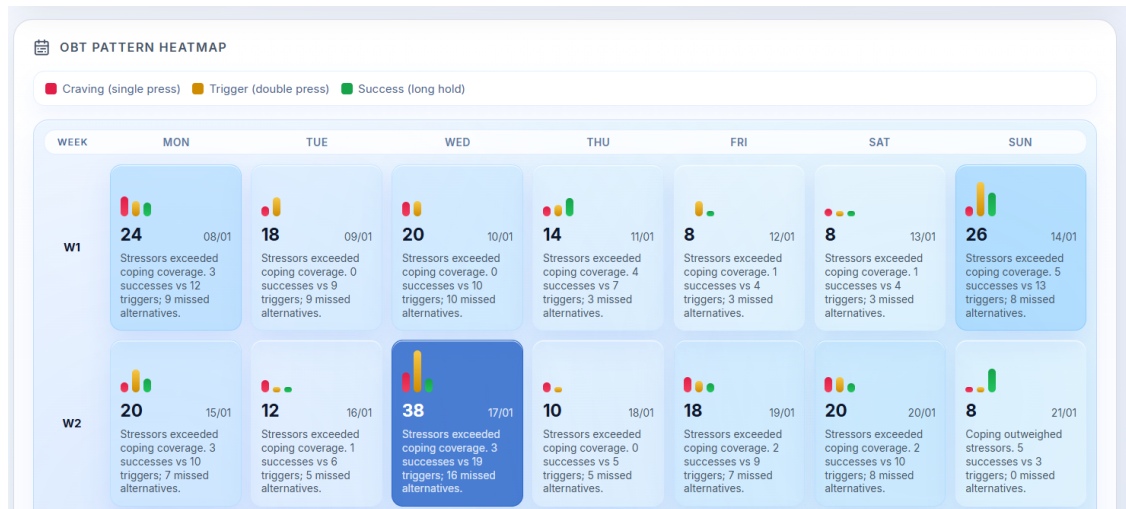


Figure 4.15: Sprint 5 - Heatmap. Optimized Heatmap representing a true intensity map. Changed the color scheme to a light to deep blue according to per-day log volume, while category bar plots maintain vivid, consistent colors. This was deemed necessary following discrepancies found during user tests. Similarly, the bar plots were corrected to better represent the daily data. Also added pagination to simplify this view and feature.

The Weekly Timeline Highlights have been redesigned with a coherent blue gradient and standard typography, thereby reducing visual noise while preserving rule-related cues (Figure 4.16). Text contrast was standardized across the platform, with light text reserved exclusively for instances where it was deemed necessary on darker heatmap tiles. In all other instances, dark text was maintained to ensure optimal legibility. Concurrently, report generation was constrained by more authoritarian LLM prompts that enforced concise, non-redundant bullet points and a single actionable recommendation to streamline clinical narratives. Collectively, these changes deliver a presentation that is more clinically neutral, modern, and interpretable, while reducing informational redundancy and enhancing the clarity of intensity patterns.



Figure 4.16: Sprint 5 - Weekly Timeline Highlights. The newest and final version of this weekly timeline feature now features a coherent blue gradient and standard typography. Coloring was adjusted to better match the new clinical feel the platform was portraying. The core feature remained the same, with each week containing highlights on logs, success rate, protocol, and clinician notes.

In order to rework the organization of the dashboard, patient selection was moved into the profile card as compact chips, keeping dataset switching while removing the extra *Other Patients* container, Figure A.6.

Firstly, it is important to note that the deterministic aggregation layer had been developed to compute peak log week and weekly log totals in addition to trigger-based peaks. This development had the capacity to resolve questions that so far could not be answered, for example, questions concerning the number of logs for a given period. Furthermore, it enables insights into log volume to be provided alongside metrics focused on triggers. Secondly, the report highlights were rebalanced to focus on daily average activity and the toughest day, while maintaining the success rate and most-used rule as the primary indicators. The total logged events and the peak craving window were relocated to the *Interpretation & Next Steps* section, thereby enhancing the clinical significance of the headline panel without discarding the contextual totals.

Conversational accuracy was further optimized by supplying the full dataset and dashboard payload in supplemental context chunks, explicitly designated as authoritative in the system prompt. This multi-feed grounding reduced model uncertainty in follow-up questions that require longitudinal references. The changes mentioned earlier enhanced the accessibility of clinically meaningful metrics, thereby reducing vagueness in analytical responses and aligning the report's focus with actionable patterns, such as the typical daily load and the highest-strain days.

In addition to the changes that were implemented in response to the feedback received at the conclusion of the previous iteration, the final session conducted within this sprint revealed flaws that required implementation in order to demonstrate the platform as a more complete solution. This way, the final days of development were dedicated to implementing these modifications and ensuring that the platform approached a proper minimum viable product.

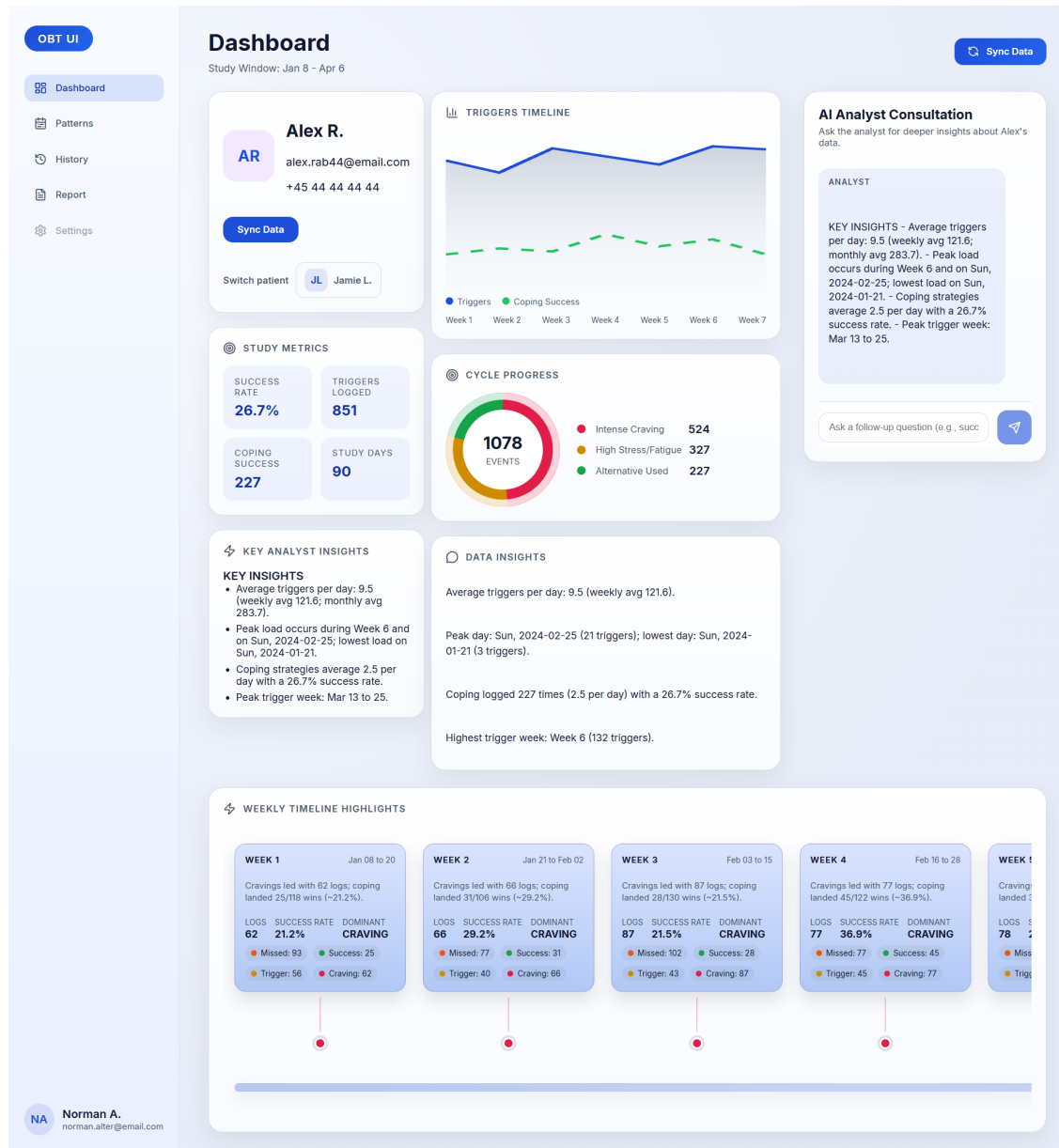


Figure 4.17: Final Dashboard. The Final dashboard was comprised of nine sections. Three sections were dedicated to graphs showing progress and overalls, which included the Triggers Timeline, the Cycle Progress, and the Weekly Timeline Highlights. The Study metrics section focused on showing deterministic metrics, while the Key Analyst Insights and the Data Insights showed LLM-based narratives. Lastly, a chat container was the most direct LLM interaction within the platform, encompassing standard features to basic LLM interaction and patient management (top left), which allowed for changing patients and datasets.

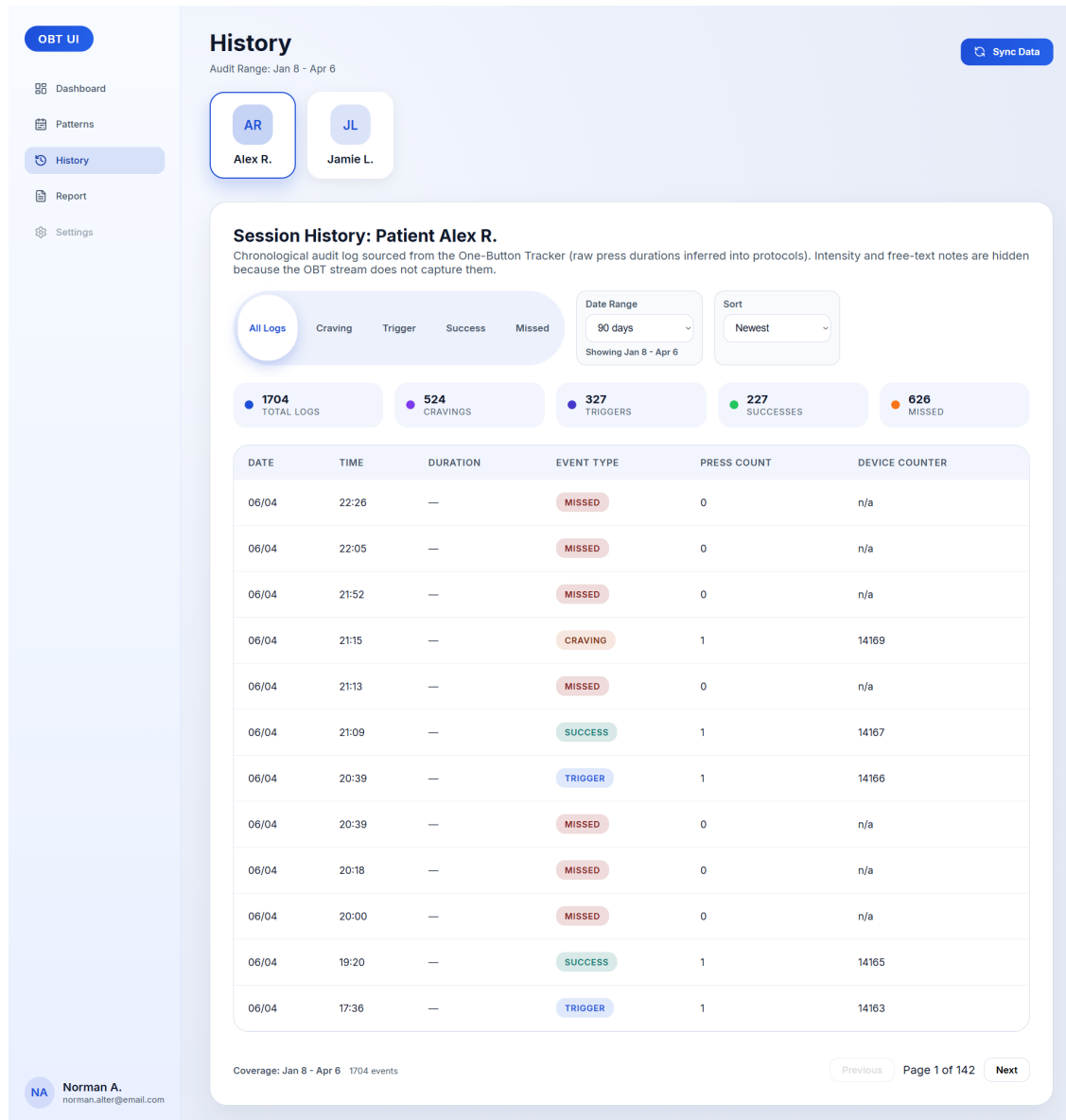


Figure 4.18: Final History. Final view built upon all the changes from the sprints. Fixed the filtering options of rule types and date ranges, as well as the sorting mechanics. Also contains pagination to help better visualize and scroll through the dataset. Overall, it's a standard feature for lists and datasets, which was kept as a low priority but is reliable.

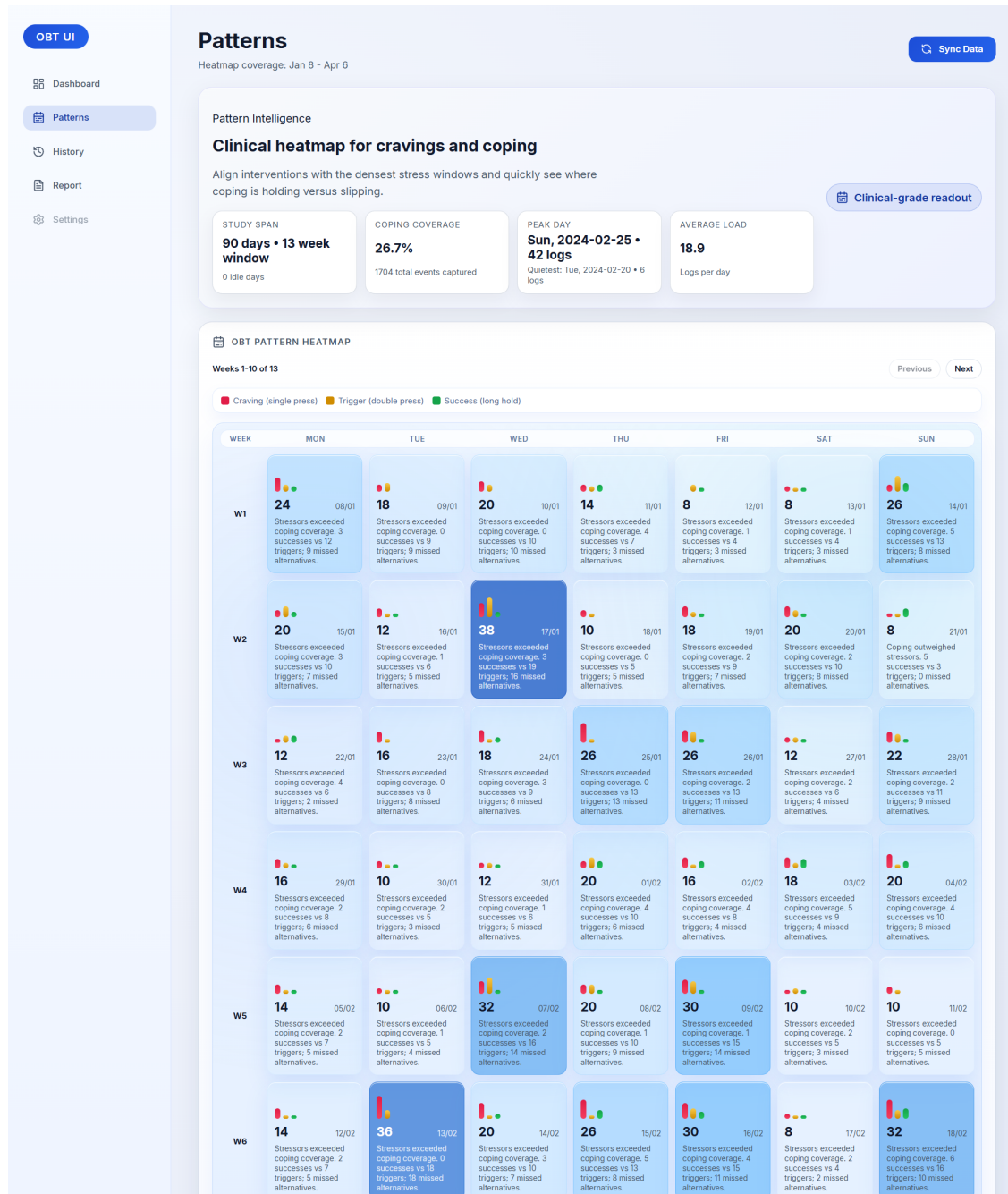


Figure 4.19: Final Patterns Feature. True Heatmap with detailed information regarding the daily logs via text and bar plots. All the known and discovered issues were fixed and tested, and the feature showed reliability and approval in the user sessions and psychologist interviews. The feature functions as a heatmap of the rules used, while the bar plots show which rules were used on a given day. This is important for users and psychologists to be able to have an overall feel of a week or month and be able to lead the conversation towards specific instances. Full Heatmap view in Figure A.7.

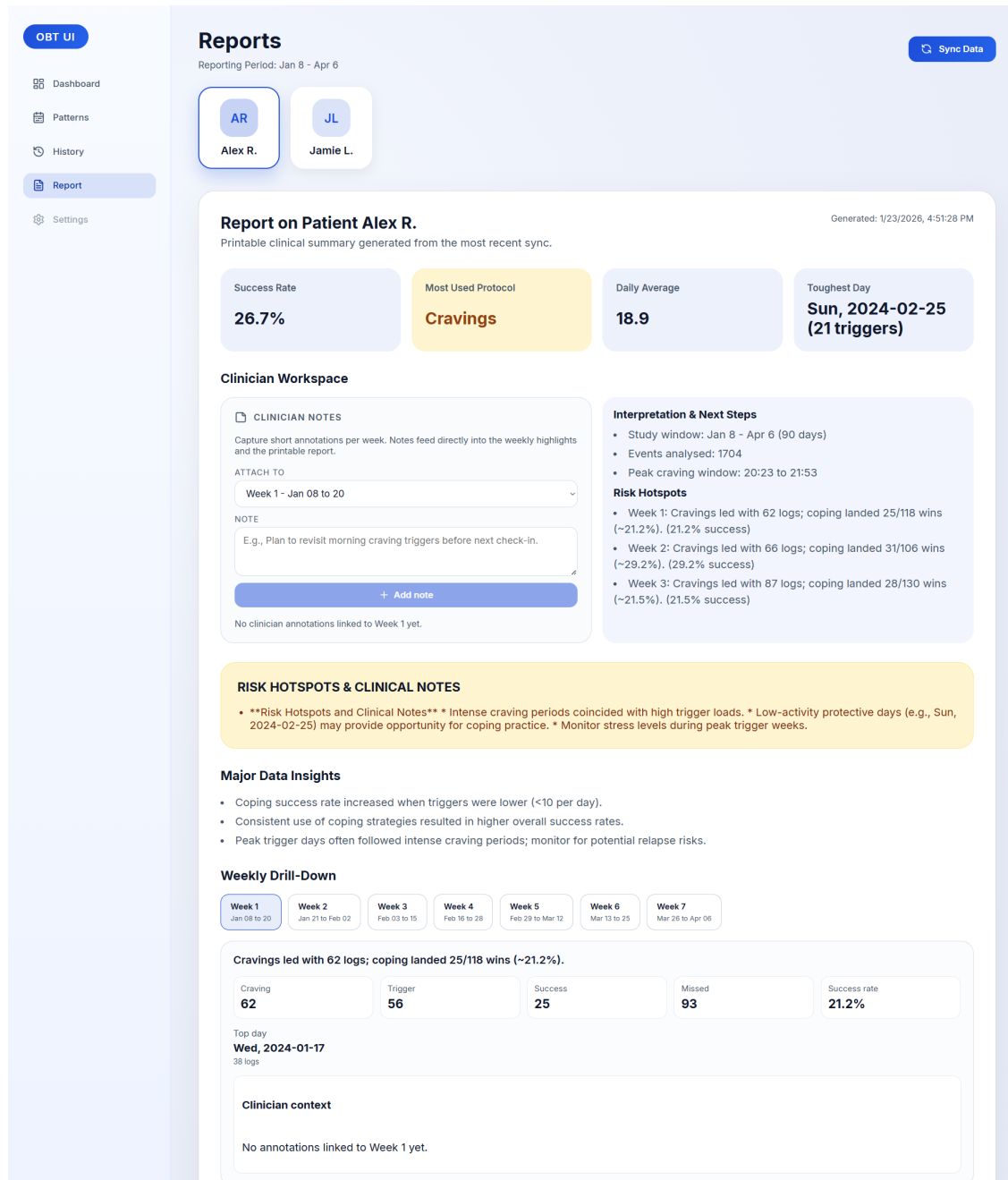


Figure 4.20: Final Report Feature. In the final development stage, this feature developed into encompassing deterministic metrics, user information through the dataset, clinician notes, and LLM narrative insights as well for patient protocol and documentation. This page serves as a summary of the patient, ready to be used during the therapeutic process. Full Report view in Figure A.8.

In the end, the final application follows a layered architecture that separates data processing, context assembly, prompt orchestration, and model interaction. At the core is a pipeline that ingests One-Button Tracker press logs, transforms them into a structured protocol, computes derived metrics, and then uses those artifacts to drive both a dashboard summary and a conversational interface. This design allows the system to control model output through a combination of deterministic pre-processing and carefully structured context payloads, reducing reliance on ad hoc prompting.

Data ingestion begins with a pre-processing stage that converts raw press streams into protocol-aligned events. The script derives patient-specific thresholds and then groups press events into semantically meaningful categories such as *intense craving*, *high stress*, and *coping success*. Alongside these categories, the pipeline computes study-window aggregates, such as total events, trigger counts, averages per day/week/month, peak, and low trigger days. These numerical summaries form a stable base on which all further reasoning is based.

On top of this foundation, the system builds a context assembly layer that integrates multiple representations of the same data for different model surfaces. The context builder produces a narrative state summary for human-readable grounding, a structured JSON snapshot for deterministic grounding, and a preprocessing summary that exposes the threshold logic used in event classification. This multi-view representation is explicitly built into the prompt block and is treated as authoritative in later interactions. The design pushes for traceability: every analytic statement in the model's output is intended to be grounded in the provided metrics rather than inferred heuristics.

Prompt engineering is applied as one of the lowest layers of control, specifying role, format, and response constraints. The system prompt clearly defines the model's responsibilities, style constraints, and safety boundaries. However, the distinguishing feature of the final architecture is that prompt engineering is not the sole control mechanism. Instead, it is wrapped by context engineering: the model is given a rich, structured, and redundant context payload that applies the instructions. In other words, prompt engineering defines *how* the model should respond, while context engineering defines *what* it is allowed to respond from.

For conversational follow-ups, the system contains an explicit context map that includes the system prompt, a compact dashboard snapshot, and extra context chunks when the data exceeds a fixed size. Chunking is performed at the context level, not the prompt level, ensuring that additional context is treated as authoritative extensions rather than optional reference text. This design supports bounded response lengths and mitigates hallucination risk by reiterating that only the provided metrics should be used. The retrieval hook exists as a placeholder for validated resources but is intentionally disabled in the current implementation to preserve experimental control.

In summary, the final architecture contains a dual strategy. Prompt engineering imposes behavioral constraints and output formats, while context engineering functions as a structured representation of the study data that the model must use. This separation yields a system that is more robust and traceable than a pure prompt-based baseline, because the model's reasoning is anchored in deterministic preprocessing and explicit context assembly rather than implicit prompt interpretation alone.

5 Results

5.1 Introduction

As the development process reaches its conclusion, the results from surveys, assessments, interviews, and user testing sessions are extremely important for assessing the project's viability. The current chapter presents the findings from the present study and discusses them.

As previously stated in the third chapter, the results obtained are the product of both qualitative and quantitative methods. This chapter is oriented towards the results and accordingly places greater emphasis on quantitative methodologies, including surveys and assessments. However, it should be noted that it also incorporates a psychologist's interview. Since the user test sessions were predominantly focused on the user experience and the platform's viability, these sessions were introduced and thoroughly delineated in the fourth chapter, as they directly influenced the alterations and decisions made during the development phase.

5.2 Participant Demographics

According to the survey, most users were young adults. As per Figure 5.1, **58%** of the people who answered were between 25 and 34, **33%** were between 18 and 24, and only **7%** were over 35. The number of men and women is roughly the same, with **53%** of respondents identifying as female and **47%** as male. The data shows that around **75%** of the subjects say they are from Portugal, **22%** say they are Denmark-based, and a smaller number say they are from the Netherlands or Slovakia. Around **51%** of all regions are in Setúbal, indicating that the local network is reflected there.

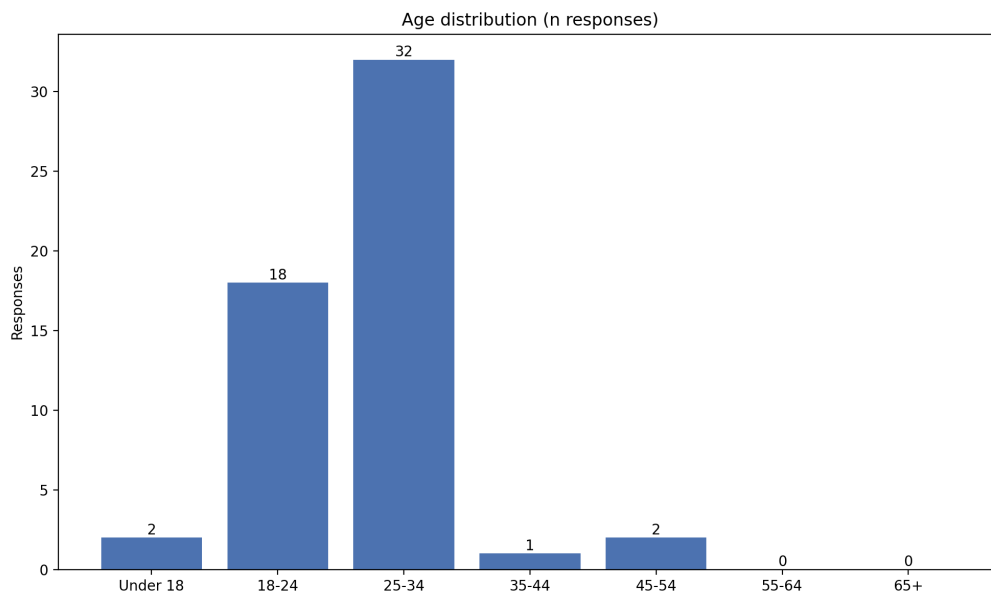


Figure 5.1: Initial User Survey - Age Ranges. A histogram demonstrating the demographic breakdown of the initial user study, showing a sample predominantly composed of young adults. The distribution is heavily skewed towards people who are traditionally more accustomed to digital means, which correlates with the high technology comfort scores recorded in the same survey. This demographic profile supports the decision to approach the MVP with a modern, responsive web application.

Regarding their backgrounds, they were distributed across a variety of sectors, with a notable concentration in the fields of technology and engineering, roughly **25%**, health-care, approximately **16%**, students consisting of **15%**, and business and operational roles, approximately **13%**. The remaining occupations were distributed more uniformly across creative and service-oriented sectors. Additionally, 5 respondents were engaged in training to become psychologists or were already working as psychologists. Furthermore, the data indicated that **73%** of individuals had already sought assistance from a psychologist or therapist, with **64%** of these individuals attributing their decision to seek help primarily to their life experiences. When inquired about their motivations for mental health, the respondents selected an average of almost 4 life challenges, with the most frequently selected categories being personal growth **69%**, anxiety **65%**, stress management **65%**, relationship issues **56%**, depression **51%**, and trauma **45%**.

5.3 User Habits & Preferences

As is common among young adults, this group was found to demonstrate a high level of technological comfort. The mean comfort rating was 4.45, with a median of 5, indicating high satisfaction. The market share of smartphones was distributed as follows: **56%** of smartphones are iPhones, while **44%** are Android devices. The data-tracking habits of respondents varied, with **16%** tracking daily, **18%** tracking weekly, **25%** tracking monthly, **16%** rarely tracking, and **24%** never tracking, which indicated a significant segment of respondents who express enthusiasm for the concept but have not yet adopted it as a routine. The most prevalent adoption impediments, Figure 5.2, pertained to value proof, with effectiveness (**67%**), accuracy (**58%**), cost (**56%**), privacy (**51%**), and both security and ease of use (**44%**) being the main concerns. Concerns regarding text analysis included issues of intrusiveness, temporal expense, and the potential for over-analysis. The importance of simplicity was emphasised by the mean score of 4.24/5, with **85%** of respondents awarding a higher rating. Despite the high level of affinity for high-tech solutions, the importance of data-driven insights is only moderate with an average score of 3.55 out of 5, Figure A.9, and **53%** of respondents scored 4–5, leading to the belief that, in order to convince skeptics, concrete therapeutic benefits must be articulated.

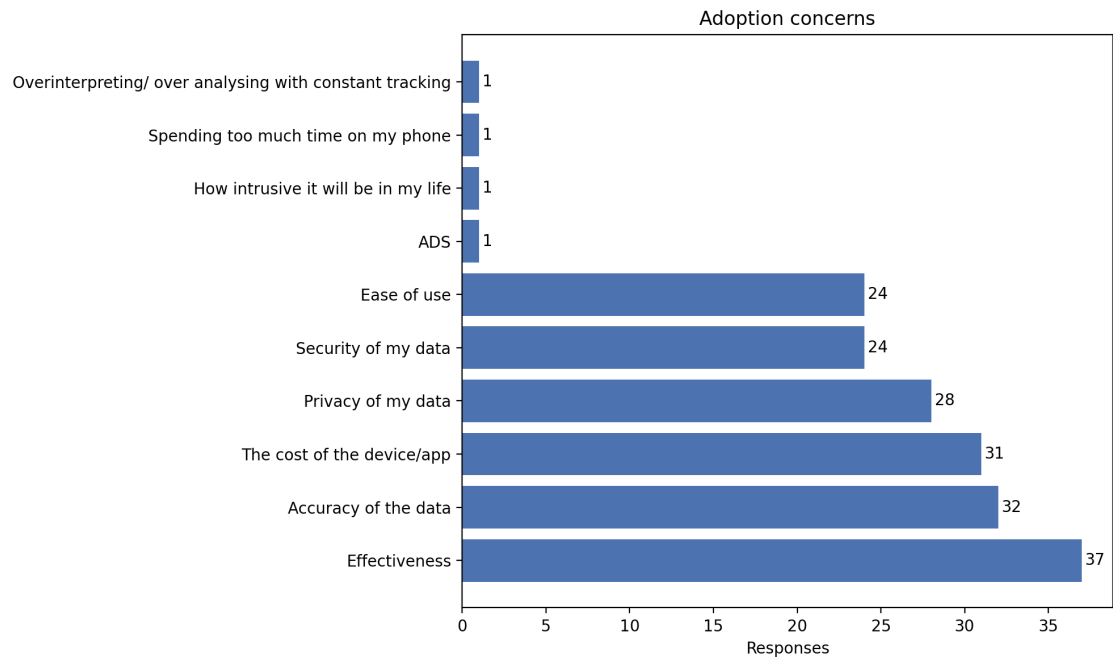


Figure 5.2: Initial User Survey - Adoption Concerns. A ranked bar chart displaying the main concerns users face when adopting new personal health technologies. While privacy and security remain significant concerns, the data reveals that users prioritize effectiveness and accuracy above all else, which pushes for prioritizing LLM optimization factors and architectural decisions.

Regarding data privacy, respondents were willing to share protected data, provided trust is established. The collected data indicated that more than **89%** of respondents reported being comfortable sharing information regarding stress, sleep, mood, and physical activity. In other categories, including work-life balance (**84%**), diet (**78%**), and social interactions (**76%**), were also deemed to be acceptable for disclosure. The insights desired by the subjects of this study, as shown in Figure 5.3, include goals such as identifying stress

triggers (**75%**) and understanding mood swings (**64%**). While the provision of an easy data packet for therapists (**51%**), the tracking of progress (**49%**), and the linking of mood with sleep (**35%**) were mentioned, even though to a lesser extent.

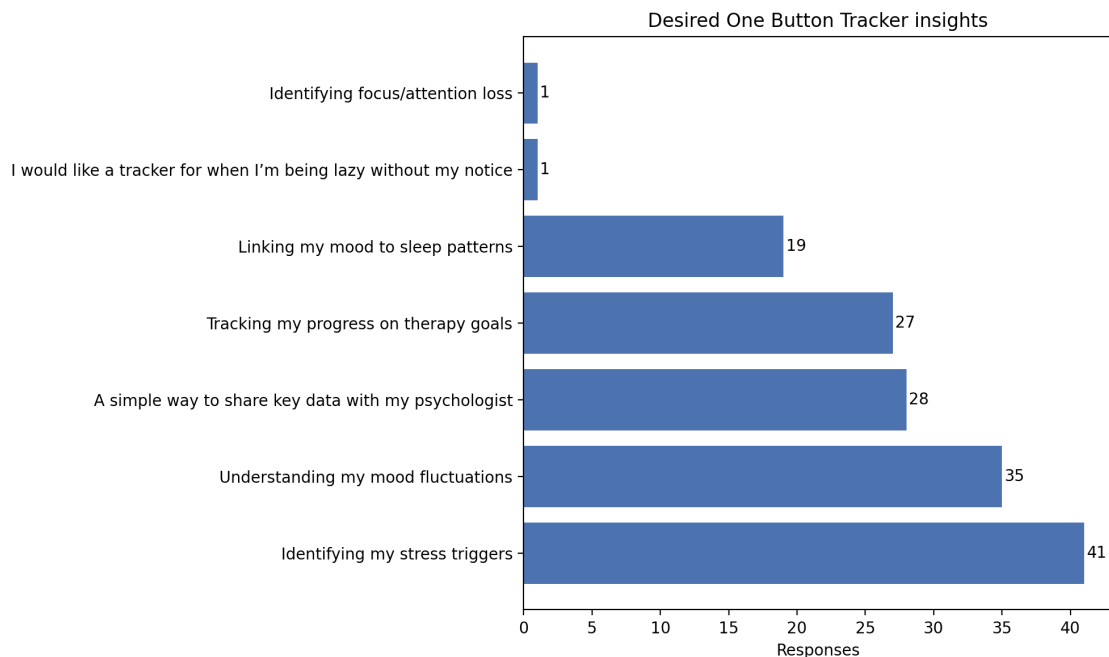


Figure 5.3: Initial User Survey - Desired Insights. Classification of the specific types of insights users hope to derive from the platform. This study showed a clear hierarchy of needs, with users initially prioritizing casual analysis.

Furthermore, insight delivery preferences favored self-serve tools. The survey results indicate that the majority of respondents, **58%**, expressed a desire for an app dashboard. In addition, **22%** of respondents preferred a report that could be shared, while **11%** favoured direct integration with EHRs. A smaller proportion of respondents, approximately **1%**, expressed a preference for emails or conversational user interfaces. Support for both self-tracking and clinician-ready exports would be sufficient to satisfy the majority of use cases.

As mentioned before, even though it was not exactly within the scope of this project, the survey also aimed to assess commercial viability. It was shown that the tendency to pay for a product or service is demonstrably influenced by its price point. In this survey, a **51%** cap at €20, **27%** would only utilize it if it were complimentary, **16%** would pay between €21 and €50, and a mere **6%** would pay in excess of €50.

A total of **53%** of the participants volunteered to be testers, while **28%** provided their contact information for project participation in the future.

5.4 Early Stage Feedback

In these early sprints, the focus was mainly on qualitative feedback, for which part of the received feedback was revealed during the fourth chapter, specifically section 4.2.1. Nevertheless, this questionnaire also opened the door to investigating LLM integration. Thus, after the prototyping sections, the next step focused on exploring the integration of the LLM, specifically the incorporation of a chat-like feature, which testers exhibited varying degrees of uncertainty about or even a tendency to avoid using. Nevertheless, this reluctance was accompanied by a consensus on the intuitiveness of this feature for facilitating follow-up inquiries about the data.

In light of the expressed opposition to incorporating a chat-like feature, the testers were asked about their conceptualization of how the interaction with the LLM would unfold. The testers' responses indicated a preference for a more summary-focused or report-oriented feature, despite variations in how they presented such summaries. A number of testers proposed a more proactive approach, involving brief suggestions or notifications similar to those used by Duolingo, which aligns with studies on the importance of push notifications but is outside the scope of this project [46, 47]. On the other hand, some testers preferred a more comprehensive, in-app presentation of data information.

Regarding early LLM output, the study's results were found to be of interest to psychologists. From the psychologist's perspective, the inquiries were primarily designed to convey information through simple requests for patterns, enhanced data explanations, and results. This approach serves to preserve the psychologist's role in action, with the objective of providing actual assistance to the patient. Nevertheless, from the patient's perspective, the testers were more available to address queries regarding strategies and appropriate courses of action in specific circumstances. However, they consistently maintained a neutral position toward the trained LLM.

Upon being presented with these preliminary key insights, the testers demonstrated a slightly more favorable utilization of the LLM, which exhibited a discernible correlation between the rules and the data. The testers agreed that the insights demonstrated high accuracy in interpreting the expected patterns and proposed unbiased recommendations for the designated user in the hypothetical scenario. This occurred subsequent to the testers adopting the role of psychologists once more. This inquiry proved instrumental in showing the subsequent actions a psychologist might undertake upon reviewing this synopsis. In the same tone, when presented with the Q&A session, the answers to this Q&A session were very positive across all themes: completeness, detail, suitability, and usefulness for clinical consultation.

5.5 Prompt Engineering Platform

During these stages, the main focus was on user experience and platform development, while concurrently investigating the LLM optimisation. Consequently, user test sessions were conducted in order to address the platform’s user experience and LLM quality assessment. The LLM’s integration and output were the primary focus of this assessment. The subsequent section will concentrate on the LLM assessment, as the user feedback was already detailed in the fourth chapter, under section 4.3.1.

5.5.1 LLM Quality Assessment

The LLM was queried on six headline questions, and the answers thus obtained were then compared to the deterministic baseline derived from a created Python script to be assumed as ground truth.

Table 5.1: Early Assessment: Ground Truth vs. LLM

Metrics	Ground Truth	LLM Output
Most Used Rule	Intense Cravings	Intense Cravings
Intense Cravings	282	205
High Stress Events	214	160
Coping Successes	182	Not available
Total Events	992	892
Success Rate	36.7%	41.0%
Peak Window	08:08 to 09:38	Not available
Study Duration	84 days	90 days
Peak Day	2024-01-18 with 8 triggers	2024-01-10 with 5 triggers

Regarding accuracy, the LLM often failed to identify items that were checked. The study incorrectly reported total events as the raw press count 892, exaggerated the study’s duration, showing 90 instead of 84 days, and omitted the peak window, classifying it as *‘not available’* despite its availability within the context. The most-used rule was identified as *‘intense craving’*, which had 282, which was indeed the only correct metric. Regarding the success rate, it was recomputed from a subset (roughly 41%) rather than using the provided 36.7%.

As for hallucination and drift patterns, responses exhibited a combination of real numbers, for example, 282 against 205, with fabricated roll-ups, for example, weekly sums and 90-day time span, and there was an explicit disregard of metrics that were supplied, such as the peak window and the total event count. In several cases, it substituted raw processing counts as 892 presses for processed event totals, which were 992, and invented ungrounded percentages.

Furthermore, despite using deterministic settings and injecting ground-truth JSON and structured data, the model failed to reliably reproduce the provided values. Instead, it attempted to infer or recompute, resulting in systematic numeric drift and intermittent *"not available"* deflections where data was present.

In summary, under the early tested prompt and grounding configuration, the LLM did not replicate the basic metrics and exhibited repeated numeric errors, indicating low fidelity to the supplied study data for these headline questions. Given the importance of reliability and precision in the medical context, the current state of the LLM shows to be unfavorable. This way, during the following stages, the focus was more on the importance of *prompt engineering* and how the LLM's output can be optimized by optimizing the input as well.

5.6 Final Hybrid Platform

In this final stage of development, three primary methods were employed to evaluate and experiment with the system. Including the standard user sessions displayed in the fourth chapter, during section 4.4.3, and the quality assessment, which was also featured in earlier iterations, an interview with a psychologist was also conducted in order to obtain more in-depth feedback from an expert in the field.

5.6.1 LLM Quality Assessment

The results were derived from an evaluation log, which contains 184 assessment items spanning dashboard, report, history, and chat outputs. The distribution across platform surfaces was as follows: dashboard containing 134 items, report had 28 items, history containing 5 items, and, lastly, chat containing 17 items. Each item is assigned a check type, thereby distinguishing between structured, automatically verifiable fields and qualitative or freeform responses. The evaluation set was dominated by structured checks based on a newly generated dataset. A total of 36 items were designated as exact checks and 118 as object checks, including headline metrics, timelines, weekly summaries, and heatmap entries.

Table 5.2: Late Assessment: Ground Truth vs. LLM

Metrics	Ground Truth	LLM Output
Most Used Rule	Intense Cravings	Intense Cravings
Intense Cravings	524	524
High Stress Events	327	851
Coping Successes	227	227
Total Events	1704	1704
Success Rate	26.7%	26.7%
Peak Window	20:23 to 21:53	20:23 to 21:53
Study Duration	90 days	92 days
Peak Day	2024-02-25 with 25 triggers	2024-02-25 with 25 triggers

In both bundled and direct metrics, the model responded extremely accurately, so no metrics were mismatched.

However, the remaining 30 items consisted of qualitative and free-form elements. Thirteen qualitative checks were conducted, in addition to seventeen free-form checks facilitated via the chat function. These items capture narrative insights, action suggestions, and conversational answers that require manual judgment rather than direct equality comparison. A manual review of these freeform elements was conducted, and it was observed that the model provides highly accurate feedback across almost all features under very controlled conditions. In the rare cases when the model struggled, it was due to arithmetic issues or to combining metrics, as seen in table 5.2.

This indicates the model's improved reliability, as generally LLM's expected answers matched the processed data, even for more narrative-type answers previously deemed less reliable.

5.7 Psychologist Interview

The iterative development and evaluation of the Decoding the Click platform helped identify key findings across main categories regarding the integration of Large Language Models (LLMs) into mental health self-tracking. These findings summarize the results of the initial user surveys, the technical performance assessments, and the expert clinical validation.

5.7.1 Expert Profile

The interview was conducted with Patricia W. Marcelino, a practicing clinical psychologist trained in Évora and holding a master's degree in clinical psychology. She completed a professional internship at a psychiatric hospital in Leipzig, working primarily with anxiety disorders and personality disorders. Her professional experience includes both school and clinical settings, working with children, adolescents, and adults. She follows a psychodynamic therapeutic approach and holds a postgraduate qualification in child and adolescent psychopathology. She provides clinical services in Portuguese, English, and German, working with patients from multiple countries, and also has experience as a German language instructor.

5.7.2 Feature Review and Clinical Feedback

Metrics Dashboard

The psychologist emphasized that averages computed over clinically meaningful windows, meaning weekly and monthly, and daily when appropriate, are significantly more useful than cumulative totals over long periods. Peak indicators were identified as particularly valuable, as they allow joint exploration with the patient of contextual events associated with increased symptom intensity.

Cumulative totals were considered to have limited clinical utility. On the other hand, identifying days with the lowest incidence was highlighted as important for understanding protective factors and effective coping strategies. Visualizing the use of strategies, as defined in the protocol, was also considered clinically valuable for assessing self-regulation and coping behavior.

Heatmap Visualization

Daily heatmaps were regarded as highly valuable for analyzing temporal patterns and linking critical days to external stressors or events. As this functionality does not rely on the LLM, it was recommended that it be maintained and enhanced, particularly by improving the inspection of specific days and visualizing associated averages and peaks.

Raw Event Logs

Raw event lists were considered of lower priority for routine clinical analysis, with aggregated views preferred. However, access to filtered raw data was deemed useful for investigating specific patient-reported events, provided appropriate temporal filters are available.

Reports

Weekly reports and key indicators, metrics like averages, peaks, and strategy usage, were considered appropriate. However, the psychologist noted that repeated information across the dashboard and report sections reduces clarity and usability. It was recommended that the section previously titled *Interpretation and Next Steps* be restructured to focus explicitly on *risk hotspots*, avoiding repetition of metrics already presented.

The inclusion of clinical notes written by the psychologist was strongly supported, as this aligns with standard clinical practice and facilitates longitudinal tracking. It was also recommended that protocol description and documentation be removed from the report itself and instead provided through other contextual help mechanisms, as such information is already familiar to clinicians.

Insights by Time of Day

For coffee dependency, time-of-day analysis was considered to have limited relevance, with greater emphasis placed on daily and weekly patterns. However, for certain disorders, such as self-harm, specific hours may be clinically significant due to factors such as isolation or ritualized behavior. Consequently, time-based analyses should be configurable according to disorder and clinical context.

Model-Generated Recommendations

Model-generated recommendations were viewed as potentially useful in some contexts as support for clinicians, but they should remain optional and never replace professional judgment. Direct exposure of such recommendations to patients was discouraged, as it may be triggering in certain cases. If used, this should be governed by clear policies and safeguards.

Risks, Limitations, and Future Opportunities

Key risks identified include the operational cost of continuous LLM usage, privacy challenges when transitioning to hosted environments, and the difficulty of supporting a wide range of disorders without increasing system complexity. The clinical validity of LLM-generated insights was also highlighted as requiring ongoing validation.

Future opportunities include the introduction of a patient diary feature, configuration of platform modules by disorder, multilingual support, enterprise wellbeing applications, and a clearer separation between deterministic metric pipelines and LLM-driven interpretative analysis with appropriate clinical governance.

5.8 Key Findings Summary

The iterative development and observational evaluation of the *Decoding the Click* platform demonstrated its relevance in categories of findings regarding the integration of large language models into mental health self-tracking. The findings presented synthesise the results from the initial user surveys, the technical performance assessments conducted throughout the development sprints, expert clinical validation, and user feedback.

5.8.1 Reliance on Large Language Models

Testing demonstrated that LLMs, specifically those used in this study, exhibit significant limitations when performing arithmetic operations on raw longitudinal data. While comparing against the deterministic ground truth revealed persistent patterns of *numeric drift* and *hallucination* when the model was tasked with calculating cumulative totals or success rates directly from raw JSON logs. This finding validates recent literature on the limitations of probabilistic models in mathematical reasoning [10]. Consequently, the research establishes that, for clinical reliability, a supported architecture is required. The system must strictly decouple quantitative processing from qualitative reasoning, using deterministic algorithms for all statistical metrics while reserving the probabilistic LLM solely for narrative generation and contextualization.

5.8.2 Context Engineering Superiority over Prompt Engineering

Results from the platform's prompt engineering phase indicated that standard techniques, such as zero-shot or few-shot prompting, were insufficient for managing the complex longitudinal state of a patient. The quality of clinical insight was found to be less dependent on the instruction phrasing and more dependent on the structure of the input data [14]. The implementation of a *Dynamic Context Assembly*, where raw data is preprocessed into a structured *Context Bundle* containing weekly highlights, trends, and definitions before inference, significantly reduced model uncertainty. This transition from static prompting to dynamic context engineering was essential for ensuring that the model's outputs remained grounded in the specific temporal reality of the patient's week, preventing the *lost-in-the-middle* phenomenon often observed in long-context windows [14].

5.8.3 Clinical Utility and Risk Hotspots

Feedback from the psychologist interview highlighted a critical distinction between data availability and clinical utility. The study found that clinicians do not require a textual reproduction of raw data. Instead, the system provides the highest therapeutic value when it identifies anomalies as *Risk Hotspots*, which are specific temporal windows or contexts where coping mechanisms fail. The expert validation confirmed that the platform's ability to isolate Peak Trigger Days and Lowest Incidence Days directly supports the therapeutic dialogue, whereas generic summaries were deemed clinically negligible. Furthermore, user testing revealed that visual artefacts, such as Heatmaps and Timelines, must be strictly aligned with the LLM's narrative. Discrepancies between visual indicators of high-stress periods and the LLM's textual summary were found to lead to immediate user distrust, reinforcing the need for the single-source-of-truth architecture established in the final prototype.

6 Discussion

6.1 Introduction

The main objective of this thesis was to study the viability of large language models in efficiently translating One Button Tracker data into meaningful insights, thereby addressing the interpretive gap in mental health self-tracking. The results obtained from the iterative development sprints and the final expert validation provide strong evidence regarding the architectural requirements for such a system. The findings indicate that while LLMs are powerful narrative engines, they are fundamentally unreliable as standalone analytical tools for longitudinal health data. This introduces the necessity to transition from the conventional practice of prompt engineering to a hybrid or fully context-engineered architecture. In this chapter, we undertake a comprehensive examination of the outcomes to assess the feasibility and viability of the LLM functioning autonomously, the platform as a standalone entity, and the potential for LLM integration.

6.2 Interpretation and Contextualization of Findings

6.2.1 The Necessity of a Specialized Architecture

During the conducted manual quality assessments, a major finding was the model's inability to perform accurate arithmetic on raw log data. Despite using low-temperature settings to minimize randomness, the Llama 3 model exhibited persistent numeric drift when asked to calculate total event counts or derive success rates from raw JSON logs. It hallucinated weekly totals or failed to account for missing data points, a phenomenon that directly supports the limitations identified by Schulhoff (2024) [10] in their systematic survey of prompting techniques. They classify arithmetic and exact symbolic reasoning as known weaknesses of probabilistic models, which are designed for pattern completion rather than deterministic calculation.

When it comes to clinical AI, this failure indicates a specific architectural requirement, which is the separation of roles. The success of the final prototype, which achieved 100% accuracy in structured fields during the final review, was not due to better prompting, but due to the implementation of a hybrid architecture. By unburdening all quantitative analysis and metrics to a deterministic Python pipeline by adding a data pre-processing stage and reserving the LLM solely for qualitative summarization, the system avoided the risk of hallucination. This finding challenges the trend in some recent literature that advocates for a more LLM reasoning [3]. The findings of this study indicate that, in certain environments, such as a clinical setting, LLMs should be used exclusively as interpreters of pre-validated data rather than used as calculators.

6.2.2 From Prompt Engineering to Context Engineering

During the investigation and development, the carried out research hypothesized that refining the system instructions, using prompt engineering, would be sufficient to guide the model's output. However, as the dataset grew in complexity and size, the model began to struggle and suffer from known limitations, as it was attending to recent data while ignoring older patterns. This observation aligns with the theoretical framework proposed by Mei (2025) [14], who argues that static prompting is insufficient for complex, state-dependent tasks.

To resolve this, the decision was made to transition to a context engineering methodology, built upon the existing prompt engineering structure. Instead of treating the context window as a passive container for raw logs, the system implemented what is defined as a *Dynamic Context Assembly* function ($C = \mathcal{A}(c_{metrics}, c_{history}, c_{knowledge})$). By assembling a context bundle containing pre-processed weekly highlights, specific clinical definitions, and compressed historical summaries, the system was able to stabilise the model's output. This reinforces the claim that for longitudinal health monitoring, the quality of the clinical insight is determined not by the wit of the prompt text, but by the structural engineering of the information payload [14]. This effectively solved the consistency issues observed in Sprint 3, ensuring that the AI Analyst feature maintained a consistent *memory* of the patient's progress without exceeding the model's effective context window.

6.2.3 Bridging the Interpretation Gap

The main hypothesis of this thesis was to assess if the hybrid, context-engineered system could bridge the gap between raw OBT data and therapeutic insight. The findings confirm that utility is actually bi-directional, meaning that the system must reduce cognitive load for the clinician while simultaneously preserving organisation and providing meaningful reflection for the user.

Clinical Utility

The expert interview with the clinical psychologist provided crucial insight into the *nature* of the interpretation gap. The findings revealed that clinicians do not require a reproduction of the raw logs. Instead, they require the identification of anomalies. The psychologist explicitly devalued cumulative metrics, such as total counts, in favour of *hotspot* and *risk areas*. This way, the focus should be on specific temporal windows where the patient's coping mechanisms consistently failed.

This also pushes for the architectural shift from simple summarisation to *context engineering* mechanics. By filtering raw behavioural signals through a deterministic processing layer before LLM interpretation, the system acts as a noise-reduction filter. It transforms high-frequency data, which can be overwhelming in its raw form, into a more accessible narrative with metrics like *Peak Trigger Days* [17]. This confirms that the value of data processing and AI in this context lies not in generating advice, which carries the risk of hallucination and is not suggested by a psychologist, but in reducing the cognitive load required to identify patterns, and, this way, allowing the therapist to focus on the therapeutic alliance rather than data decoding [3].

User Utility

For the patient, utility is not defined only by diagnostic precision, but also by the sense of organization and self-reflection throughout the tracking process. Results from the initial user surveys and the literature study on the OBT indicate that the act of tracking serves as a *mental switch*, allowing users to express subjective experiences such as cravings or intrusive thoughts in the moment [4].

In this study, the results suggest that the platform positively supports *Personal Science*, meaning the practice of using self-tracking to gain self-knowledge [6]. Users in the initial survey prioritized trigger identification and mood understanding over purely clinical reporting. The proposed solution not only supports this by providing users with immediate visual feedback, heatmaps, and timelines that mirror their reality, but also supports integration with psychologist input, not replacement. The platform also supports the transition of the user from a passive subject to a more active participant, consistent with previous studies [5], which found that the physical act of clicking provides a tangible sense of control over internal states.

Data as a Boundary Object

Ultimately, the platform also advocates for OBT data as a *boundary object* that facilitates communication between the patient's experience and the therapist's clinical framework [8, 4].

For the patient, the data serves as a validation of their lived experience. However, for the clinician, it serves as objective evidence of the patient's patterns. The platform's final iteration, developed directly from user and clinician feedback, followed ethical safety guidelines for AI in healthcare [3]. This design choice prioritizes the therapeutic alliance over direct and automated intervention, ensuring that the AI supports the relationship rather than replacing the therapist's judgment. By grounding the LLM's narrative in the patient's specific history through dynamic context assembly, the system ensures that the *story* told by the data remains recognizable to the user and clinically useful for the psychologist.

6.2.4 Viability of an AI-based platform

The development and evaluation of the platform have demonstrated that the viability of such a system relies heavily on a dedicated architecture. The question of viability must be analyzed by distinguishing between the platform's deterministic capabilities, its interpretative requirements, and the technical viability of local usage.

Viability Without AI: The Deterministic Foundation

A considerable portion of the platform's features are not dependent on the chosen LLM but rather on traditional data processing. When the psychologist interview and user testing were conducted, it was confirmed that features such as the *Heatmap*, *Weekly Timeline*, and *Metric Aggregations* are indeed quite valuable, even though they are purely deterministic. These components rely on standard data science operations such as filtering, clustering, and counting, which can be executed with perfect accuracy using traditional algorithms.

Attempting to replace these functions with an LLM is not only unnecessary but, in fact, detrimental. As observed in the project's quality assessments and previous studies, LLMs can exhibit numeric drift and probabilistic instability when performing arithmetic tasks, leading to hallucinated totals or inaccurate event counts. Therefore, the platform could be highly viable without AI for displaying data without additional insights or narratives. This would rely solely on these features to address the core research problem, the *Interpretation Gap*. However, a platform without AI insights would be mostly considered a standard data platform. The solution does not simplify it into a clinical narrative or allow dynamic interrogation, thus alleviating the cognitive load on the clinician but not completely removing it [8].

Viability With AI: The Role of the Analyst

The addition of an LLM is justified not by its ability to calculate, but by its ability to *contextualize*. Although a data scientist could manually script rules to generate static text, such rule-based systems might lack the flexibility to handle the complexity of longitudinal therapy and adaptability to different patient datasets.

The LLM serves two specific roles that deterministic code cannot so easily replicate. Not only does it help to transform abstract patterns into a coherent clinical summary that connects multiple data points into a narrative regarding the patient's weeks, but it also helps the clinician to ask follow-up questions. The latter requires a system capable of semantic understanding and context engineering techniques, which a static script may not provide.

Nevertheless, the platform using an LLM is only viable with a dedicated architecture built around it using a context engineering approach and specific data pre-processing. In other words, using deterministic code for the *what*, specifically for handling metrics, and probabilistic AI for the *why*, dedicated to interpretation.

Viability of Local Inference

Finally, the technical architecture and decisions during development push for non-massive cloud infrastructure dependency on clinical-grade inference [15]. Which, even if not within the main scope of the thesis, supports the architectural decision to prioritize data sovereignty over the convenience of managed APIs.

Local execution ensures that possible sensitive Protected Health Information never leaves the infrastructure. This architectural isolation is critical for following strict regulatory frameworks such as GDPR without the need for complex third-party data processing agreements [16, 48]. By eliminating external data transmission, the system effectively avoids the risk of data leaks associated with cloud-based LLM providers [12].

Additionally, by eliminating per-token API costs and network latency, local inference establishes a predictable, flat-rate operational cost model ideal for clinical budgeting [49]. Moreover, recent advancements allow high-performance models, as the Llama 3, to run efficiently on standard consumer hardware with limited capacities [50, 51]. This opens the door for the platform to scale to individual or smaller clinics without depending on corporate-grade compute clusters.

In conclusion, a purely AI-driven platform is not viable due to reliability issues, and a purely deterministic platform needs additional optimization in order to properly address the interpretation gap. A locally hosted, hybrid platform pushes for an optimal balance of clinical utility, technical reliability, and data privacy while keeping deterministic metrics and LLM-driven narratives.

6.3 Implications of Research

The findings of this study go beyond reinforcing the platform's technical viability. In fact, we explore a practical framework for integrating AI into mental healthcare that preserves professional roles and patient trust. The validation of the hybrid, context-engineered platform carries distinct implications for both clinical practitioners and patients. This opens the door to a possible One Button Tracker solution for therapeutic integration.

6.3.1 For Psychologists

The most significant implication for clinicians is the shift in role from decoding data to learning and analyzing patterns. Current literature identifies the cognitive load of interpreting high-frequency longitudinal data as one of the main barriers to the adoption of digital health tools [8, 17]. By tackling the translation of raw logs into narrative metrics, the proposed system effectively takes on this burden.

As observed in the expert validation and supported by previous studies [4], when data is pre-processed into accessible narratives, it functions as a shared reference point that helps communication. This allows the therapist to spend less session time establishing *what* happened and more time exploring *why* it happened, deepening the therapeutic conversation.

The results that a context engineering architecture eliminates issues like numerical drift push for acceptance regarding AI reliability. By proposing that quantitative metrics be derived deterministically while qualitative insights are generated probabilistically, this research offers a blueprint for safe AI adoption. Psychologists can rely on the system's statistics with the same confidence as standard psychometrics, avoiding the fear of hallucinations raised in recent surveys [3, 10].

6.3.2 For Patients

The research confirms that active self-tracking, when supported by intelligent feedback, transcends mere observation to become an active coping strategy.

Consistent with the findings of Riisager (2025) and Arendt (2021), the physical act of clicking, but now contextualised by the platform, serves as a *mental switch*, allowing patients to externalise intrusive thoughts or cravings. The platform's ability to visualize positive rule usage reinforces positive behavioral changes, transforming the data from a record of pathology into a record of agency [5].

The successful implementation of Local Inference via Ollama directly addresses the privacy issue, which was one of the main user concerns identified in the initial surveys. This implies that patients no longer need to choose between advanced AI insights and data security. The viability of running clinical-grade models on local hardware ensures that sensitive OBT data remains within the therapeutic process, gathering greater willingness to share honest, high-fidelity data [15, 6].

6.4 Strengths and Limitations

6.4.1 Strengths of the Research

The main strength of this study lies in its iterative methodology. By structuring the development into six distinct sprints, each one directly influencing the next, the project was able to evolve from a theoretical concept to a validated Minimum Viable Product (MVP) based on continuous feedback.

A significant technical contribution is the successful implementation of a hybrid architecture that separates deterministic data processing from probabilistic reasoning. By dealing with quantitative tasks through Python scripts using traditional data processing techniques and reserving the LLM for narrative generation, the system effectively avoids the *numeric drift* and *hallucination* risks common in AI [10], and especially dangerous in healthcare.

In response to the initial user surveys, which identified data privacy as one of the main considerations for adoption, this study advocates for the viability of a fully local inference stack. By using pre-trained Llama 3 models via Ollama, the platform proves that clinical interpretation can be achieved on consumer hardware without transmitting Protected Health Information to third-party cloud providers, satisfying strict data requirements [15].

Most importantly, unlike generic chatbot interfaces, the platform implements context engineering techniques. By restructuring raw logs into a context bundle before inference, the system achieves a solution that shows viability in tackling the interpretation gap. The expert validation in later sprints confirmed that this approach successfully translates behavioural signals into actionable information, aligning the technological output with actual therapeutic workflows.

6.4.2 Limitations of the Study

Despite the successful validation of the final solution, several limitations regarding the data source, evaluation scale, and technical optimisation must be acknowledged.

The platform, in its current state, was developed and tested using a synthetic dataset simulating coffee dependency. This was important to avoid privacy risks, it enabled a quicker iterative process and allowed for a controlled ground truth for testing, however it lacks the noise, irregularity, and high-stakes nature of real-world psychopathology. Consequently, the system's resilience to the chaotic data patterns often seen in acute mental health crises remains to be tested in a live environment.

In a similar vein, this research serves as a viability study and proof-of-concept, not a clinical trial. While the expert interview provided qualitative validity, the study lacks longitudinal quantitative data regarding patient outcomes. We have not yet demonstrated that using this platform reduces interpretation burden or improves patient adherence over a sustained treatment period, which would require integrating the platform into a real, controlled therapeutic process.

While the system successfully transitioned from simple prompting to Dynamic Context Assembly, it still relies on a fixed context window strategy. The current implementation does not yet utilize the optimal context management techniques, such as vector database retrieval for external clinical knowledge or sophisticated memory compression for multi-year patient histories [14]. Future iterations would benefit from abandoning residual prompt engineering that was created in the early sprints in favour of a fully context engineering optimisation pipeline to handle ultra-longitudinal data without context overflow.

6.5 Future Research

While considering the potential for the ongoing development process of the proposed solution, it is important to acknowledge the fundamental liabilities within the current platform. These liabilities need iterative refinement to ensure the optimal outcome. In order to ensure optimal results when it comes to LLM output, it is suggested that the platform undergoes a thorough restructuring process. This process must be undertaken with the explicit objective of eliminating unnecessary prompt engineering mechanics and implementing context engineering as the main methodology for optimizing the LLM within the OBT context.

An alternative approach to development should also be considered. As the results demonstrated, both users and the psychologist expressed a high level of interest in specific metrics through features that were not LLM-dependent. The proposed solution could be able to function as a data platform in isolation, without the need for an optimised structure based on a large language model. Although this solution would only be viable if the narrative features were eliminated and an early user study were to be conducted in order to more thoroughly investigate the type of information required to aid the therapeutic process and the appropriate data preprocessing.

Any further development would naturally lead to the initiation of clinical trials, which would serve to evaluate the platform not only with real data but also within a realistic therapeutic context. The main objective is to transition to a study of the user experience, including patients, psychologists and the therapeutic context. It is necessary to understand the way in which the platform would be integrated, the manner in which it would be utilized, and to undertake a thorough examination of its viability and usefulness, independent of the development direction that was previously referenced.

7 Conclusion

The main objective of this research was to investigate whether a Large Language Model integrated platform could bridge the gap in interpretation between self-tracking data and clinical insight. The development and validation of the Decoding the Click platform yielded diverse findings and opened the door to various future approaches.

The platform translates raw behavioural signals into clinical and grounded narratives, supporting the main hypothesis regarding the use of LLMs in mental health. Its main contribution is the development of a Minimum Viable Product that helps the transition of the OBT from a data logger into an active clinical decision support system. This provides a study blueprint for integrating Large Language Models with mental health data while maintaining data security.

However, testing sessions and quality assessments revealed that standalone LLMs are unreliable for quantitative analysis and consistently exhibit known issues, such as numeric drift and hallucinations, when calculating key metrics. During the study, it was established that, to guarantee the viability and clinical safety of LLMs, a specialised architecture using context engineering techniques and role separation is mandatory, with deterministic code handling metrics and LLMs handling narrative. Incorporating structured context bundles and dynamic context assembly into the core system architecture was necessary to stabilise the model's output. This validated the previously studied theoretical framework that performance in long-context tasks depends on information architecture rather than prompt phrasing [14]. Given the longitudinal nature of OBT data, this research demonstrated the ineffectiveness and flaws of prompt engineering relative to context engineering, stressing the latter as a necessary key for LLMs in healthcare.

The future of artificial intelligence in psychology is not grounded in collecting additional data. Instead, it is subjected to the interpretation of existing data to derive meaningful insights. The thesis demonstrates that technology can serve as a translator and a helper, thereby converting the digital exhaustion of daily life into the human language of therapeutic insight.

This thesis also serves as a statement that the platform validates AI as a boundary object that mediates the patient-clinician relationship. However, it never replaces it. The use of a platform dedicated to pattern data and narrative insights enables psychologists to prioritize relational therapeutic tasks, thus enhancing the human element of therapy rather than replacing it entirely. Moreover, demonstrating the viability of consumer-grade local hardware for sophisticated analysis suggests a future in which advanced, privacy-preserving diagnostic tools will be accessible to independent practitioners and smaller clinics, thereby opening access to data-driven mental healthcare.

Mental health is of incredible importance and should be regarded as a main area of health investment. It is encouraging to observe that a growing body of research and a range of tools have emerged in recent years to optimise this area, thereby making it more accessible to all. This project aligns with recent trends in analysing recent technologies and integrating them with the aforementioned research questions. In conclusion, this study, Decoding the Click, demonstrates that when constrained by appropriate engineering and grounded in a clinical context, Large Language Models can help in bridging the gap between a patient's experience and a clinician's expert understanding.

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A Appendix

Third Question: Alex is someone that not only practices football, but also reads, plays guitar and draws. However coffee craving often affect long sessions spent on these hobbies. What can I suggest him to change?

Mistral7B Response

To help Alex manage his coffee addiction and improve focus during his hobbies, I would suggest the following changes:

1. Encourage the use of pre-defined coping mechanisms more frequently, especially during high stress or fatigue periods (e.g., deep breathing, short walks, or hydration breaks). This can help break the automatic habit loop associated with coffee consumption.
2. Introduce a new OBT protocol to log Alex's state during his hobbies: 'Coping Attempt / Alternative Used During Hobby'. This will help identify if and when coffee is used as a coping mechanism during these activities, allowing for more targeted interventions.
3. Encourage Alex to take regular breaks during long hobby sessions to practice relaxation techniques or engage in other activities that promote focus and reduce stress. This can help minimize the need for coffee as a crutch.

Figure A.1: Second LLM Interaction Example of Psychologist-Model Interaction. This is the last question shown in the conducted user survey. This question shows the LLM adoption for a direct approach when prompt to. This was expected since in these early interaction the study on tones, roles and speech constraints was still being concluded.

```
1 bundle = simulate_patient_activity(PATIENT_ID)
2 bundle["prompt_block"] = compose_prompt_block(PATIENT_ID, bundle)
3
4 fallback = build_fallback_dashboard(bundle)
5 dashboard, llm_used = merge_dashboard_with_llm(fallback, llm_payload)
6
7 response_dashboard = {
8     "patientProfile": PATIENT_PROFILE,
9     "otherPatients": OTHER_PATIENTS,
10    "triggerTimeline": dashboard["trigger_timeline"],
11    "cycleProgress": dashboard["cycle_progress"],
12    "dataInsights": dashboard["data_insights"],
13    "timelineEvents": dashboard["timeline_events"],
14    "calendarHeatmap": dashboard["calendar_heatmap"],
15    "headlineMetrics": dashboard["headline_metrics"],
16    "studyPeriod": {
17        "start": bundle["start_date"],
18        "end": bundle["end_date"],
19        "days": len(bundle["daily_records"]),
20    },
21    "generatedAt": generated_at,
22    "dataSource": "llm" if llm_used else "synthetic",
23 }
```

Listing A.1: Bundle and Prompt Construction during the Prompt Engineering Phase. This is an example of the bundle constructed and how the API consumes treated data within our backend

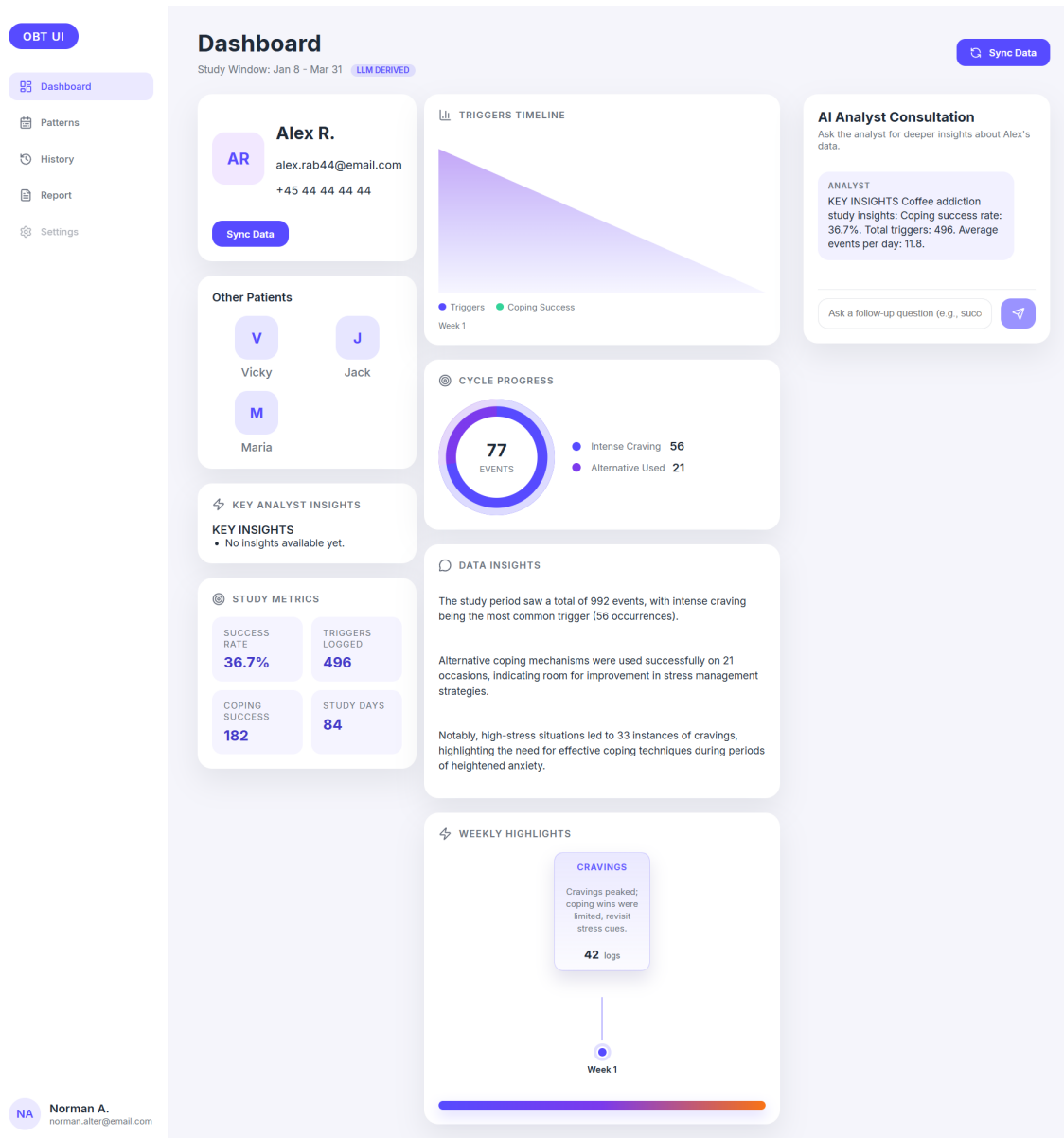


Figure A.2: Sprint 3 Development - Dashboard. Dashboard view during Sprint 3, still neutral and clinical colors around the central blue color. Contained all the features present in the latest iteration, nevertheless data was not properly consumed, insights not loaded properly and overall needing major fixes.

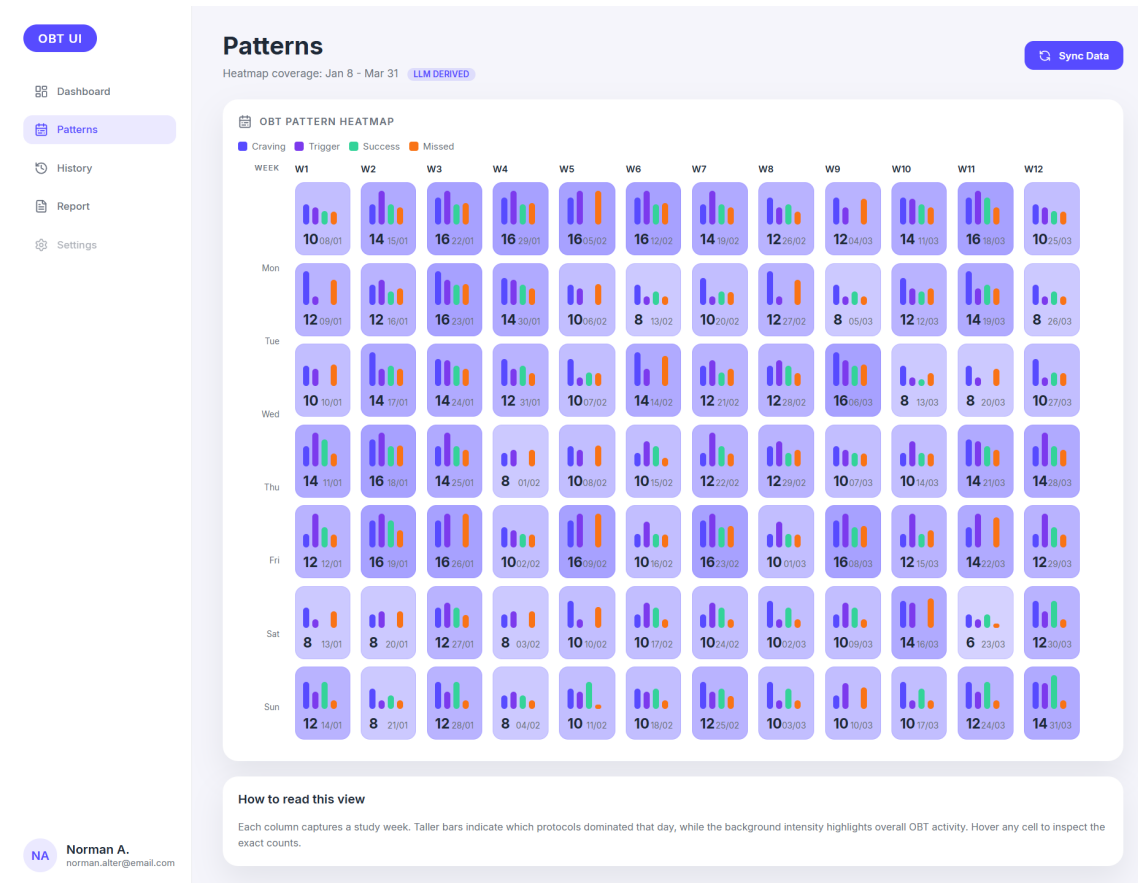


Figure A.3: Sprint 3 Development - Patterns/Heatmap Feature. Heatmap feature in it's initial stage of development. Color scheme eventually was changed and the data expanded. Bar Plots not properly reflective of daily data.

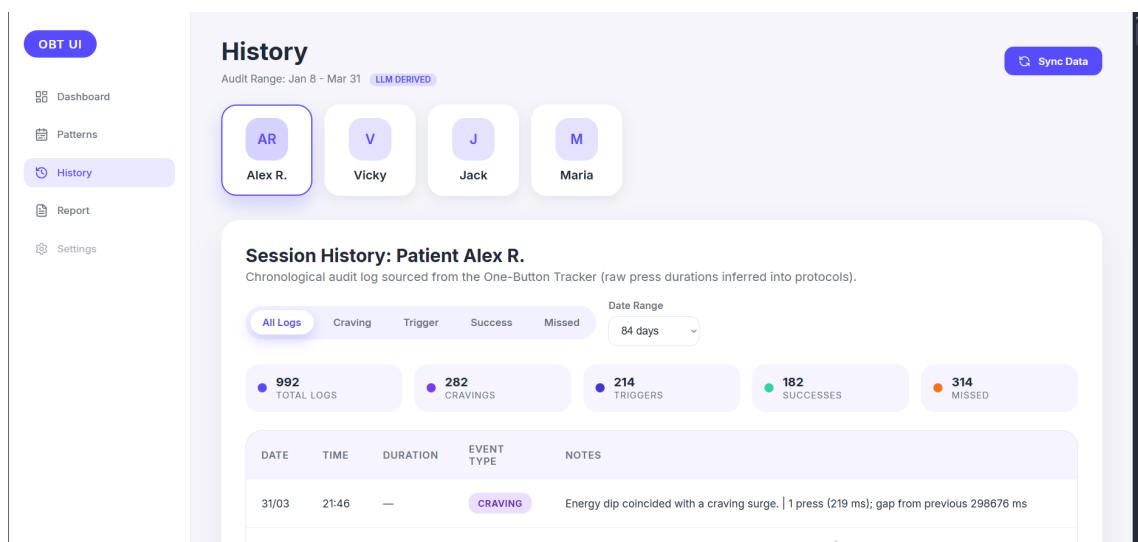


Figure A.4: Sprint 3 Development - History Feature. History view during Sprint 3, which, feature wise, was closer to the latest iteration, missing pagination, new filters and sorting. During this stage, only the Alex R. dataset existed thus the other patients served as placeholders.

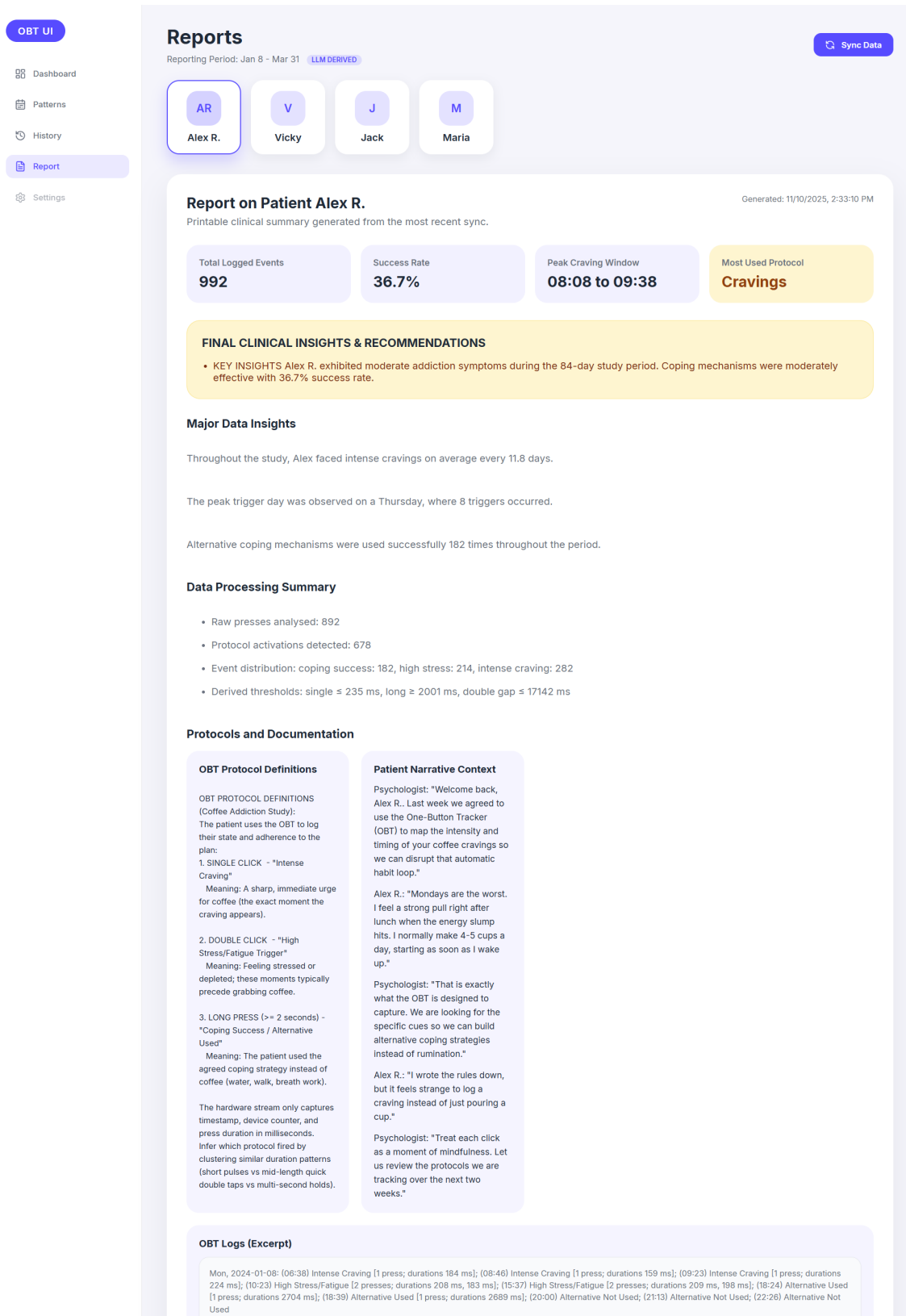


Figure A.5: Sprint 3 Development - Report Feature. During Sprint 3, the Report was being developed to be the concise and direct summary. However, still missing the clinician notes and prioritizing of different narrative and metrics derived from the psychologist interview and test sessions.

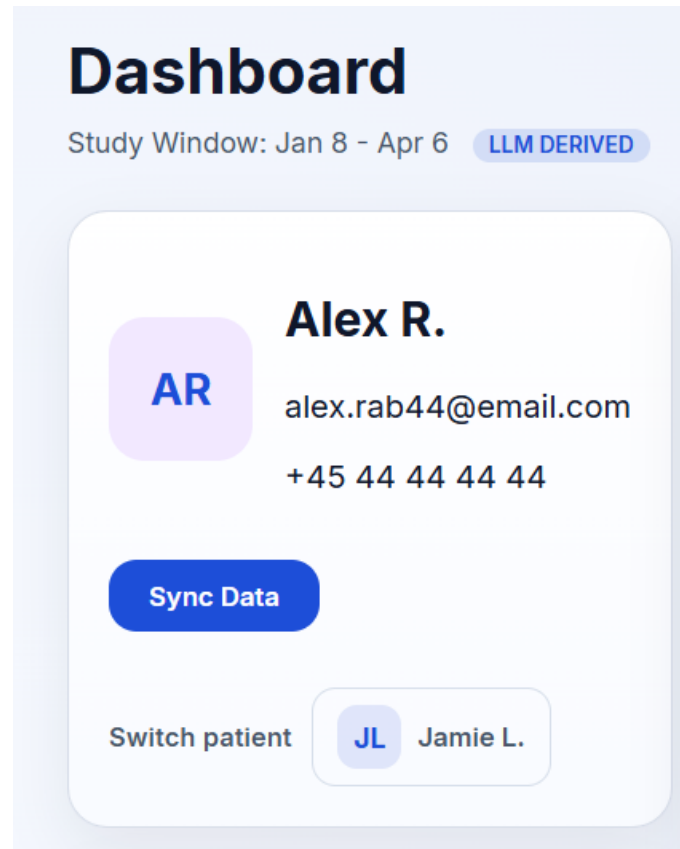
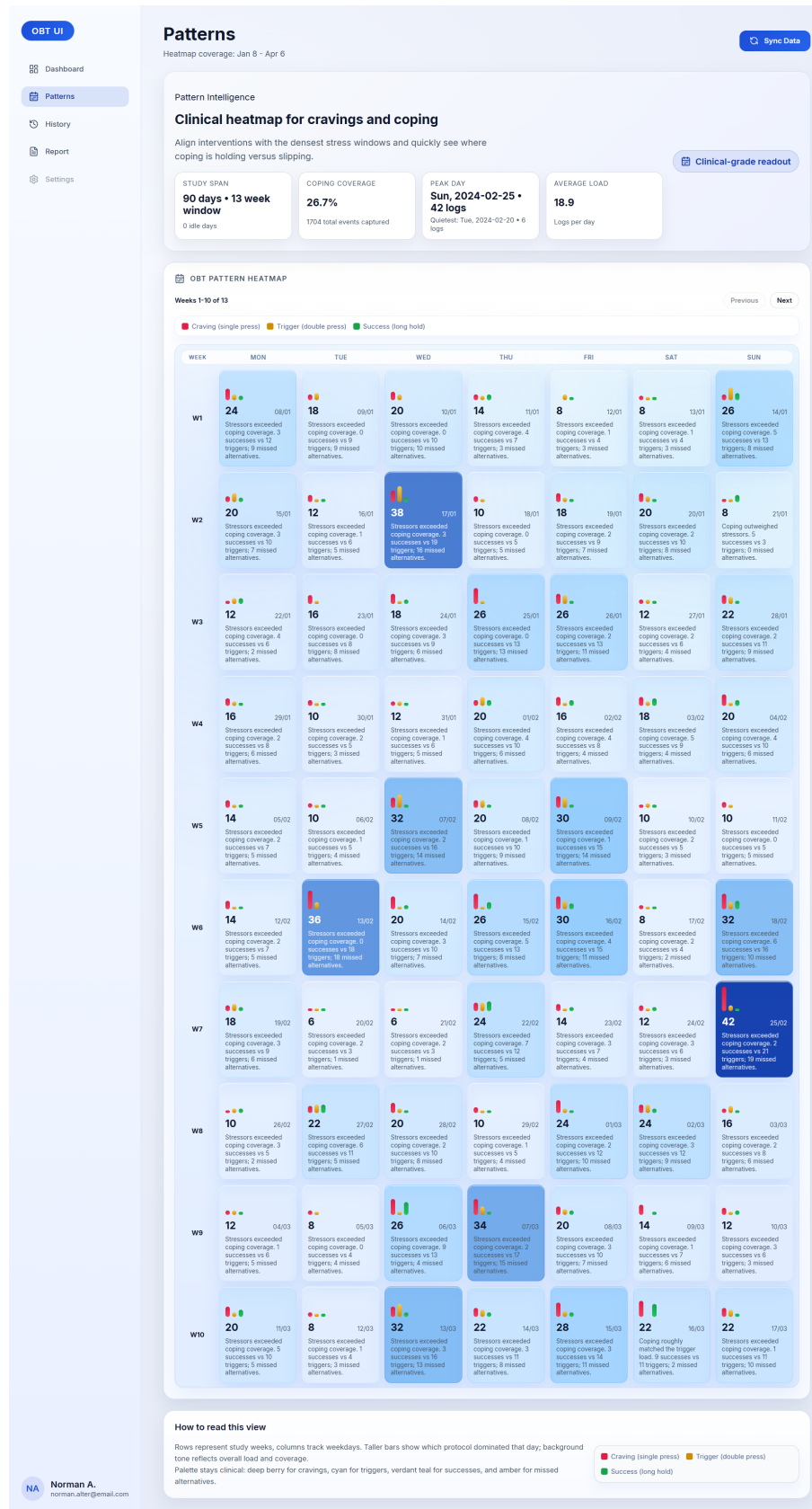


Figure A.6: Sprint 5 - Patient Selection. Introduction of a second dataset, to test the adaptability of the platform when faced with a dataset with less protocols. By clicking the other patient, the platform would load the necessary information for that dataset.



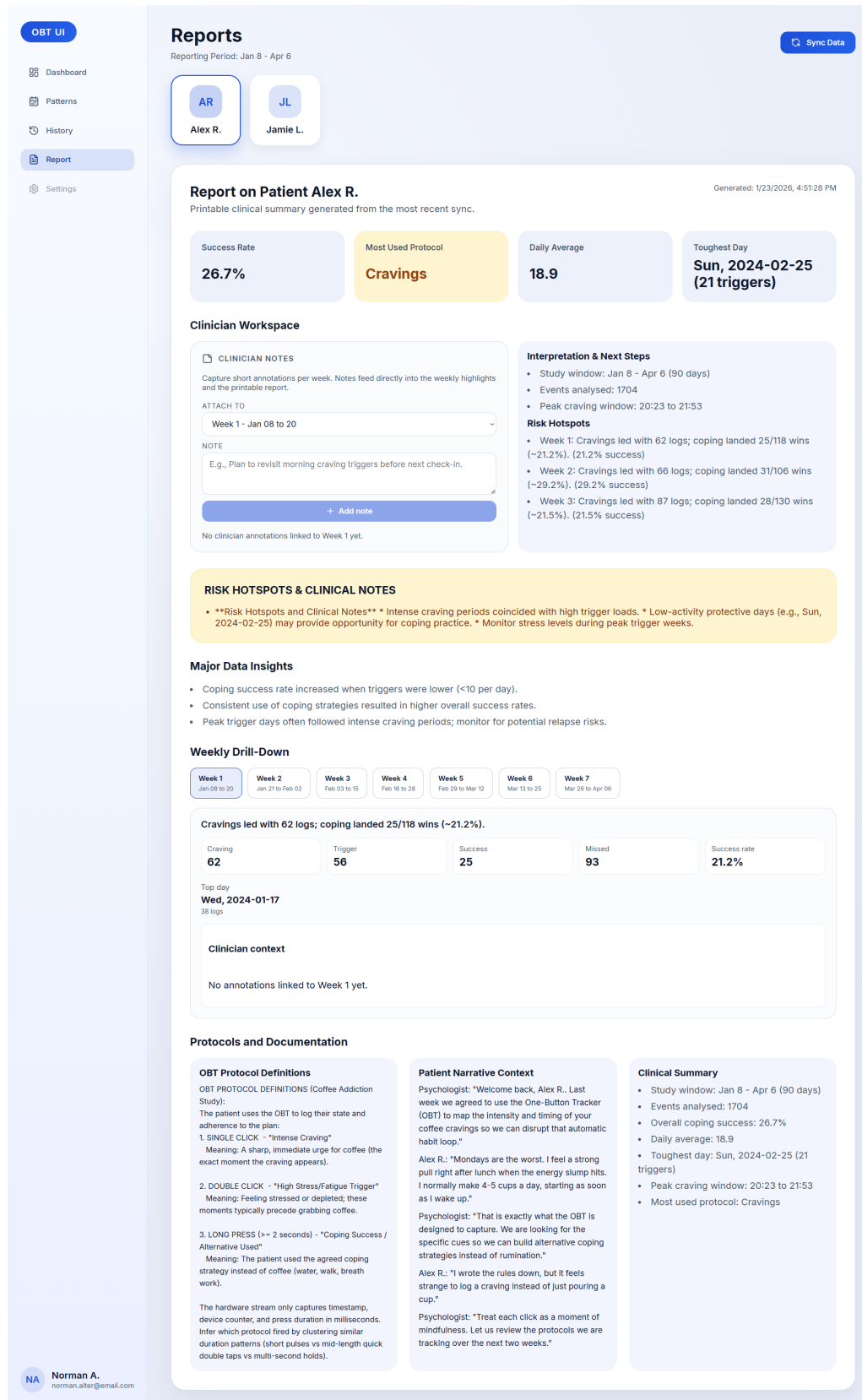


Figure A.8: Final Report View. Report on Alex R. with prioritized information detailed in the psychologist interview and the notes feature

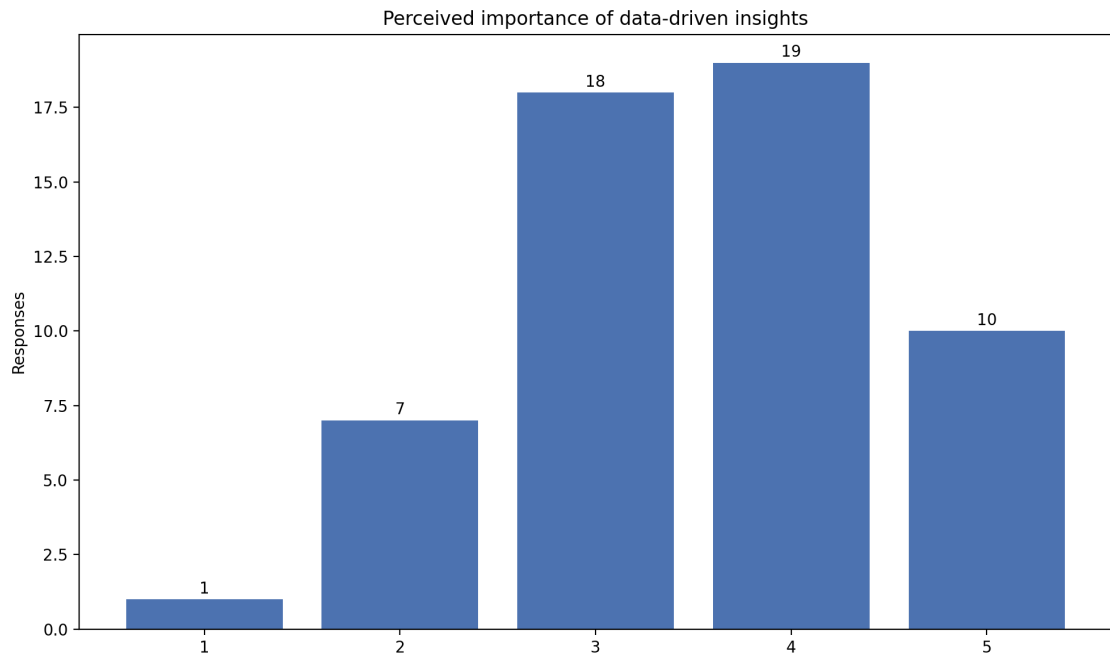


Figure A.9: Initial User Survey - Data Importance. Histogram on data-driven insights importance. Skewed to slightly favor insights.



Figure A.10: Initial User Survey - User Distribution. Representation of the participants demographics. Extremely biased to three main areas: Setúbal, Lisbon and Copenhagen. Given the sample size, the two answers from The Netherlands and Slovakia where not visible in this representation.

B Appendix - Sprint Documentation

First Sprint

Technical University of Denmark
Afonso Cunha - s233177

16. februar 2026

1 Planning and Objectives

Sprint 1: Initial Development and Prototyping

Duration: September 22, 2025 – October 6, 2025

Sprint Objective:

The primary objective of this sprint is to establish the foundational components of the project's data-to-insight pipeline. This includes the generation of synthetic data, the development of preliminary user interface prototypes, and the initial selection of a suitable machine learning model.

Key Activities & Deliverables:

- Data Generation
 - **Activity:** Generate a synthetic dataset to simulate a plausible user-tracking scenario (e.g., coffee addiction) using a state-of-the-art Large Language Model (LLM).
 - **Deliverable:** A documented synthetic dataset and the prompt used for its generation.
- Prototyping & UX/UI Design
 - **Activity:** Develop preliminary user interface prototypes in Figma based on the insights gathered from the initial user questionnaire. This activity will ensure that the design aligns with user needs from the project's outset.
 - **Deliverable:** A Figma prototype of the one-button tracker interface and the psychologist's dashboard.
- Model Investigation & Integration
 - **Activity:** Conduct a preliminary investigation of suitable pre-trained models. This initial iteration will prioritize models with strong computational performance and architectural simplicity to establish a Minimum Viable Product (MVP).
 - **Deliverable:** A technical report summarizing the selected model's characteristics and its initial integration plan.

Rationale:

This sprint is designed to validate core assumptions about the project's technical feasibility and user interface. By using synthetic data and pre-trained models, we can rapidly prototype and test the fundamental data-to-insights pipeline while de-risking the project for future, more complex iterations. The emphasis on performance over clinical accuracy at this stage is crucial for ensuring a functional system from which to build upon.

2 Development

As mentioned on the previous section, the primary objective of Sprint 1 was to establish the foundational components of the project's data-to-insight pipeline, encompassing synthetic data creation, preliminary user interface design, and the integration of a stable, local LLM execution environment.

2.1 Model Investigation and Integration

Initial exploration suggested using Hugging Face models with CPU offloading for local inference. This approach was abandoned due to anticipated performance bottlenecks (latency and resource constraints) incompatible with rapid, iterative testing. The toolchain was decisively shifted to leverage Ollama as the local execution environment, hosting the Mistral 7B parameter model (GGUF quantization).

This architectural decision provided two key advantages:

1. **Efficiency:** It allows for high-speed, local inference execution, decoupling the project from cloud resource dependencies.
2. **Scalability:** It creates a stable baseline environment against which all future fine-tuned models can be directly compared. The Python ollama client was successfully integrated to manage all API communication.

2.2 Data Generation, Structuring and Contextualization

This activity was critical for validating the technical feasibility of the data-to-insights pipeline.

2.2.1 Synthetic Dataset Generation

The initial plan called for using a state-of-the-art LLM to generate a synthetic dataset simulating a plausible user-tracking scenario (e.g., coffee addiction). While the core dataset elements were later generated programmatically via a Python script to ensure precise structure and adherence to OBT logging format, this script successfully produced the three required interdependent components:

- **Conversation Context:** A narrative text block defining the patient's background and therapeutic objective.
- **OBT Protocol Definitions:** The core dictionary defining the meaning of the three input types (Single Click, Two Consecutive Clicks, One Long Click).
- **Simulated Click Logs:** Synthetic two-week log of time-stamped clicks, simulating real-world patient adherence.

2.2.2 Deliverable Rationale

Generating this documented, synthetic dataset allowed for quick prototyping and testing of the LLM's analytical capabilities dependency on a real clinical data, effectively de-risking the project for future, more complex iterations.

2.3 Prototyping and UX/UI Design

Based on insights gathered from initial user questionnaires, preliminary user interface prototypes were developed using Figma.

2.3.1 Deliverables

- **Figma Prototype (One-Button Tracker Interface):** A low-fidelity wireframe illustrating the essential input mechanism and user feedback for the patient-facing component.
- **Figma Prototype (Psychologist’s Dashboard):** A low-fidelity layout defining the structure for the clinical dashboard, focusing on data visualization areas, key insights and the interactive chat interface with the AI analyst

2.3.2 Rationale

The emphasis at this stage was on low-fidelity design to rapidly validate core assumptions about user flow and information architecture, ensuring the design aligns with user needs early in the project before significant development resources were allocated

2.4 Interactive Baseline Inference Implementation

The final deliverable of the sprint was the creation of a functional, interactive Python application to execute the Zero-Shot test and demonstrate the LLM’s ability to maintain context over a multi-turn consultation.

The primary challenges and resolutions in implementing the interactive chat logic involved prompt engineering and parameter tuning:

- **Boilerplate Repetition**
 - The model repeatedly generated its initial KEY INSIGHTS and structured prompts in every turn, failing to transition into a direct Q&A mode.
 - The SYSTEM_PROMPT was refactored into explicit Phase 1: Initial Analysis (one-time execution) and Phase 2: Interactive Consultation (multi-turn Q&A) instructions.
- **Truncated Responses**
 - LLM answers were cut off mid-sentence, indicating an incomplete generation.
 - The *num_predict* parameter in the *ollama.chat()* call within the iterative while loop was increased from a default low value to 512 tokens, ensuring sufficient response length.

The resulting application successfully performs the initial analysis and maintains the data context across user-asked questions, providing a solid, functional baseline for all subsequent development, particularly the fine-tuning experiments planned for Sprint 2.

3 Results

3.1 Phone Prototypes

As previously stated, the emphasis in this section was on establishing a baseline, encompassing both the prototype and the model. Consequently, we developed prototypes that would establish a solid foundation for a standard application, following a structure analogous to that of fitness and health tracking applications. The resulting prototypes as they are on Figure 1.

Upon presenting the prototypes, the testers were asked about the clarity of the interface’s design and labels in communicating the sections and navigation. The responses indicated a favorable reception. Nonetheless, in an effort to encourage the provision of more open-ended feedback, the testers were asked regarding the nature of the information or feedback they would



Figure 1: Low-Fidelity Interface for Phone

anticipate seeing on this layout. A close examination of the responses reveals an overarching emphasis on the presentation of data patterns and the manifestation of mood fluctuations. In these responses often is mentioned the need of the graphs being accompanied by a report and/or a summary of the data being visualized. Lastly, a comment was made regarding the significance of a neutral background in encouraging users to remain within the application and to continue inputting additional information. This is a crucial aspect to consider in future development, as it exemplifies the importance of user experience.

Interestingly, when asked about desired changes, the testers gave minimal responses besides the renaming of a label and the addition of more detail and interactivenss with the graph.

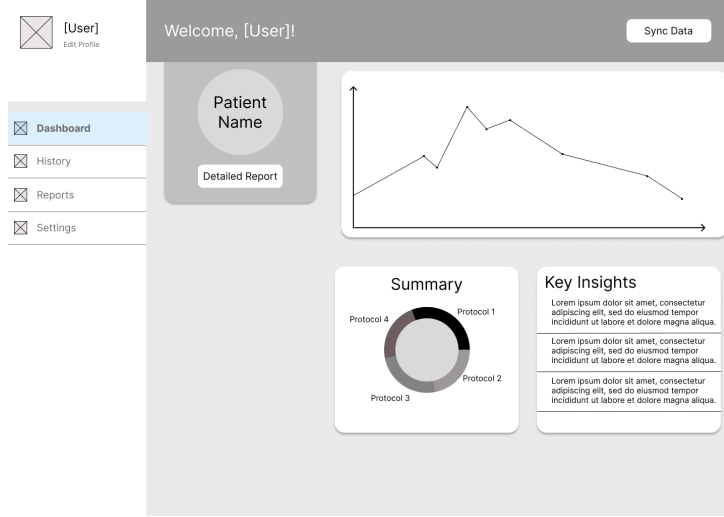
3.2 Dashboard Prototypes

Regarding the dashboard prototypes, seen in Figure 2, the testers where asked similar questions as they were for the phones prototypes. These questions ranged from direct, linear-scaled answered, questions that focused on the layout and portray of information but also open-ended questions.

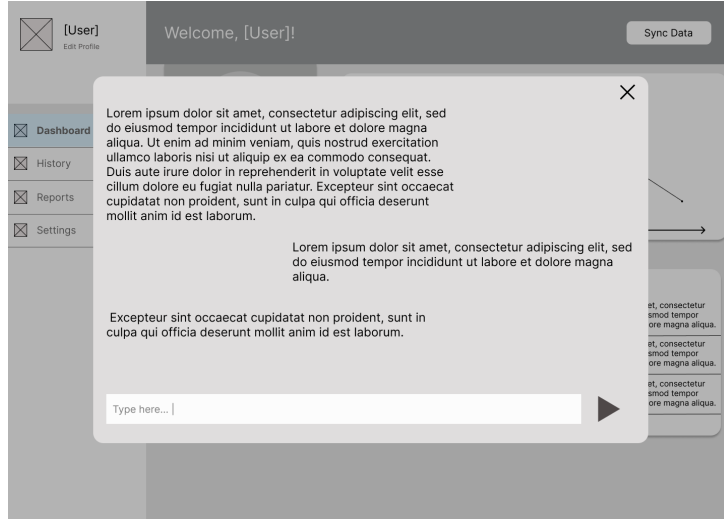
In a manner consistent with the configuration of the phone interface, testers expressed their approval of the direction and organization of the information presented. Nevertheless, as was the case previously, the testers were queried about their preferences and potential modifications to the layout. They indicated a need for the ability to switch between patients, a feature that was not a primary focus. Additionally, they proposed enhancements to the alerts, number limits, and the overall comprehensiveness of the data and information displayed.

A new question posed to the testers pertained to the type of charts they would find appealing. They exhibited a pronounced preference for timeline charts, with a small mention of a heatmap.

On the next step, when queried about the integration of the LLM, and more specifically, the incorporation of a chat-like feature, the testers exhibited a degree of uncertainty or even a



(a) Home Page



(b) History Page

Figure 2: Low-Fidelity Interface for Desktop

propensity to refrain from utilizing such a feature. Nevertheless, this reluctance was accompanied by a consensus on the intuitiveness of such a feature when it comes to the facilitation of follow-up inquiries concerning the data.

In light of the expressed opposition to the incorporation of a chat-like feature, the testers were asked about their conceptualization of the manner in which the interaction with the LLM would unfold. The testers’ responses indicated a preference for a more summary-focused or report-oriented feature, despite variations in their approach to the presentation of such summaries. A number of testers proposed a more proactive approach, involving brief suggestions or notifications similar to those utilized by Duolingo. On the other hand, some testers expressed a preference for a more comprehensive, in-app presentation of information regarding the data.

3.3 LLM Baseline Model

In this final section, we sought to perform a quality assessment of the Key Insights and engage in a limited interaction with a tuned LLM within the context of our OBT. In this study, a hypothetical scenario was devised to illustrate the effects of coffee addiction. The patient, designated "Alex R.," was created to represent an individual grappling with a significant dependency

on coffee. In this initial sprint, the data created for the LLM and used for its fine-tuning was generated through the implementation of a Python script.

In this particular instance, the data structure exhibited slight discrepancies compared to the authentic OBT data. Nevertheless, the primary objective was to assess the efficacy of the insights derived from the scenario-generated data in a clinical setting and their potential application in clinical contexts.

This way, we presented the three defined protocols for the first fake scenario:

1. **Single Click** - Intense Craving
2. **Two Consecutive Clicks** - High Stress/Fatigue Trigger
3. **One Long Click (more than 2 seconds)** - Coping Success / Alternative Used

In this section, we sought to provide a more comprehensive illustration of the type of agreement and its application in a clinical setting. However, despite the provision of extensive context, the testers still exhibited reluctance in utilizing an LLM, even with fine-tuning to therapy and OBT data context.

Nevertheless, in order to further investigate how a possible LLM integrity could occur, we inquired once again, but now with the scenario and protocols context, about what kind of inquiries would they do to the LLM, both as an psychologist and a patient. The results of the study were found to be of interest from the perspective of psychology. From the psychologist's perspective, the inquiries were primarily designed to convey information through the simple request of patterns, enhanced data explanations, and results. This approach serves to preserve the role of the psychologist in action, with the objective of providing actual assistance to the patient. Nevertheless, from the patient's perspective, the testers were more available to address queries regarding strategies and appropriate courses of action in specific circumstances. However, they consistently maintained a neutral stance with regard to the trained LLM.

Finishing the final assessment, we wanted to test the output of the model. This way, we presented the initial key insights generated from our trained model. Below, here are some of the initial insights the LLM has given.

Start of LLM Output

KEY INSIGHTS:

1. Alex R.'s coffee cravings are frequent and persistent, with an average of 4-5 cups daily as reported. The OBT logs show that cravings occur multiple times a day across all days, peaking on Mondays and Tuesdays.
2. Stress and fatigue seem to be significant triggers for Alex's coffee consumption, as indicated by the high number of 'High Stress/Fatigue Trigger' clicks. These triggers are particularly prevalent in the morning (7 AM - 10 AM) and late afternoon (3 PM - 6 PM).
3. Coping mechanisms used to address coffee cravings are sporadic, with only occasional 'Coping Success / Alternative Used' clicks. The use of these alternatives could be increased for more effective habit disruption.

These insights suggest that Alex R.'s coffee addiction is a recurring issue throughout the day and is likely triggered by stress or fatigue. Encouraging the use of pre-defined coping mechanisms may help break the automatic habit loop associated with coffee consumption.

End of LLM Output

Upon being presented with these preliminary key insights, the testers demonstrated a slightly more favorable utilization of the LLM, which exhibited a discernible correlation between the protocols and the data. The testers agreed that the insights exhibited a high degree of accuracy in their interpretation of the expected patterns and proffered equitable recommendations for the designated user in the hypothetical scenario. This occurred subsequent to the testers adopting

the role of psychologists once more. This inquiry proved instrumental in showing the subsequent actions a psychologist might undertake upon reviewing this synopsis.

Lastly, we showed a test Q&A session taken from this scenario and interaction with the LLM and asked, for the last time, about the completeness, detail, suitability and usefulness for clinical consultation. The answers to this Q&A session were very positive.

Here is the test Q&A:

First Question: Should I, as the psychologist, suggest a 4th or a 5th protocol for Alex?

Mistral7B Response:

Based on the data provided, it appears that Alex R. is using the pre-defined OBT protocol effectively to log his coffee cravings and associated triggers (stress/fatigue). However, the use of alternative coping mechanisms seems sporadic.

To encourage more frequent use of these alternatives, I would recommend introducing a new protocol:

4. THREE CONSECUTIVE CLICKS: 'Unsuccessful Coping Attempt / Alternative Not Used' Meaning: The patient attempted to use a pre-defined coping mechanism but was unsuccessful in preventing coffee consumption (e.g., deep breathing did not reduce craving). This will help identify patterns and areas for improvement in the development of more effective coping strategies.

Second Question: On our new session, Alex has told me that, on Mondays and Wednesdays, a cup of coffee helps him focusing on football practices in the end of the day. What does the data say about this?

Mistral7B Response:

The OBT logs show that Alex experiences high stress/fatigue triggers throughout the week, with a slight peak on Mondays and Tuesdays. However, there is no specific pattern indicating that he uses coffee to help him focus during football practices at the end of the day (17:00-19:00) on Wednesdays or any other days.

To better understand if this habit exists, it would be helpful to ask Alex about his coping mechanism usage during those specific time slots or consider adding a new OBT protocol to log his state during football practices (e.g., 'Coping Attempt / Alternative Used During Football Practice'). This additional data could help pinpoint whether coffee consumption is indeed an effective coping strategy for Alex during these practices.

Third Question: Alex is someone that not only practices football, but also reads, plays guitar and draws. However coffee craving often affect long sessions spent on these hobbies. What can I suggest him to change?

Mistral7B Response

To help Alex manage his coffee addiction and improve focus during his hobbies, I would suggest the following changes:

1. Encourage the use of pre-defined coping mechanisms more frequently, especially during high stress or fatigue periods (e.g., deep breathing, short walks, or hydration breaks). This can help break the automatic habit loop associated with coffee consumption.
2. Introduce a new OBT protocol to log Alex's state during his hobbies: 'Coping Attempt / Alternative Used During Hobby'. This will help identify if and when coffee is used as a coping mechanism during these activities, allowing for more targeted interventions.
3. Encourage Alex to take regular breaks during long hobby sessions to practice relaxation techniques or engage in other activities that promote focus and reduce stress. This can help minimize the need for coffee as a crutch.

Second Sprint

Technical University of Denmark
Afonso Cunha - s233177

20. februar 2026

1 Planning and Objectives

Sprint 2: UI Development, High-Fidelity Design and Data Structuring

Duration: October 7, 2025 – October 24, 2025

Sprint Objective: The primary objective of this sprint is to transition from conceptual validation to foundational technical implementation. This involves converting low-fidelity designs into actionable mid-to-high fidelity prototypes, establishing the front-end development framework (React), and defining a rigorous, production-ready data structure to facilitate clinical context for the LLM.

Key Activities & Deliverables:

- Prototyping & UX/UI Refinement
 - **Activity:** Refine the low-fidelity Figma wireframes based on feedback from the user questionnaire (Q1). This includes defining the visual style (color, typography, components) and designing key interactions, such as the visual feedback mechanism for the OBT and the final layout for the primary data visualization.
 - **Deliverable:** A Mid-to-High Fidelity Figma Prototype of the Psychologist's Dashboard, demonstrating defined visual aesthetics and specific interaction states (e.g., successful log feedback, active chat panel).
- UI Development & Framework Setup
 - **Activity:** Initiate coding of the Clinician Dashboard using React and Tailwind CSS. This involves setting up the core, single-file component structure, implementing a responsive layout for insights, visualizations, and the chat interface, and creating the placeholder logic for communicating with the local LLM endpoint (Ollama).
 - **Deliverable:** The Initial React Dashboard (App.jsx), a functional front-end application displaying the core responsive layout and including the non-functional skeleton code for asynchronous LLM API calls and chat history management.
- Data Structuring & LLM Input Refinement
 - **Activity:** Finalize the internal data model to ensure it accurately mirrors the data produced by a real OBT device. This involves formally defining the schema for OBT Protocol Definitions (click type, label, category) and changed how they are logged.
 - **Deliverable:** A documented Structured OBT Data Model that defines the format for protocols and logged events. This structure must be easily serializable (e.g., to CSV or JSON) for consistent input into the LLM prompt context.

Rationale:

This sprint focuses on technical foundation and design maturation. By building the UI structure and simultaneously finalizing the data model, we ensure the front-end design can accurately accommodate the real data outputs, and the LLM will receive clean, structured input, which is essential for improving analytical quality and moving beyond the baseline model.

2 Development

This development cycle focused on stabilizing the core application structure and elevating the user interface (UI) from basic wireframes to high-fidelity, component-based prototypes. The work successfully established a robust design foundation and expanded the application's core feature set.

2.1 Architectural Focus and Achievements

The primary architectural success was the establishment of a single-file, multi-page application structure. By leveraging modern web frameworks React and Tailwind (Meta Platforms, Inc., 2025, Tailwind Labs, 2025), we ensured that the entire client-side interface is self-contained and highly maintainable.

2.1.1 Core Achievements

- **Component-Driven Development:** All UI elements, including headers, metric cards, and navigation, were formalized into reusable components based on the defined design blueprint (Indigo primary, Yellow accent).
- **Multi-View Expansion:** Two critical clinical views were designed and prototyped, moving beyond the initial Dashboard scope:
 - History View: Focused on data density, scannability, and client-side filtering controls to manage a large chronological log of OBT events.
 - Report View: Established a clean, print-friendly structure with clear section headers, designed to be used as a final clinical document.
- **Design Consistency:** The transition to high-fidelity prototypes ensures that the application adheres to strict aesthetic guidelines (shadows, rounded corners, responsive layout), making the final product ready for integration into a clinical setting while still following modern design guidelines (8Designers, 2023, March 18) to optimize user experience. .

2.2 Technical Challenges and Pending Integration

While the visual and structural goals were met, several technical challenges and dependencies remain that must be addressed in the next phase to ensure full application functionality.

2.2.1 Visual Dependency

A significant portion of the Dashboard remains reliant on visual placeholders. We successfully defined where the data charts should go (e.g., Event Distribution, Weekly Trend), but the actual implementation of complex data visualization libraries (like D3 or Recharts) is still pending [Ref1].

- **Impact:** The immediate clinical utility of the Dashboard is currently limited to textual analysis until the interactive data visualization layer is fully integrated.

2.2.2 Mock API Layer

To facilitate simultaneous frontend development, we implemented a mock API layer within the frontend code. This layer successfully simulates the expected structure of responses from the Flask/Ollama backend [Ref2]

- Challenge: This reliance means the critical LLM communication pipeline has not yet been stress-tested with real-world latency. We need dedicated integration time to verify robust error handling, loading states, and the efficient passing of large context payloads to and from the backend server, a continuation of the Zero-Shot baseline work [Ref2].

2.2.3 Server-Side Health Check

The current frontend implements basic logic to confirm server readiness but requires refinement. The next phase must focus on implementing an authoritative, low-latency health check mechanism to provide the clinician with clear feedback if the local Ollama instance or Flask server is offline.

3 Results

The primary result of this development cycle was the successful elevation of the core application structure from low-fidelity wireframes to a high-fidelity, component-based prototype. This establishes the final visual language and structural framework for the production application, adhering strictly to the Mid-to-High Fidelity Design Criteria [Ref1].

3.1 Application Visual Language

The final design successfully implements the required color palette and styling for a professional clinical tool.

- Primary Color: All key interactive elements, navigation, and user-generated content are shades of violet-blue with RGB values *574bff* as seen in Figure 1.
- Aesthetics: All main content containers use extra-large rounded corners and deep shadows for a clean, modern aesthetic.

3.2 Structural Expansion: New Clinical Views

The core result is the addition of two vital clinical views, providing a complete structural framework for data review and documentation.

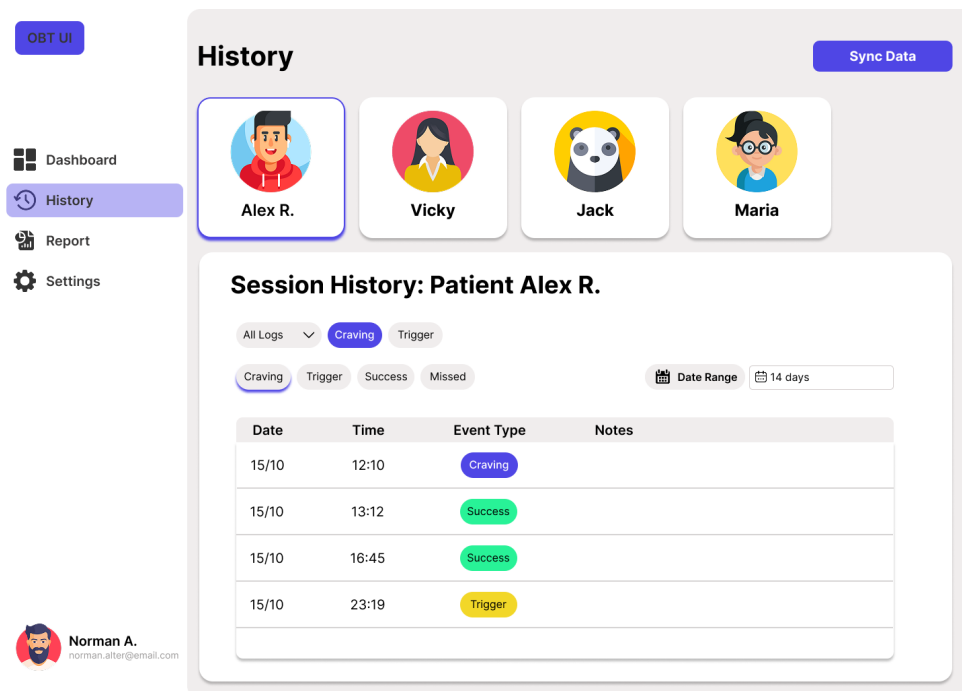
3.2.1 History View: Raw Log Auditing

The History View, Figure 2, provides the clinician with a chronological, auditable record of every event logged by the patient's One-Button Tracker (OBT). The design prioritizes immediate readability and filtering flexibility.

- Readability: Events are clearly categorized using colored badges (e.g., Green for 'Success', Indigo for 'Craving').
- Functionality: Controls for Event Type and Date Range filtering were integrated, allowing the clinician to narrow the focus for specific diagnostic periods.



Figur 1: Home Page



Figur 2: History

3.2.2 Report View: Printable Clinical Summary

The Report View, Figure 3, finalizes the data analysis into a concise, professional, and easily sharable document. This view is designed to be the final output for archival or patient-facing review.

- **Key Metrics:** The design prominently features high-level metrics (Total Logged Events, Success Rate, Peak Craving Window) using large, colorful cards.

- **Actionable Insights:** The LLM-generated analysis is positioned centrally under the header "FINAL CLINICAL INSIGHTS & RECOMMENDATIONS" and highlighted with the Yellow accent to ensure maximum clinical focus.
- **Documentation:** All supporting data, including the OBT Protocol Definitions and Patient Narrative Context, is included for complete archival documentation.

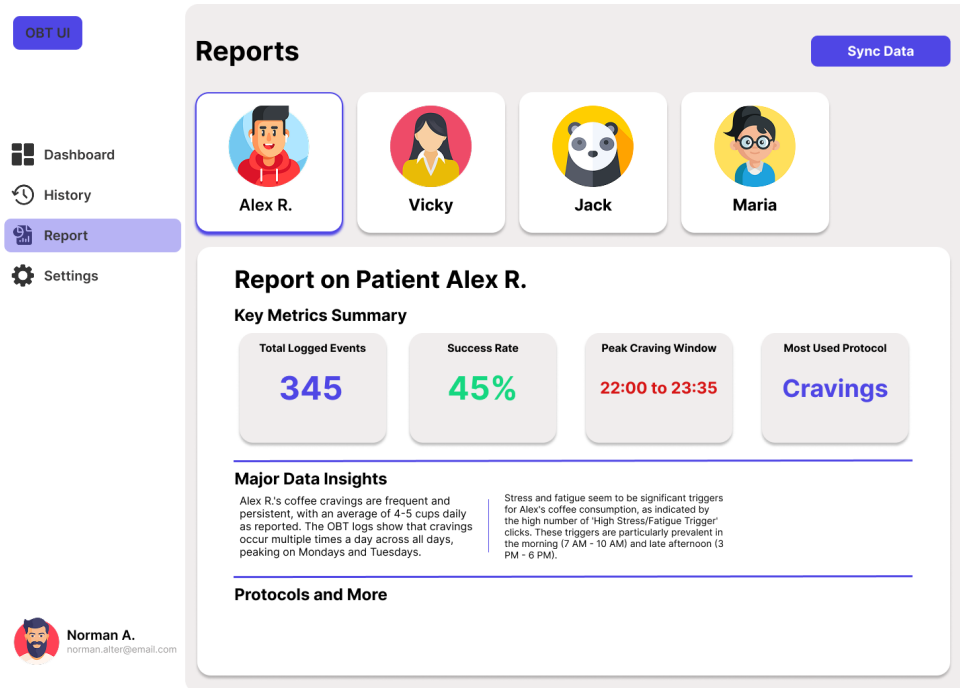


Figure 3: Reports

3.3 MVP Creation

Following up the results of the first sprint, one of the key achievements for this sprint was to construct a working Minimum Viable Product that would serve as the basis for the rest of development. This way, we built a full-stack prototype that follows the dashboard, 1, while wiring every surface to LLM-derived insights. The React/Vite front end reproduces the sidebar, patient card, trigger timeline, donut, narrative blocks, and chat console. Each component now hydrates from live JSON delivered by the Python service, so the visuals refresh when new simulations run or follow-up questions are asked. Engineered a Flask backend that (1) simulates 14 days of One-Button Tracker logs, (2) aggregates the counts into patient metrics (totals, success rates, weekly segments, timeline events), (3) queries the local llama3 model through Ollama for structured dashboard prose, and (4) exposes /start_analysis and /query endpoints consumed by the UI. A deterministic fallback preserves functionality if the LLM is unavailable.

Also worked on providing conversational analysis by persisting LLM context per session. After the initial call, clinicians can ask follow-up questions via the chat panel; the backend injects the latest metrics and raw log excerpts into each prompt so the model's responses stay grounded in the generated data. In the end, the implementation was validated locally: linted the React codebase (npm run lint), confirmed Ollama connectivity, installed Python dependencies, and ran the Flask server alongside the Vite dev server.

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Third Sprint

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18. februar 2026

1 Planning and Objectives

Sprint 3: Development, Data Restructuring and Testing Sessions

Duration: October 27, 2025 – November 12, 2025

Sprint Objective: The objective of this sprint is to expand the developed Minimum Viable Product (MVP) in order to incorporate the history and report view, with all Large Language Model (Large Language Model) functionality. This objective builds upon the dashboard implementation that was constructed during Sprint 2. As the present undertaking reaches its conclusion, it is imperative to initiate the next phase, which entails the refinement of the data structures to ensure a more precise alignment with the authentic OBT data. This process will be accompanied by rigorous experimentation to assess the boundaries of the LLM. In the context of the assessment process, this sprint will be of significant value, as its objective is to develop equitable and rational one-on-one evaluations and assignments for the web application under construction.

Key Activities & Deliverables:

- Development
 - **Activity:** Construct the additional views and features of the app that we build according to feedback got from the initial surveys and previous OBT works.
 - **Deliverable:** Two separate views/sections built upon the current structure that tackle a simplified data visualization and access and a quick and direct report on the data.
- Data Restructuring
 - **Activity:** Finalize the internal data model to ensure it accurately mirrors the data produced by a real OBT device. This involves formally defining the schema for OBT Protocol Definitions (click type, label, category) and changed how they are logged.
 - **Deliverable:** A documented Structured OBT Data Model that defines the format for protocols and logged events. This structure must be easily serializable (e.g., to CSV or JSON) for consistent input into the LLM prompt context.
- Test Sessions
 - **Activity:** Conduct one to one sessions for the main developed features. Have a user seat and use the platform, trying to achieve a set of designated tasks and ensure a comfortable and fair environment for feedback
 - **Deliverable:** A testing framework that is replicable and usable for future sprints and allows for further and deeper testings of the platforms features.

Rationale:

This sprint focuses on expanding the platform to two other sections and properly testing it. The goal is to have a replicable test protocol that allows the gathering of unbiased but informed feedback. Lastly it'll finally tackle and adapt the coffee addiction scenario to a data structure that properly mimics that of the real OBT data pushing the limits of the LLM's feature when it comes to translation and pattern recognition.

2 Development

The current development cycle concentrated on the expansion and exploration of features that are LLM-focused or based. This approach builds upon the structure established in the previous sprint, integrating our LLM with features that were identified as pivotal in the initial user surveys or prior OBT studies. Furthermore, a pivotal step that was incorporated into this sprint was the construction of an in-person testing framework. The purpose of this framework was to obtain detailed information and feedback on the features that were being developed.

2.1 Feature Development

The front-end feature set was extended to facilitate clinician cognition during evaluative walk-throughs. Within the application's source code, the Weekly Highlights module was reimaged as a gradient-based temporal rail. In this new iteration, each week is represented by a floating capsule that encodes the dominant behavioural category, an LLM-generated micro-insight, and the relative intensity of logged events through vertical scaling. Complementary style primitives were introduced in the *index.css* file in order to maintain consistent color semantics, stem geometry, and hover affordances. Concurrently, a high-density OBT heatmap was created and, although initially was incorporated into the dashboard, later on it was migrated from the congested dashboard column into a new "Patterns" workspace, accessible via the navigation schema. This dedicated page employs a compact, scroll-friendly rendering of daily activity, accompanied by interpretive copy that primes clinicians prior to a more thorough inspection. Collectively, these feature additions translate the previously developed analytics into narrative artefacts that can be exercised during think-aloud evaluations.

In the end of this sprint, the refined UI components provide discrete checkpoints for measuring comprehension. The dedicated Patterns page isolates the high-volume heatmap so moderated sessions can focus on pattern recognition without layout interference from other cards. Finally, the backend's enriched payloads and consistent LLM context make it possible to script reproducible scenarios: every usability test can replay the same synthetic study window, ensuring that observed behaviours are attributable to interface affordances rather than fluctuating data quality.

2.2 Data Restructure

With the introduction of real obt data, on this sprint build directly on the data pipeline established earlier. Scripts under `app/scripts` now synthesise 84 days of button presses that include realistic double-click gaps and long-press durations, persist the raw stream to *coffee_obt_press.csv*, and convert it into a clustered, threshold-aware JSON representation at a processed version *ofcoffee_obt_press.json*. The Flask backend (`server.py`) consumes this processed artefact, deriving aggregated metrics, weekly segments, and per-day summaries that feed the new timeline and Patterns view. Moreover, the preprocessing synopsis—cluster centroids, inferred thresholds, category counts—is injected both into the LLM prompt (via `compose_prompt_block` and `build_chat_system_prompt`) and the API payloads returned by `start_analysis`, ensuring that every visual element on the front-end can be traced back to the structured dataset produced in this restructured flow.

2.3 Testing Framework

With the analytical foundation and feature surfaces aligned, the project now supports a clearer testing protocol for upcoming user studies.

In the course of this sprint, the objective was to devise a viable testing session that could be utilised on subsequent occasions and thus establish a foundation for the subsequent in-person tests. The experimental phase of the study involved the administration of tests to students from a range of backgrounds in individual sessions. It is regrettable that, due to limitations of time and distance, the present sprint was not subjected to evaluation by psychologists or psychology students. The necessity for local sessions as opposed to online surveys is primarily due to the fact that all components of the web application, inclusive of the LLM, are executed locally. This decision was made on the basis that the deployment of the web application was considered to be beyond the scope of this thesis.

It is important to note that the sessions were conducted on a one-to-one basis. In these sessions, the participants were tasked with exploring the app, with the objective of assessing the quality of the features being developed. Prior to the commencement of the session, each participant was furnished solely with the requisite context to utilise and navigate the application, with no additional input provided to circumvent the introduction of potential biases. The context that was established was simple. The participants were guided through the process of analysing a hypothetical scenario, namely Alex R.'s (fictitious persona) coffee addiction, as if they were psychological analysts. The subjects were informed of the nature of the data that the OBT was collecting, as well as the protocols defined with Alex R. for the study of his disorder. Subsequent to the briefing, the participants were granted the opportunity to utilise the web application and evaluate its features, which was followed by a think-aloud session.

Task List:

- Task 1: Review the key and data insights
- Task 2: Ascertain which week presented the most significant challenges for Alex.
- Task 3: Ascertain the day that proved most challenging for Alex.
- Task 4: Investigate the list of cravings and logs.
- Task 5: Analyse the dashboard and reports metrics.
- Task 6: Interact with the LLM for more information.
- Task 7: Please provide suggestions for the next sessions with Alex.

3 Results

3.1 Data Restructure

The re-engineered pipeline established a single source of truth that improves traceability and interpretability. Because the backend now reads directly from *coffee-obt_press.json*, as seen in the example below taken directly from the *server.py* file, every figure shown in the dashboard, this meaning, weekly trigger volumes, coping success counts, peak-day narratives, can be traced back to a documented pre-processing step that details cluster means and threshold decisions. This transparency also propagates into the conversational system: the same structured summary primes the LLM, so textual insights reference empirically grounded numbers rather than ad hoc heuristics. Additionally, the JSON schema's inclusion of daily and weekly slices eliminated redundant computations inside the Flask service, decreasing response latency and simplifying future validation since analysts can compare API fields against stored aggregates without replaying the clustering logic.

```

1 bundle = simulate_patient_activity(PATIENT_ID)
2 bundle["prompt_block"] = compose_prompt_block(PATIENT_ID, bundle)
3
4 fallback = build_fallback_dashboard(bundle)
5 dashboard, llm_used = merge_dashboard_with_llm(fallback, llm_payload)
6
7 response_dashboard = {
8     "patientProfile": PATIENT_PROFILE,
9     "otherPatients": OTHER_PATIENTS,
10    "triggerTimeline": dashboard["trigger_timeline"],
11    "cycleProgress": dashboard["cycle_progress"],
12    "dataInsights": dashboard["data_insights"],
13    "timelineEvents": dashboard["timeline_events"],
14    "calendarHeatmap": dashboard["calendar_heatmap"],
15    "headlineMetrics": dashboard["headline_metrics"],
16    "studyPeriod": {
17        "start": bundle["start_date"],
18        "end": bundle["end_date"],
19        "days": len(bundle["daily_records"]),
20    },
21    "generatedAt": generated_at,
22    "dataSource": "llm" if llm_used else "synthetic",
23 }

```

Listing 1: API consumption of treated data

3.2 Feature Development

The interface refinements materially enhanced how clinicians can read and act upon the structured data. The new timeline rail presents seven weeks of behavioural shifts within a single viewport, seen in the end of 1, allowing users to contrast craving-dominant periods against coping-dominant ones without scrolling. Capsules expose the model-generated micro-insights inline, improving narrative continuity between quantitative spikes and qualitative interpretation.

Meanwhile, relocating the dense heatmap, 2 to the Patterns page relieved visual pressure on the primary dashboard and created a focused environment for pattern analysis, complete with guidance text describing how to interpret stacked bars and background intensity. Together, these results tighten the coupling between the analytics generated by the restructured pipeline and the cognitive tasks clinicians must perform, producing a demonstrably clearer surface for the planned user evaluations.

Despite the fact that the report and history views were principally developed towards the conclusion of Sprint 2, it was during Sprint 3 that these received greater attention. Consequently, they were reworked and better aligned with the proposed information to be displayed and tested during this sprint, achieving the views seen in 3 and 4.

3.3 Testing

Following the assembly of a handful of users for the purpose of evaluating the platform, the framework delineated in the development section was observed. Feedback was solicited for each task.

Task 1 – Review Key/Data Insights: Participants indicated that the interface initially felt confusing despite projecting a “medical” visual language; orientation required a short acclimation period. After familiarisation, however, they judged the presentation to be highly informative, noting that every tab corresponded clearly to its intended functionality and collectively conveyed the expected feature set.

Task 2 – Ascertain the Most Challenging Week: Users appreciated the richness of the History tab but observed that Weekly Highlights and the dashboard timeline rendered only two weeks of data, preventing a complete comparison across the study period. They proposed

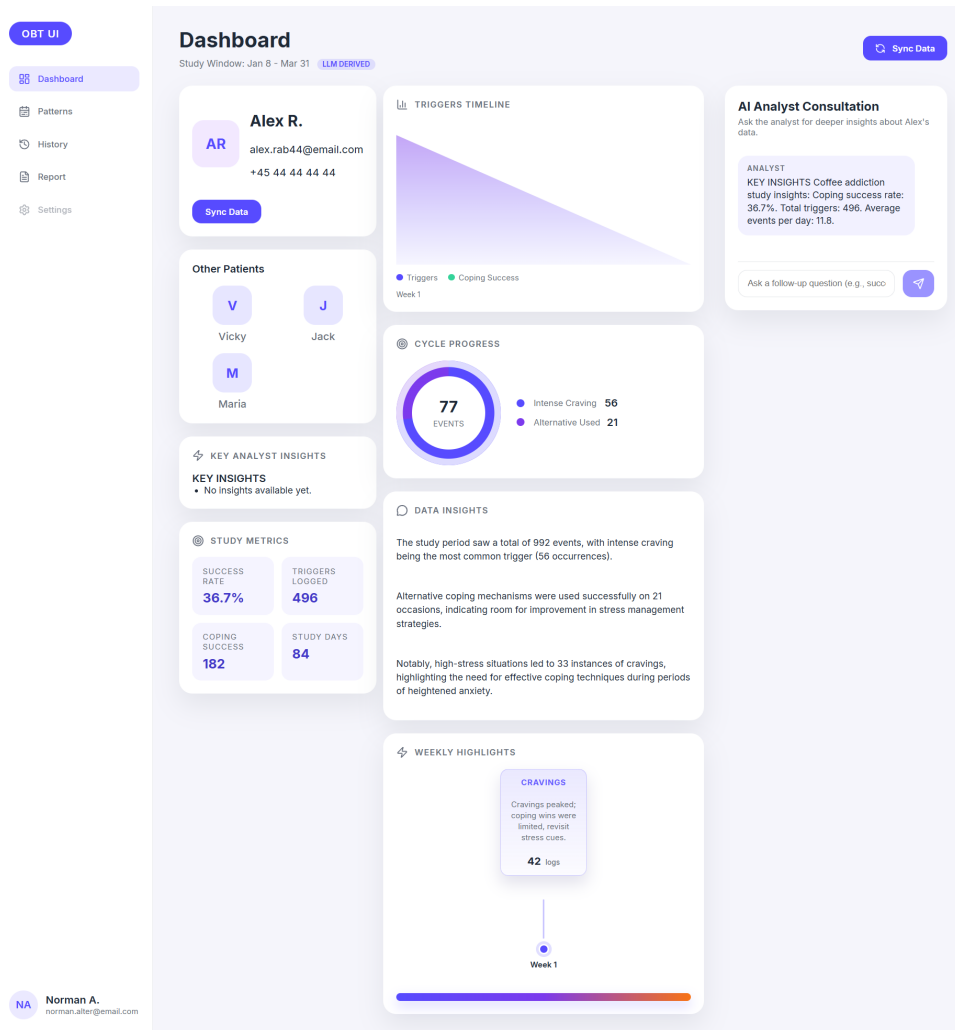


Figure 1: Dashboard

replacing the repetitive text-based craving insights with a comprehensive pie or iconographic representation that uses colour (and hover-triggered details) to encode weekly differences, thereby exposing the full dataset in a single view.

Task 3 – Ascertain the Most Challenging Day: Evaluators reported that the colour scale in the heat map provided minimal perceptual distance between “difficult” and “good” days; they recommended adopting a green-to-red palette with greater dynamic range. Combining the heat map with the chat interface was considered effective for clarifying the worst days, yet participants asked for explicit documentation of how the visualization defines a “bad” or “successful” day—whether the classification depends on aggregate protocol counts or solely on missed/negative events—which they flagged for review in the next sprint.

Task 4 – Investigate Cravings and Logs: Feedback was largely positive, but clinicians criticised the current date-range selector as non-informative and advocated truncating the logs into paginated views. They also requested sorting capabilities (e.g., newest, oldest, intensity-based) to make the long list manageable within clinical workflows.

Task 5 – Analyse Dashboard and Report Metrics: Although the key insights and headline metrics were deemed accurate and useful, users detected apparent inconsistencies that suggested some dashboard cards might not be loading the complete dataset. For the reporting workspace, they asked for the ability to drill into week-specific reports rather than only reviewing an aggregate. Additionally, they recommended omitting technical artefacts such as derived thresholds from clinician-facing narratives, since those details hinder interpretability.

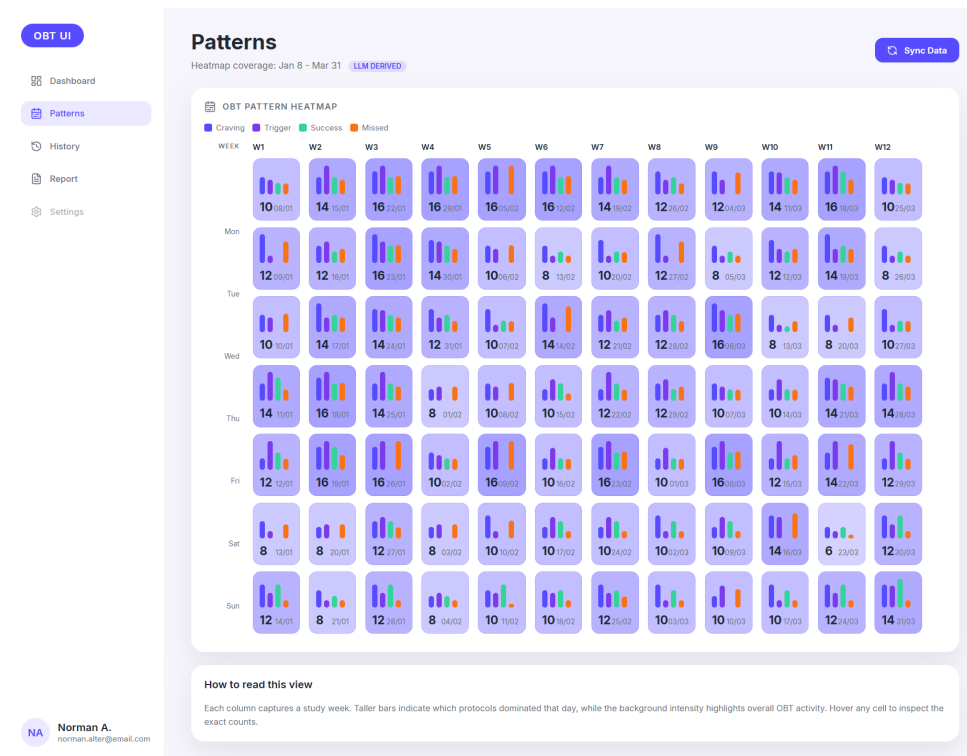


Figure 2: Daily Patterns Heatmap

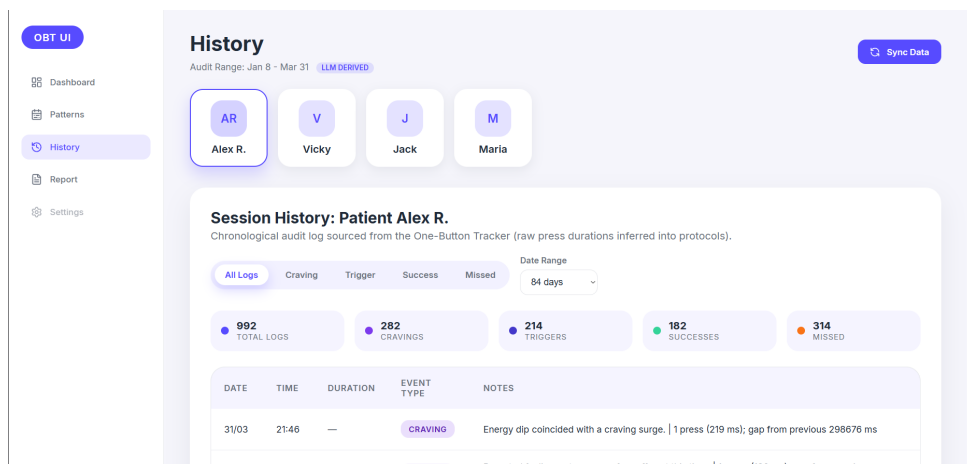
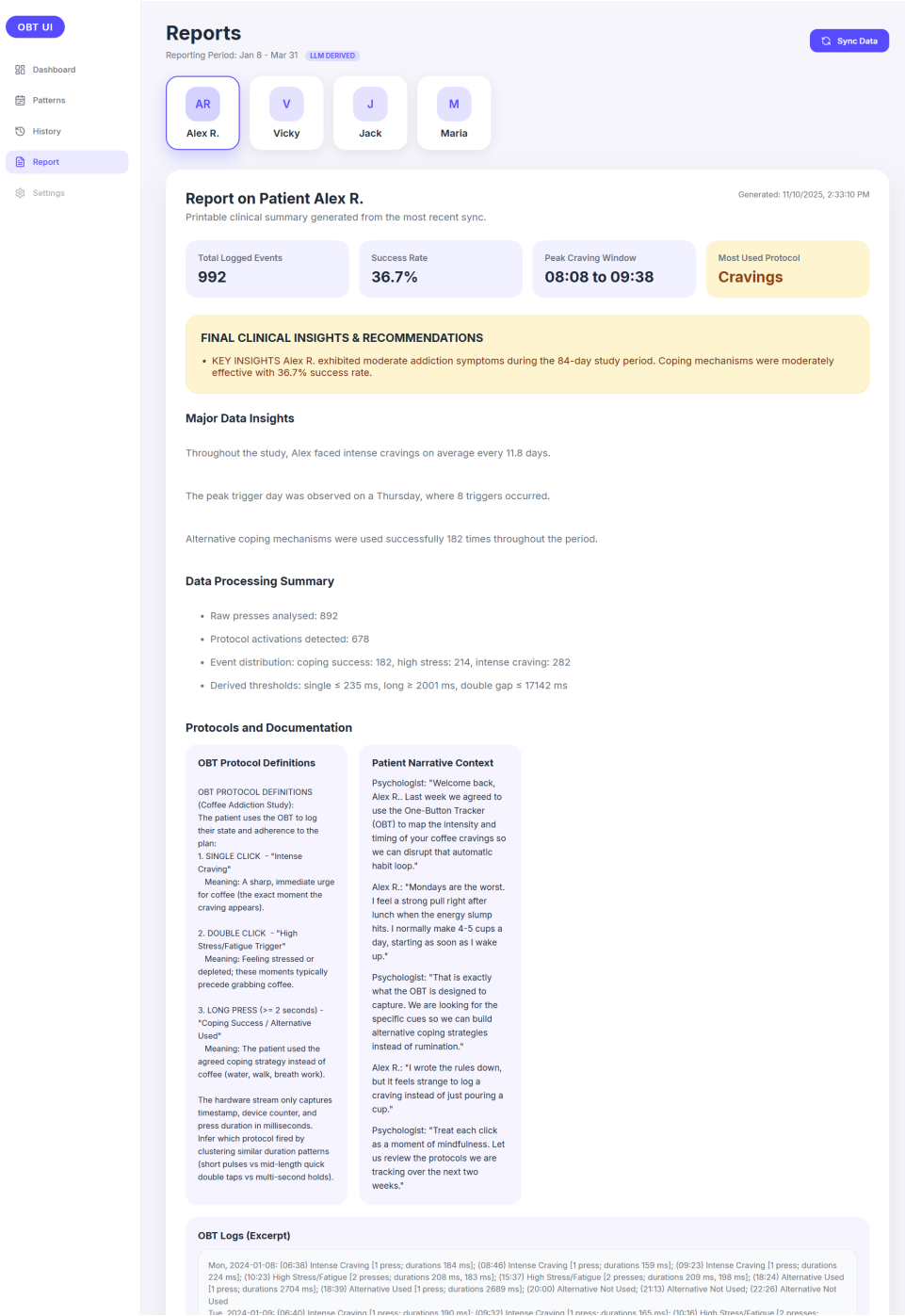


Figure 3: History

Task 6 – Interact with the LLM: The conversational tool delivered substantive answers, yet participants characterised the responses as overly long and conversational. They preferred concise, clinically worded guidance. The chat container should remain fixed in height with internal scrolling anchored to the newest message so the page does not grow excessively. Several clinicians also suggested tuning the tone to be more direct and time-efficient.

Task 7 – Provide Suggestions for Upcoming Sessions: Despite the improvement areas noted above, users found the system highly supportive for preparing future consultations with Alex. They also proposed a forward-looking enhancement: enabling psychologists to append notes or contextual annotations that would feed the visualisations and reports, thereby closing the loop between clinical reasoning and the platform’s analytics.



Fourth Sprint

Technical University of Denmark
Afonso Cunha - s233177

17. februar 2026

1 Planning and Objectives

Sprint 4: Development, Data Restructuring and Testing Sessions

Duration: November 13, 2025 – December 3, 2025

Sprint Objective: The objective of this sprint goes in two different directions: firstly, to expand the developed features from previous sprints; and secondly, to conclude any pending investigations and external sources. In the aftermath of receiving feedback during the testing phase of the previous sprint, the developed features have been the recipients of crucial feedback on how to improve and have been assisted in creating priorities to determine future work and areas for improvement. Nevertheless, a matter of concern that emerged during the test sessions pertained to the issue of data privacy and protection. The objective of this sprint is to investigate the constraints imposed by government legislation and psychological principles that may potentially influence the features under development. This investigation will be given particular consideration in the context of LLM utilisation. Moreover, the present sprint will address and conclude the preliminary user analysis initiated at the project's early stages. The objective is to investigate the potential correlation between users, constraints and biases that may influence the testing process and, consequently, the development direction.

Key Activities & Deliverables:

- Feature Development
 - **Activity:** Review of the feedback received from Sprint 3's test sessions is required, leading to the implementation of any changes deemed necessary or significant within the scope of this project and sprint.
 - **Deliverable:** The platform under discussion is intended to be more developed and concise, with the core features being clinically relevant for usage. The primary objective is to eliminate incoherencies and to rectify errors and mistakes that have been previously identified.
- Data Constraints Investigation
 - **Activity:** A thorough review of the investigation conducted on psychology and government laws and rules is necessary. This review should then inform the constraints on features and development.
 - **Deliverable:** A documented investigation into the potential constraints that should be taken into consideration with regard to the scope of the project development and the future programme of work.
- User Investigation

- **Activity:** It is imperative that the user data that has been collected is finalised, analysed and contextualised. This should be undertaken in conjunction with an examination of the cultural differences and local politics that may introduce a bias with regard to adherence to a LLM-based platform in the mental health context.
 - **Deliverable:** A thorough user data analysis connecting with the project’s literature and context.
- Test Sessions
 - **Activity:** Repeat the conducted one to one sessions for the main developed features, similarly to the previous sprint while still aiming to keep the core testing framework to later compare how the user feedback deviated.
 - **Deliverable:** Another set of user responses focused on the presented features and the usability of the platform and how viable it is.

Rationale:

As mentioned before, the objective of this sprint is twofold: firstly, to establish connections between ongoing investigations; and secondly, to finalise these investigations, with the aim of determining their potential influence on the project’s development, whilst concomitantly implementing the requisite changes. In light of the constructive criticism proffered in response to the preceding sprint, it is imperative to acknowledge the continued significance of the core technical development, which will continue to dominate a substantial proportion of the sprint. It is, however, equally crucial to recognise the imminent culmination of two significant investigation domains. These domains encompass the legal and ethical frameworks surrounding data privacy, as well as the psychological underpinnings and the intricacies of user background and their propensity towards adherence and interest in this project’s context. Ending this sprint, testing sessions will be conducted utilizing the framework constructed in the preceding sprint.

2 Development

2.1 Feature Development

During the developmental phase, the platform was iterated based on user feedback, with a particular emphasis placed on enhancing visibility and ensuring optimal workflow integration. The weekly timeline was restructured into a true multi-week strip with a blue to purple to orange gradient, restored stems/dots, and richer per-week cards that include ranges, dominant protocols, success rates, per-protocol counts, and clinician notes. Guardrails were implemented to ensure that LLM payloads cannot collapse coverage. The weekly highlights were incorporated into the timeline tiles while maintaining pie summaries. Furthermore, cycle progress and headline metrics are now ensured to always reflect the full dataset via stricter fallback validation in comparison to LLM merge validation. The history view has been enhanced with the introduction of filtering, flexible date ranges, intensity scoring, sortable/paged tables, and badges, with the objective of enhancing the navigability of extensive logs. Reports are now published on a weekly basis, accompanied by detailed analysis, insightful commentary, and clear definitions of protocols. This approach ensures that only relevant information is disseminated, effectively filtering out superfluous technical details. The patterns were reoriented to a weeks-on-rows configuration and weekdays-on-columns configuration, with the heatmap palette aligned to the timeline gradient so that low-volume/okay days were represented by cool colours and high-volume days by warm colours. This was supported by quality badges and explanatory copy. The LLM consultation was constrained to concise, clinical replies, with a fixed-height chat container and aligned input/footer, and clinician annotations are locally persisted and surfaced in timelines and reports. It is

evident that deterministic fallbacks and stricter merge logic contribute to enhanced resilience in scenarios where AI output is partial.

Nevertheless, following a review of the content on the platform, it was determined that the content was still deemed unsatisfactory. Consequently, minor changes were deemed necessary prior to commencing the respective test phase. This approach enabled the refinement of clinician-facing surfaces, with the objective of enhancing readability and ensuring visual alignment with the platform’s deep-blue identity. The dashboard chat was constrained within its card (no input overflow) while maintaining the fixed-height consultation flow. The weekly timeline retains the blue to purple to orange bar signals, but now extends the gradient rail to cover all tiles. The layout of the patterns underwent modification, with the week-row/day-column configuration retained, but with a shift in the colour scheme of the container to a chromatic scale of ice to slate to deep-platform-blue gradient. The bars, however, continued to utilise the warm/cool gradient for the purpose of per-protocol counts. The text was fixed to a dark neutral colour for the purpose of enhancing contrast, and the quality badges remained unchanged. The notes taken by clinicians have been relocated from the dashboard to the report’s ”Clinician Workspace”, where they can be edited and paired with hotspots and next-step summaries. In addition, the report has been modified to include structured insights rather than raw log excerpts. A realignment of history controls was undertaken, with the objective of achieving greater consistency. This realignment entailed the tightening of date-range styling. Palette will henceforth utilise the platform’s deep blue as the darkest heatmap stop to ensure chromatic consistency with the rest of the UI.

2.2 LLM Quality Assessment

As previously stated, one of the objectives of this sprint is to evaluate the extent to which it can provide precise data, given that a platform in a healthcare context with erroneous data can be highly dangerous. The objective of this approach was to establish a replicable pipeline for the purpose of evaluating the fidelity with which the LLM reflects the One-Button Tracker (OBT) study data. This methodology is designed to distinguish between deterministic ground truth and model behaviour, thereby facilitating the auditing of deviations.

In order to circumvent the occurrence of discrepancies resulting from duplicated logic, it was imperative to anchor all elements to the existing server pipeline. This process consequently reuses the same aggregates that are consumed by the UI, including the study window, totals, segments, timelines, daily records, event log, processing meta, and textual prompt block. The processed dataset is derived from a JSON file that has been created and used to simulate the coffee addiction situation. This is the same dataset that has been fed to the LLM for the dashboard.

For the deterministic ground truth (No LLM) a script was created that obtains the patient data and computes the necessary metrics to compare in the current test. These metrics are metrics also shown in the platform, all across the different features, which include:

- The Study Window Itself
 - Amount of days that were part of the study.
- General Measurements
 - such as total events, total triggers (cravings and stress), coping success, success rate (in percentile), intense craving, high stress.
- Peak Craving Window
- Peak Trigger Day
- Most-used Protocol

- Risky Weeks
 - Showing a small analysis on which weeks show the more logs

This data is saved unto a JSON file and printed for manual review and comparison, as seen in.

In order to test the LLM, a script was created for the purpose of collecting LLM answers and manually reviewing them. In order to maintain the integrity of the model, it transmits the identical complete prompt block that is utilised by the application, along with a ground-truth JSON comprising pivotal metrics. In view of the unsatisfactory outcomes that were being achieved, the option of the complete structured bundle was subsequently incorporated (comprising totals, trigger timeline, segments, daily records, weekly highlights, calendar heatmap, processing meta, logs and the event log, with an optional cap). In this section, the default temperature setting was another factor subjected to testing. In order to ensure reproducibility, this setting was set to 0. Given that the environment in question is one that is associated with a clinic, it was imperative to eradicate randomness. However, it should be noted that the script loads the same data bundle that is consumed by the app. In addition, if configured, it requests the LLM for dashboard JSON. In the event of this failing, the script falls back to synthetic metrics. For each question, the system prompts the user, followed by grounding messages. The system then records the ID, question, expected, LLM answer, model, and context snapshots.

Reproducibility controls include a fixed temperature of 0, prompt block inclusion by default, structured data on by default, and an adjustable event-log cap (`-max-events`, default 0 to include all). The harness does not modify source data; it merely reads from the processed dataset and writes outputs to `app/outputs/`. The objective of this study is to utilise the deterministic baseline in comparison with the model outputs in order to quantify adherence, hallucinations and numeric drift. The ratings CSV assists in the summarisation of model accuracy by question type and serves as a guide for prompt or model adjustments without altering the underlying data pipeline.

2.3 Testing Framework

Building on the testing framework constructed on sprint 3, an experiment was similarly conducted trying to assess the current quality and viability of the platform and respective output. This way, in order to test the new features, a new task was added to address the notes features and make sure, like last iteration, this experiment could guide the users through all corners of the platform. Thus, these were the conducted tasks for this experiment.

Task List:

- Task 1: Review the key and data insights
- Task 2: Ascertain which week presented the most significant challenges for Alex.
- Task 3: Ascertain the day that proved most challenging for Alex.
- Task 4: Investigate the list of cravings and logs.
- Task 5: Analyse the dashboard and reports metrics.
- Task 6: Interact with the LLM for more information.
- Task 7: Please provide suggestions for the next sessions with Alex.
- Task 8: Add you own note, in the role of psychologist, to week 5.

2.3.1 LLM’s Quality - Manual Assessment

In order to assess the fidelity of the language model in relation to the One-Button Tracker (OBT) study data, a two-track evaluation harness was engineered. This harness separates deterministic ground truth from model behaviour, while reusing the application’s existing data pipeline. Firstly, the baseline was anchored on the app’s own pre-processing by invoking the same data pipeline. This ensured that all downstream comparisons reference the same aggregates consumed by the UI. A deterministic checker was developed to compute canonical metrics directly from the processed OBT dataset. These include the study window, total events and triggers, coping successes, success rate, peak craving window, most-used protocol, peak trigger day, and leading trigger weeks. The script in question emits a JSON baseline and performs no mutations on source data.

Secondly, a grounded LLM assessment tool was constructed, with the model being queried using the same prompts and data as employed by the application, under conditions of strict control. The harness supplies the full prompt block, a ground-truth JSON of key metrics, and, by default, a structured bundle covering totals, timelines, segments, daily records, weekly highlights, a calendar heatmap, processing metadata, raw logs, text, and the event log. The sampling temperature was fixed at zero with the objective of maximising reproducibility. For each question, the tool initiates a system prompt and grounding messages, records the model’s reply alongside the expected value, and writes these exchanges.

Throughout the process, the principle of reproducibility was upheld by defaulting to temperature zero, incorporating prompt and structured data, and constraining alterations to read-only interactions with the processed dataset. The resulting outputs – deterministic baselines and LLM responses – enable a side-by-side comparison to quantify adherence, identify hallucinations and characterise numeric drift. Manual ratings offer a further means of summarising accuracy by question type, thereby providing a basis for prompt or model adjustments without modifying the core data pipeline.

3 Results

3.1 User Analysis

At the inception of the present study, an investigation was conducted into the behaviours and characteristics of users. This investigation was initiated through the utilisation of an online survey, which was disseminated to the relevant population. The objective of the survey was twofold: firstly, to gain a deeper understanding of the demographic characteristics of our testers and participants within the network; and secondly, to ascertain their level of interest in the platform and their willingness to participate in the project.

Initially, by investigating the users background, it became notable that the sample skew represents more young adults as the demographic composition of the sample shows that 58% of respondents fall within the 25-34 age bracket, 33% within the 18-24 bracket, and a mere 7% are over 35, 1. The gender split is balanced, with 53% of respondents identifying as female and 47% as male. In order to fully understand possible cultural constraints, localization became quite important to learn more about. Thus, as seen in Figure 2, the data indicates that approximately 75% of the subjects self-identify as Portuguese, while 22% identify as Danish and a smaller percentage as Dutch or Slovak. Setúbal alone accounts for approximately 51% of all regions, thereby reflecting the local network.

The occupations analysed in this study are distributed across a variety of sectors, with a notable concentration in the fields of technology and engineering (approximately 25%), healthcare (approximately 16%), students (approximately 15%), and business and operational roles (approximately 13%). The remaining occupations are distributed more uniformly across creative and service-oriented sectors. Additionally, 5 respondents (9%) are currently engaged in training

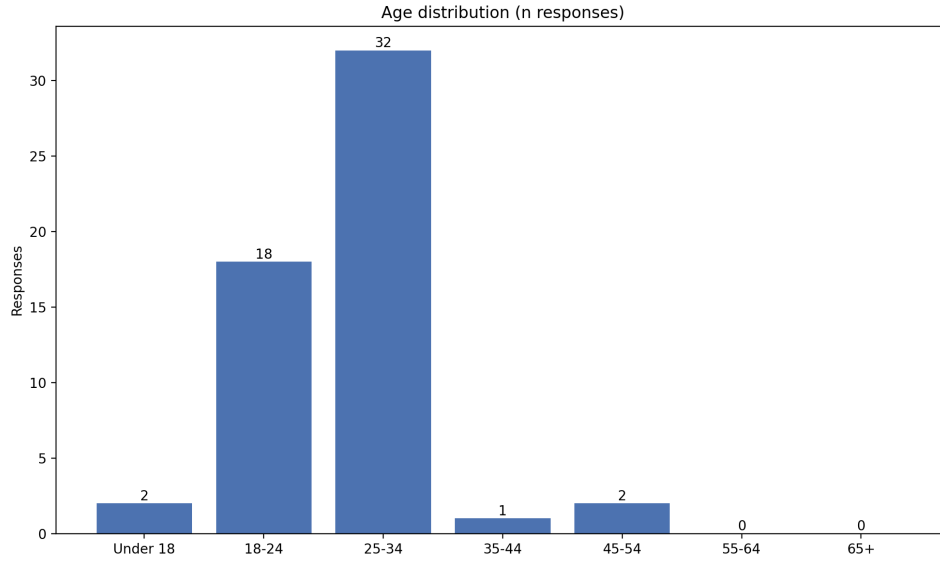


Figure 1: Initial User Survey - Age Ranges



Figure 2: Initial User Survey - Heatmap

as psychologists or working as such. Furthermore, the data indicates that 73% of individuals have already sought assistance from a psychologist or therapist, with 64% of these individuals attributing their decision to seek help primarily to their life experiences. The respondents selected an average of 3.9 life challenges, with the most frequently selected categories being personal growth (69%), anxiety (65%), stress management (65%), relationship issues (56%), depression (51%), and trauma (45%). The present study explores the relationship between technology, behaviour and concerns.

Although the background is of extreme importance, the integration of a technology into an unfamiliar environment invariably entails challenges. Consequently, the survey's objective was not only to ascertain the key insights pertinent to a platform of this nature, but also to assess the challenges associated with the integration process. Nevertheless, this group has been found to demonstrate a high level of technological comfort. The mean comfort rating was 4.45, with a median rating of 5, indicating a high level of satisfaction. The market share of smartphones is distributed as follows: 56% of smartphones are iPhones, while 44% are Android devices. The data-tracking habits of respondents vary, with 16% tracking daily, 18% tracking weekly, 25% tracking monthly, 16% rarely tracking, and 24% never tracking. This indicates a significant segment of respondents who express enthusiasm for the concept but have not yet adopted it

as a routine. The most prevalent adoption impediments, Figure 3, pertain to value proof, with effectiveness (67%), accuracy (58%), cost (56%), privacy (51%), and both security and ease of use (44%) being the primary concerns. Concerns regarding long-tail free-text analysis include issues of intrusiveness, temporal expense, and the potential for over-analysis. The importance of simplicity is emphasised by the mean score of 4.24/5, with 85% of respondents awarding a rating of 4–5. Despite a high level of affinity for high-tech solutions, the importance of data-driven insights is only moderate (average score of 3.55 out of 5; 53% of respondents scored 4–5), leading to believe that concrete therapeutic benefits must be articulated in order to convince the sceptics.

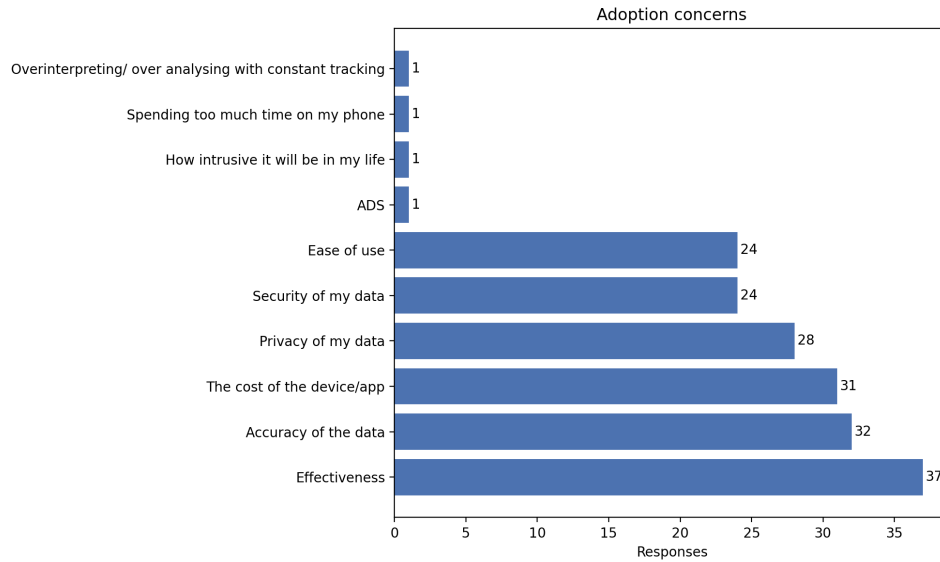


Figure 3: Initial User Survey - Adoption Concerns

Following this, it was evident that respondents are willing to share rich data, provided that trust is established. The data indicates that more than 89% of respondents reported being comfortable sharing information regarding stress, sleep, mood, and physical activity. Subsequent categories, including work-life balance (84%), diet (78%), and social interactions (76%), were also deemed to be acceptable for disclosure. The subjects selected approximately six data types on average (range 1–8). The insights that are desired by the subjects of this study, as seen in Figure 4 can be categorised as follows: the identification of stress triggers (75%), the understanding of mood swings (64%), the provision of an easy data packet for therapists (51%), the tracking of progress (49%), and the linking of mood with sleep (35%). The mean number of insight types desired by respondents was found to be approximately three.

Furthermore, insight delivery preferences are found to favor self-serve tools. The results of the survey indicate that the majority of respondents, 58%, expressed a desire for a mobile app dashboard. In addition, 22% of respondents indicated a preference for a report that could be shared, while 11% favoured direct integration with EHRs. A smaller proportion of respondents, approximately 1%, expressed a preference for emails or conversational user interfaces. The provision of support for both self-tracking and clinician-ready exports would be sufficient to satisfy the majority of use cases.

Even though, not exactly in the scope of this project, the survey also aimed to assess commercial viability. It was shown that the propensity to pay for a product or service is demonstrably influenced by its price point; a 51% cap at €1-€20, 27% would only utilise it if it were complimentary, 16% would pay €21-€50, and a mere 6% would pay in excess of €50. It is reasonable to expect the emergence of freemium/low-cost tiers or bundling with therapy services. A total of 53% of the participants volunteered to be testers, while 28% provided their contact information,

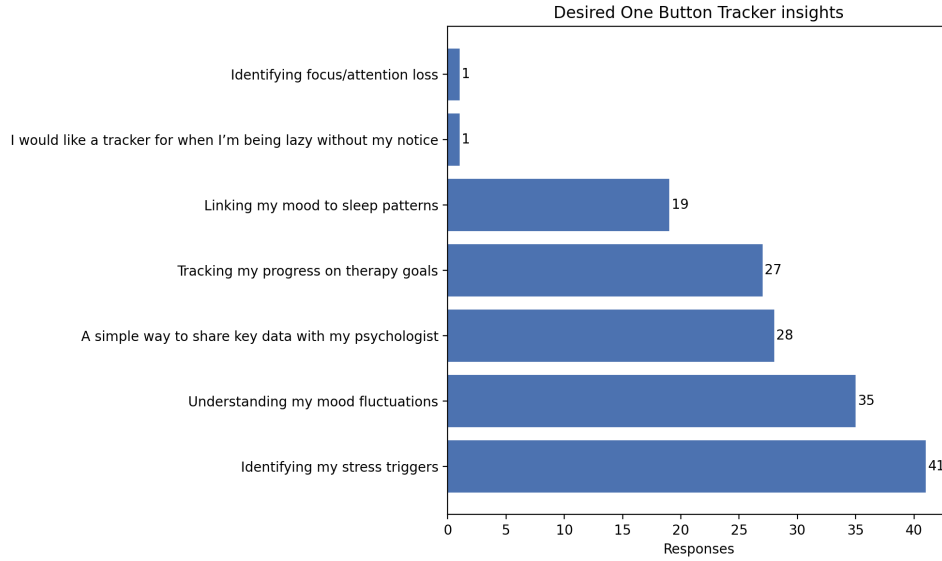


Figure 4: Initial User Survey - Desired Insights

indicating a substantial interest in the study’s findings if prompt follow-up is conducted.

The objective of the survey was to determine the boundaries of the project’s scope. This way, it became imperative to prioritise demonstrable accuracy and effectiveness, while concurrently disseminating insights that are substantiated by empirical evidence. Secondly, the design process should incorporate privacy-visible defaults and an onboarding/reporting flow that is extremely simple, in order to address the top concerns expressed. Thirdly, it is important to emphasise the manner in which data facilitates the connection between self-awareness and therapy goals, thereby assisting mid-scale (score 3) users in their progression towards adoption. Fourthly, while not in scope for this project but for future work, one should consider conducting pricing experiments that commence at no or low cost, with the objective of subsequently offering enhanced insights or clinician integrations. This is due to the fact that merely 20% of respondents exceed €20 in terms of their willingness to pay. In the process of recruiting testers, it would be recommended that the Setúbal/Portugal networks be utilised as a primary source. However, it would also be advisable to expand the outreach to diversify both geography and age, thereby ensuring an equitable representation of individuals from diverse backgrounds in subsequent iterations. This approach is recommended in order to mitigate potential regional bias that may be present in future iterations.

3.2 Privacy and Data Constraints

In contemplating the prospective implementation of the platform currently under development, it is imperative to meticulously examine the potential constraints on data privacy, particularly in the context of innovative data creation and translation in the domain of mental health. This approach, when employed to utilise LLMs in the interpretation of data for the purpose of psychological support within the European Union, reveals a more intricate web of legal and privacy constraints. A preliminary investigation was thus conducted into the potential ramifications for privacy, processing and ethical considerations, with particular emphasis on GDPR and AI-specific regulations. Subsequent to the administration of the preliminary user surveys and the conduction of the experimental sessions during the developmental process of this project, the investigation was oriented more towards the Danish, Portuguese and EU-wide regulations, given that this was the foundation for the project and the majority of the test subjects were recruited from these regions.

3.2.1 EU-Wide Basis: The General Data Protection Regulation (GDPR)

The GDPR is the primary legal framework governing personal data processing across the EU. It applies to any entity collecting, storing, or analyzing health-related data, which is classified as “special category data.” Processing such data requires a lawful basis (for example explicit consent or legal/clinical necessity), strict documentation, and discrete security measures (Union, 2021).

3.2.2 Denmark: Local Implementation

In Denmark, GDPR is complemented by the Danish Data Protection Act (Piper, 2024) which adds country-specific provisions. For instance, the Danish regime covers data of deceased persons for up to ten years and specifies requirements for data controllers (health professionals, researchers) to maintain strict records of processing activities and engage in frequent self-auditing. Where “special categories” of data—especially mental health records—are used beyond direct care (like in research or analytics), danish law often requires preapproval by the data protection authority and always relies on patient consent or clear anonymization.

3.2.3 Portugal: National Variations

Portugal’s implementation of GDPR also recognizes health data as highly sensitive, with additional local rules (LLP, 2020). Firstly the obligation to separate storage of health/genetic data from other personal data is one of the key factors as well for mandatory audit trails on health data access, and patient notification upon any access or transfer. Lastly, there are limitations on the reuse of health data for research unless vital for public health or with explicit written consent.

3.2.4 Special Requirements for Psychologists and Mental Health Apps

In both Denmark and Portugal, psychologists are bound by stringent confidentiality requirements that frequently exceed the standards stipulated by general privacy legislation. Disclosure without informed consent, except where required for safety or by law, is prohibited.

The utilisation of LLMs for the processing, summarisation, or analysis of mental health data is categorised as “high-risk” in accordance with the GDPR and the EU’s evolving AI Act (Compact, 2024). A Data Protection Impact Assessment (DPIA) is required, particularly in cases involving profiling or automatic decision-making. Systems must be designed to minimise data to the essentials, ensure transparent processing, and restrict foreign data transfers unless special safeguards are in place. Furthermore, the practice must also guarantee that if the LLM vendor is external or cloud-based, data remains inside the EEA or has equivalent protection.

In matters pertaining to consent, it is imperative that it be explicit in all instances when the processing of “special category” health data is concerned. Furthermore, it is essential that users/patients be apprised of the processing, storage, and dissemination of their data, and that they be made aware of the parties responsible for such processing. It is imperative to emphasise the significance of autonomy and the freedom to make one’s own choices. This entails ensuring that users/patients are accorded the prerogative to withdraw. In instances where consent is not forthcoming, the processing of health data is permitted exclusively under established exceptions, such as treatment or public health.

3.2.5 Record Keeping and Accountability

Since both countries follow EU’s GDPR, both require comprehensive, written logs of data flows, processing justifications, and security arrangements. Not only this but data must be minimized, this meaning that only data relevant and necessary for psychological care or research may be

retained. Furthermore, accountability is extremely essential in all levels, as such organizations must appoint a Data Protection Officer (DPO) if processing is regular or systematic.

3.2.6 New Risks and Safeguards with LLMs

Considering the platform’s features and their dependency on a LLM, this investigation also carried out to average new risks and safeguards, specially considering the epidemic of new LLMs on the market. Thus, following the constraints the other LLMs also deal with, AI systems processing health data must be transparent about how information is summarized, the extent of data exposure, and the source of all outputs. Not only this but profiling and inference are permitted only under very strict transparency and fairness rules. To end, there are growing audit requirements such as audit logs, technical and organizational safeguards, and frequent reviews for bias or security flaws.

3.3 Feature Development

The updated platform now showcases a richer, clinician-ready experience. The weekly timeline is a true multi-week strip with a blue-to-purple-to-orange gradient, stems/dots, and fully populated cards showing ranges, dominant protocols, success rates, per-protocol counts, and linked clinician notes, so weekly comparisons are immediate, as seen in Figure 5. In order to ascertain consistency weekly highlights and cycle progress now consistently cover the full dataset. This is protected by stricter LLM merge checks.

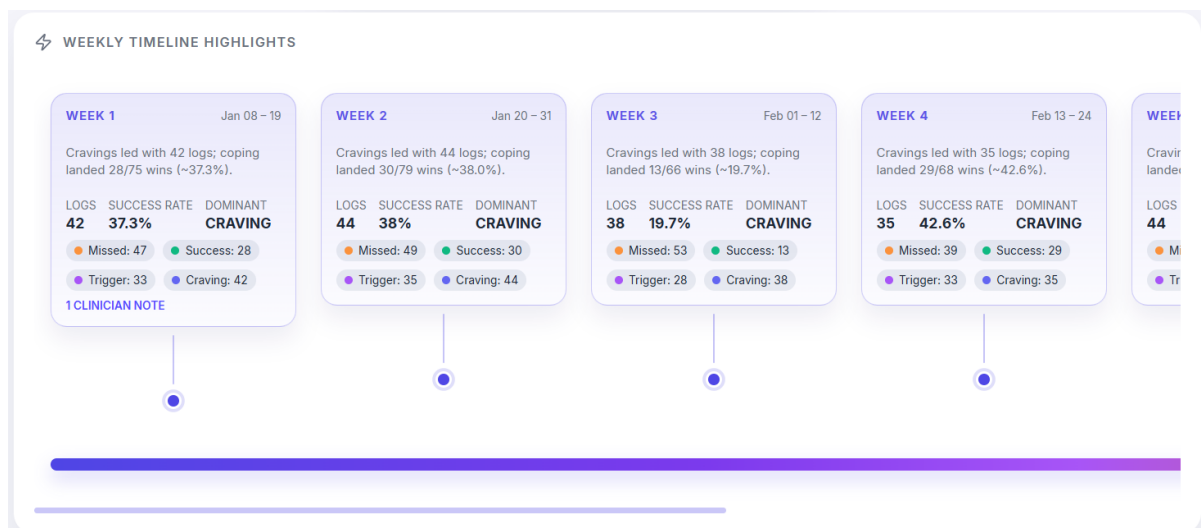
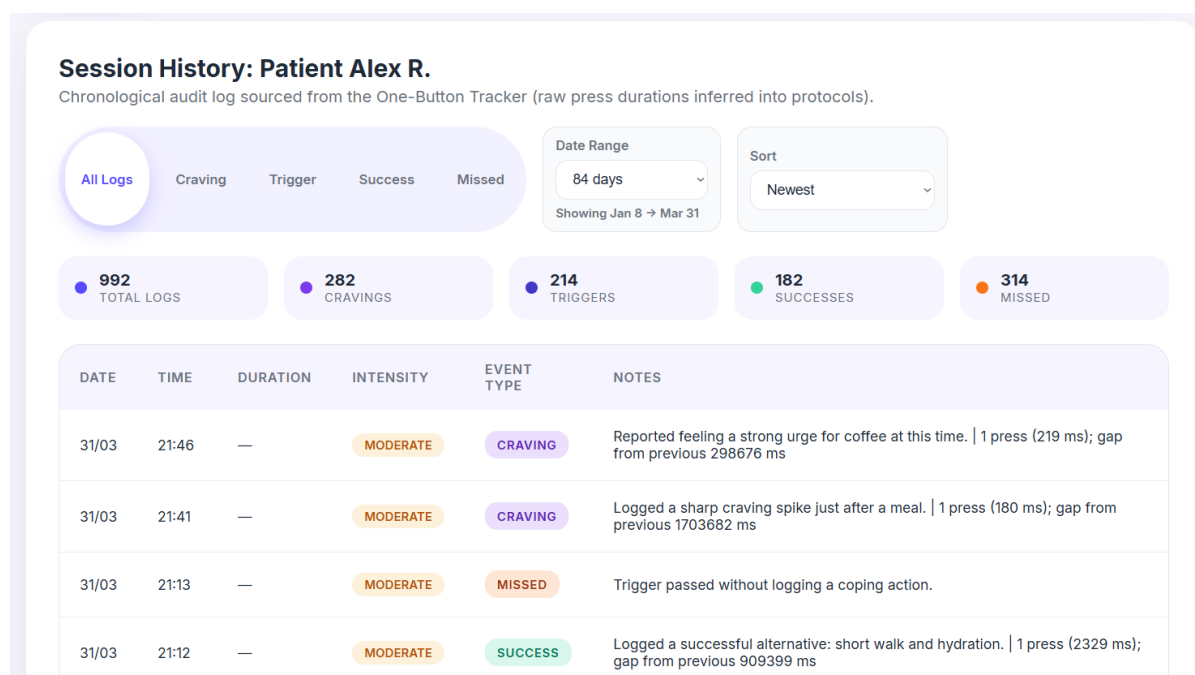


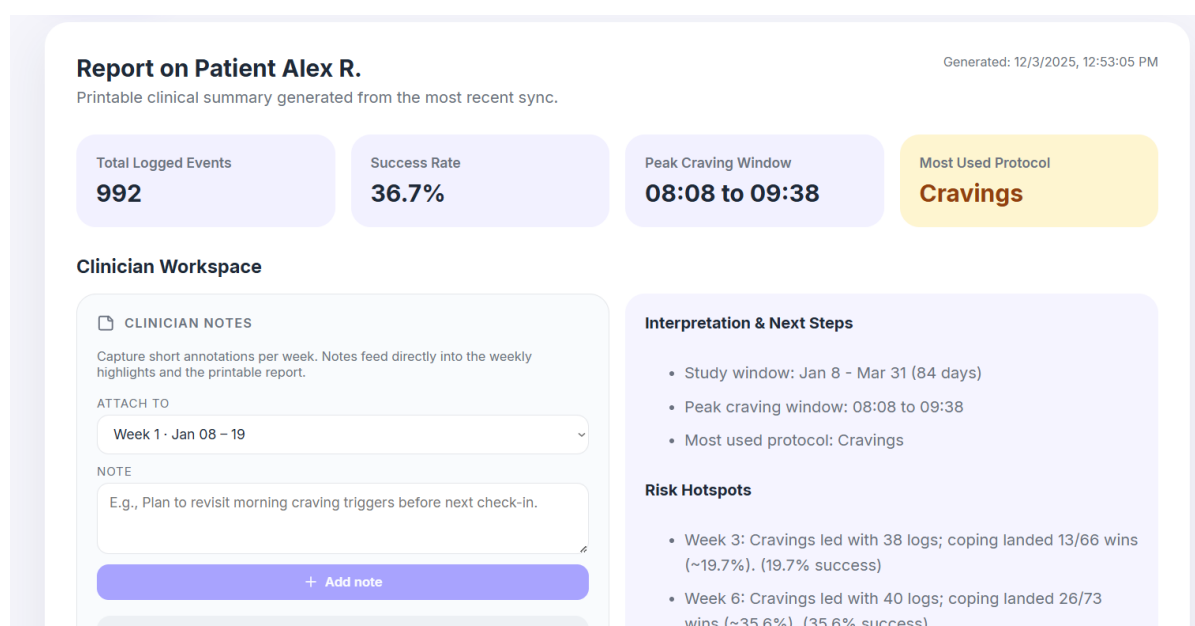
Figure 5: Sprint 4 - Weekly Timeline Highlights

The history view, Figure 6, is capable of supporting filtering, custom ranges, intensity sorting, and pagination, thereby transforming extensive audit logs into a navigable, prioritized list.

Reports, as seen in Figure 7, present weekly drill-down tabs, embedded clinician annotations, and major insights without technical clutter, while the patterns heatmap, Figure 8, mirrors the timeline palette – weeks on rows, weekdays on columns – making high-volume days “heat up” naturally. The LLM consultation is anchored to concise, clinical replies in a fixed-height chat with aligned input controls, and clinician notes persist and surface across timeline and reports. Collectively, these modifications exemplify a cohesive, feedback-driven evolution of the dashboard, history, patterns, reports, and consultation workflows.



Figur 6: Sprint 4 - History

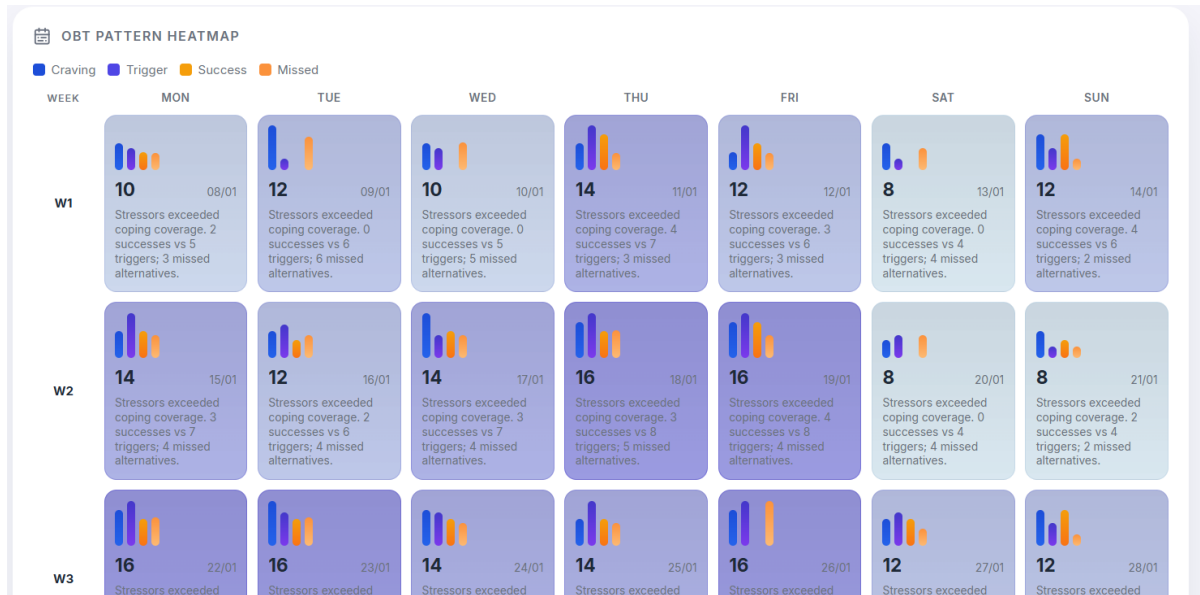


Figur 7: Sprint 4 - Report

3.4 Tests

Following the experiments conducted with a handful of users, once again the iteration concluded with thorough feedback on the platform's features.

When over viewing the dashboard and related tasks, users commented that the timeline bar, indicated with the blue to purple to orange bar indicated a level of intensity as time went by which was misleading and not relevant to the data shown in the boxes above, thus being suggested for this bar to be removed or reworked for more friendly or clinically neutral color, nonetheless, it was uniformly considered is an effective visual representation of the data handling and understanding process even making it easier to assess which weeks are harder for our fake



Figur 8: Sprint 4 - Patterns

scenario patient. Another issue occurred when interacting with the chat feature of the LLM was that whenever they would send a message not only they couldn't stop the thinking process, which is a common feature in many LLM solutions in the market, but also when the user would send the message and the LLM would respond the page's view would be pushed down, which was noted as a possible nuisance. Even with some errors, users noted that the combination of features and information presented on the dashboard made it very easy to better understand Alex's situation and respective progress throughout the 7 week relevant data.

In the patterns heatmap, users approved the chosen color palette for the mentioned feature, nevertheless still needing some tweaks mainly on the colors of the bar plots since the success and missed colors were often mistaken due to their similarity and on the most intense days the text should was too difficult to read. Furthermore, some users also considered that there is a high degree of similarity between the color of the trigger and that of the craving and, similarly, suggested that different colors would be used to facilitate differentiation. There some issues with the understanding of each protocol, and each bar plot in it, where some users needed help distinguishing between protocols and respective representations and their definitions.

Regarding the history view, one of the main concerns raised was the importance of the intensity data information and their notes. Since it's not part of the data neither it is part of the measurement capabilities of the OBT, these data columns were introduced on the previous sprint. Nevertheless, since it's not natural to the OBT data structure nor natural to the synthetic data created from OBT examples, for the intensity for example it's just showing as moderate to all logs seen thus making filtering through this data column, rather useless. Considering the non nature of the column, it was quickly noted that it wouldn't make sense since it's not part of the OBT measurements neither is viable for the LLM to take accountability for said information. Regarding the notes, user commented the need and purpose of having this information since it wasn't integrated on the information given from the OBT but rather calculated during the pre process phase.

Lastly, given the restructure nature of the changes done during this iteration the report section of the platform got the most feedback. One of the main comments was the repetition of information present on this view. This as to do with the LLM response to our data and needs for major insights which leads to repeating statistics on the data and showing them all across the view where it should be more diverse or else removed. One of possible solution rethinking prompt system and creating more specialized prompts to avoid redundancy on information, following

the approach of "prompt engineering", thus simplifying as well the clinical report. Statistics organization was also a hot topic during the experiment where in some cases users noted the need to change or reorganize how the platform portrays information. This being said, sections such as the risk hotspots were highlighted for review of the respective information mainly for the sorting of statistics, being suggested to sort by week instead of by failure rate. Regarding the new clinical notes feature, even though it was received with positive feedback, upon failure testing, it was seen that when adding multiple notes, they would start to line up below one general tab whereas it should be shown under the specific week or period that the note was created on. Furthermore, it was also noted that this feature lacked an edit option to complement the functional foundation of this section. Lastly, some areas required visual tweaks, such as the clinical summary, the interpretation and next steps areas and the risk hotspots, where the text was seen with visually odd space and alignments.

Overall, besides the feedback mentioned before, the platform received positive return when it comes to it's usage and information portrayed, specially considered the tasks and the output the LLM was giving, thus making important to focus on these small changes on the next iteration to achieve and optimize the best LLM driven platform.

3.4.1 LLM's Quality - Manual Assessment

The LLM was queried on six headline questions, and the answers thus obtained were then compared to the deterministic baseline derived from a created python script to be assumed as ground truth. The ground truth values were found as:

- Total Number of Events: 992
- Total Number of Triggers: 496
- Total Number of Successful Coping Mechanisms: 182
- Success Rate: 36.7%
- Peak Craving Window: 08:08 to 09:38
- Peak Trigger Day: Thursday 18th January 2024 with eight triggers
- Most-Used Protocol Key: Alternative Not Used (314)
- Intense Cravings Count: 282
- High Stress Count: 214

Accuracy: The LLM failed to identify every item that was checked. The study erroneously reported total events as the raw press count (892), exaggerated the study's duration (90 vs. 84 days), and omitted the peak window, classifying it as 'not available' despite its availability within the context. The most-used protocol was identified as "intense craving" (282), which was utilised in preference to the top category, designated as "alternative not used" (314). With regard to the totals, the results yielded partial and contradictory counts, with the issue of coping successes remaining unresolved. With regard to the success rate, it recomputed from a subset (roughly 41%) instead of using the provided 36.7%.

Hallucination and drift patterns: Responses exhibited a combination of real numbers, for example 282 against 214, with fabricated roll-ups, for example weekly sums and 90-day span, and there was an explicit disregard of metrics that were supplied such as the peak window and the total events count. In several cases, it substituted raw processing counts (892 presses) for processed event totals (992) and invented ungrounded percentages.

Consistency: Despite the implementation of deterministic settings (temperature 0) and the injection of ground-truth JSON plus structured data, the model failed to reliably replicate the

provided values. Instead, it endeavored to infer or recompute, resulting in systematic numeric drift and intermittent "not available" deflections where data was present.

In summary, under the tested prompt and grounding configuration, the LLM did not replicate the canonical metrics and exhibited repeated numeric errors, indicating low fidelity to the supplied study data for these headline questions. Given the importance for reliability and precision in the medical context, the current state of the LLM shows to be unfavorable. This way, during the sprint the focus will be more on the importance of *prompt engineering* and how can the LLM's output be optimized by optimizing the input as well.

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Fifth Sprint

Technical University of Denmark
Afonso Cunha - s233177

17. februar 2026

1 Planning and Objectives

Sprint 5: Development, Prompt Engineering and Psychologist Interview

Duration: December 11, 2025 – January 4, 2026

Sprint Objective: In the course of this sprint, the objective is to rework the platform in order to enhance its user-friendliness and clinical viability, in accordance with the feedback received during Sprint 4. It is evident that a significant proportion of the sprint will be dedicated to feature development. Nonetheless, during this time period, the investigation and integration of a more precise and concise prompt system to optimise LLM output will be conducted. The objective of the new system is twofold: firstly, to enhance the quality of the model's output, and secondly, to improve upon the substandard output that has been demonstrated in the platform and observed in the manual quality assessment of Sprint 4. In order to enhance the model and conduct further investigation, this sprint will also involve a comprehensive analysis of Ollama's Llama3 model. This model is utilised as the baseline and serves as the primary subject for testing. The objective of this analysis is to gain a deeper understanding of the model's structure and to identify viable alternatives. Finally, the experiment will be conducted with a psychologist in the form of an open interview. This will be done in order to analyse the participant's behaviour and adherence to the platform. This analysis will be conducted from a clinical perspective. The results of the experiment will show whether the platform is viable as a gateway to OBT data.

Key Activities & Deliverables:

- Feature Development
 - **Activity:** A review of the feedback received from Sprint 4's test sessions is required. This will inform the implementation of any changes deemed necessary or significant within the scope of this project and sprint.
 - **Deliverable:** In a manner consistent with the preceding sprint, the objective is for the platform to be more developed and concise, with the core features being clinically relevant for usage. The primary objective is to eliminate incoherencies and to rectify errors and mistakes that have been previously identified, thus bringing the solution closer to completion.
- Investigation and Implementation of Prompt Engineering
 - **Activity:** Undertake a thorough investigation into the notion of prompt engineering and its potential to enhance the efficacy of a LLM. Subsequent to this, an implementation derived from the investigation conducted on this topic should be conducted.
 - **Deliverable:** A comprehensive investigation into the subject has been conducted, with the aim of determining its significance in relation to the realisation of the optimal solution. Subsequent to the investigation, the implementation and integration of the aforementioned idea with the platform should be conducted, thereby achieving a new prompt system with which to interact and extract information from the platform.

- Deep Dive into *LLama3*

- **Activity:** Now that the majority of the features have been implemented, the focus should shift to refining and enhancing these features through minor adjustments and subtle changes. To this end, a comprehensive analysis of the Llama3 model and its underlying structure is imperative to ensure a thorough comprehension of its functionality and its aptitude for the designated task.
- **Deliverable:** This study will address the merits and drawbacks of utilising llama3 as the baseline model for this particular platform.

- Psychologist Interview

- **Activity:** An interview to be conducted with a working professional psychologist, with the aim of integrating the testing framework into the interview.
- **Deliverable:** A comprehensive interview with a practising psychologist has been conducted and documented, in order to facilitate an in-depth understanding of the integration process of the platform into a clinical context, as well as its viability and future directions.

Rationale: The objective of Sprint 5 is to consolidate and refine the platform in response to the limitations and feedback that were identified in Sprint 4. At this stage of the project, it is imperative to enhance the usability, coherence, and clinical relevance of the system to ensure its progression from an exploratory prototype to a stable and meaningful solution. Consequently, the reworking of existing features and the correction of previously identified issues is prioritised over the introduction of major new functionality.

A fundamental motivation for this sprint is the necessity to enhance the quality and consistency of LLM-generated output. The shortcomings observed in Sprint 4 highlight the importance of prompt engineering as a critical mechanism for controlling model behaviour and enhancing response relevance. The investigation and implementation of a more structured and concise prompt system is expected to result in a significant improvement in output quality and better alignment of the platform with clinical requirements.

The necessity for an in-depth analysis of the Llama3 model is derived from its function as the foundational model for the platform. It is imperative to comprehend the intricacies of the architecture, in order to ascertain its aptitude for the designated task. This comprehension facilitates a judicious evaluation, which may necessitate the contemplation of alternative models if deemed necessary.

The psychologist interview is of particular significance in this regard, as it provides essential clinical grounding. Incorporating professional feedback through an open interview, this sprint assesses behavioral adherence, clinical integration, and overall viability, thereby determining whether the platform can function as a credible gateway to OBT data.

2 Development

2.1 Feature Development

The Patterns heatmap has been modified to function as an intensity map. The cells are now normalized against the minimum and maximum daily totals, and utilise solid light-to-deep blue fills with auto-contrast text. Additionally, the vivid per-protocol bars remain visible. The colour used to denote the trigger was changed to a distinct dark yellow, with the aim of avoiding confusion with blues. The colours used to indicate craving, success and missed shots remained red, green and orange, respectively. The heatmap container is now aligned with the platform's card styling, characterised by a uniform white-to-soft gradient and consistent borders and shadows.

The dataset was then subjected to a process of randomisation, resulting in a synthetic distribution spanning 90 days, characterised by variable peaks and an occurrence of 4–23 events per day. This distribution also exhibited a category mix of approximately 524 cravings, 327 triggers, and 227 successes. The randomisation process was implemented to ensure that peak days or windows would not be repetitive, thereby facilitating the validation of insights across varying load conditions. This approach maintains the readability of the user interface (UI), ensures the consistency of the colour system, and ensures the data is realistic for testing purposes.

2.2 Interview Preparation

To assess and prioritize the information quality and contextual requirements of Large Language Model (LLM) outputs in a clinical setting, a semi-structured interview study was designed. The purpose of this study is to inform the design of a Context Engineering pipeline by grounding system requirements in expert clinical judgment.

2.2.1 Objectives

The primary objectives of the interviews are to:

- Evaluate the perceived usefulness and clarity of LLM-generated outputs in a clinical context;
- Identify gaps, ambiguities, or misinterpretations in model-generated information;
- Elicit and prioritize clinicians’ information needs when interacting with LLM-supported tools;
- Assess trust, transparency, and ethical concerns related to the use of LLMs in mental health practice.

2.2.2 Participant Profile

The main participant is a clinical psychologist with not only professional experience in psychological assessment and therapeutic intervention but also in a company setting. This information is collected to contextualize responses and account for variation in clinical perspectives.

2.2.3 Interview Structure

The interview follows a semi-structured format and is expected to last approximately 45–60 minutes. The interview is organized into four phases:

1. A brief introduction outlining the study goals, the role of the LLM, and ethical safeguards;
2. Presentation of one to two short exemplar LLM-generated outputs relevant to the study context;
3. Guided discussion using semi-structured questions across predefined thematic areas;
4. A wrap-up phase focused on reflection, prioritization of information needs, and overall impressions.

2.2.4 Key Question Areas

The interview guide is organized around the following thematic areas:

Usefulness The participant is asked which aspects of the LLM outputs are actionable in practice, which are overly generic, and which clinically relevant details are missing.

Accuracy and Clinical Alignment This section probes perceptions of clinical correctness, potential risks of bias or harm, and indicators of hallucination or unsupported claims.

Clarity and Presentation Questions focus on readability, terminology, structural organization, and preferred output formats (e.g., bullet-point summaries, highlighted risk indicators).

Contextual Requirements The participant is asked what background information would be necessary to trust or effectively use the outputs, such as patient history, therapeutic goals, or situational constraints.

Information Prioritization Here, was asked to identify their top three information needs, distinguish between essential and optional information, and express tolerance for uncertainty in model outputs.

Workflow Integration This theme explores where in the clinical workflow LLM outputs would be beneficial or disruptive, and whether they are perceived as saving or adding time.

Safety and Ethics Participant discusses acceptable boundaries for LLM use, required disclaimers or supervision, and concerns related to data sensitivity and patient safety.

Improvement Suggestions The participant is invited to suggest improvements, including preferred prompt structures, system controls, and feedback mechanisms.

2.2.5 Elicitation Techniques

To encourage concrete and reflective responses, multiple elicitation techniques are employed. The participant are asked to provide examples of particularly effective or ineffective outputs, rank information needs by importance, and respond to prompts such as “*What would need to change for you to use this tomorrow?*” and “*What risk concerns you the most in using this output?*”.

2.2.6 Data Collection

Interviews are audio-recorded with informed consent. Timestamped notes are taken to link specific participant comments to corresponding prompts or LLM outputs. Verbatim quotations related to trust, accuracy, and usability are captured, and relevant non-verbal cues are noted where appropriate.

2.2.7 Analysis Plan

Interview data is analyzed using thematic analysis. Responses are coded into predefined categories—usefulness, accuracy, clarity, contextual needs, workflow fit, and safety—while allowing for emergent themes. These themes are subsequently mapped to prioritized system requirements, with particular attention to divergent or conflicting perspectives among participants.

2.2.8 Success Criteria

The study is considered successful if it yields:

- A clearly defined set of high-priority information needs;
- Identified failure modes and associated risks in LLM outputs;
- Concrete preferences regarding output structure and formatting;
- Clinician-informed mitigation strategies for ethical and safety concerns.

3 Results

3.1 Feature Development

In this development iteration, the visualization layer was re-engineered with the objective of enhancing clinical readability and consistency. The Patterns heatmap now functions as a true intensity map: tile backgrounds transition from light to deep blue according to per-day log volume, while category bar plots maintain vivid, consistent colours, as in Figure 1. The utilisation of a unified colour scheme across the heatmap, timeline, and cycle progress (where craving is represented by red, trigger by cyan/blue, success by green, and missed by orange) served to eliminate ambiguity across different views.

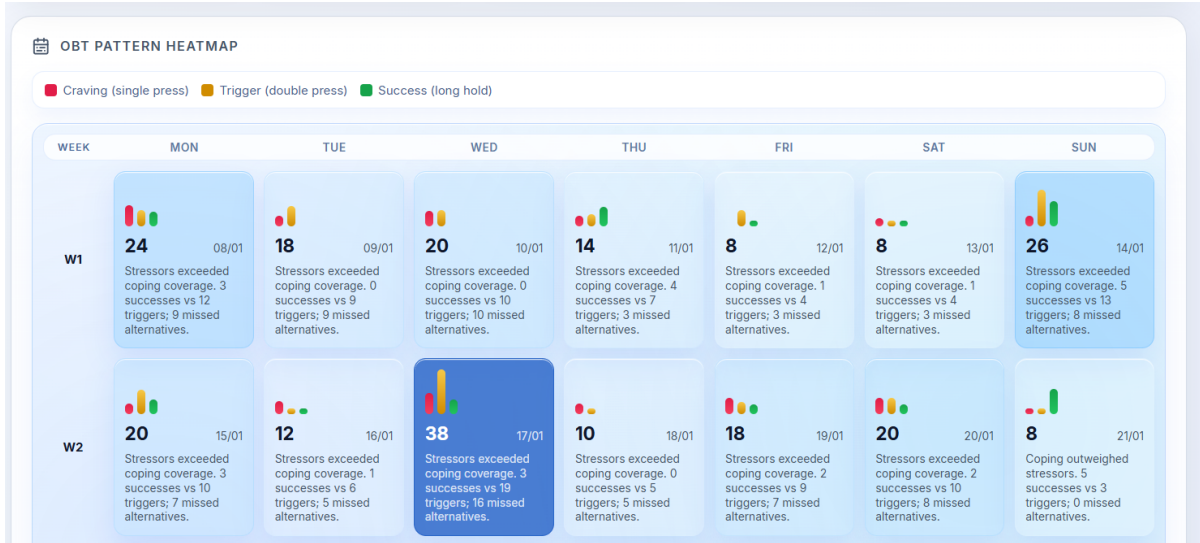


Figure 1: Sprint 5 - Heatmap

The Weekly Timeline Highlights have been redesigned with a coherent blue gradient and standard typography, thereby reducing visual noise while preserving at-a-glance protocol cues, Figure 2. Text contrast was standardised across the platform, with light text reserved exclusively for instances where it was deemed necessary on darker heatmap tiles. In all other instances, dark text was maintained to ensure optimal legibility. Concurrently, the generation of reports was constrained by more prescriptive LLM prompts, which enforced concise, non-redundant bullet insights and a single actionable recommendation to streamline clinical narratives. Collectively, these changes deliver a presentation that is more clinically neutral, modern, and interpretable, while reducing informational redundancy and enhancing the clarity of intensity patterns.

In order to rework the organization of the dashboard, patient selection was moved into the profile card as compact chips, keeping dataset switching while removing the extra "Other Patients" container, Figure 3.

3.2 Expert Interview and Platform Evaluation

3.2.1 Participants

The interview involved two participants:

- **Speaker 1 (Product and Engineering):** Responsible for presenting the platform, outlining its technical architecture, and defining the scope of the project.
- **Patrícia Marcelino (Clinical Psychologist):** A practising clinical psychologist trained in Évora, with a master's degree in clinical psychology. She completed a professional internship at a psychiatric hospital in Leipzig, working primarily with anxiety disorders and



Figure 2: Sprint 5 - Weekly Timeline Highlights

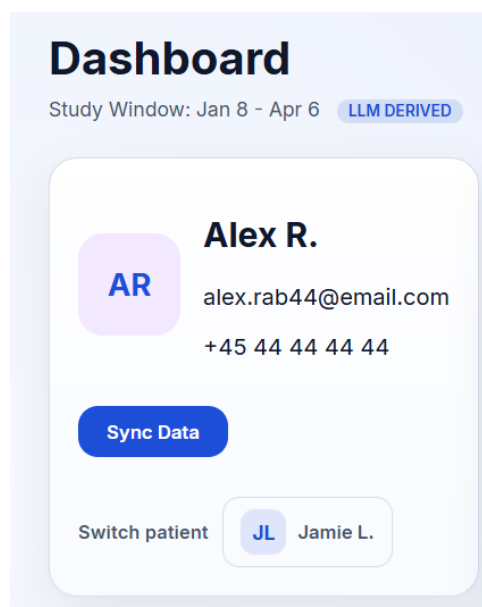


Figure 3: Sprint 5 - Patient Selection

personality disorders. Her professional experience includes both school and clinical settings, working with children, adolescents, and adults. She follows a psychodynamic therapeutic approach and holds a postgraduate qualification in child and adolescent psychopathology. She provides clinical services in Portuguese, English, and German, working with patients from multiple countries, and also has experience as a German language instructor.

3.2.2 Project Overview

The platform under evaluation is designed for the collection and interpretation of mental health data through a simple input device (a “clicker”) configured jointly by the patient and the psychologist. Each click corresponds to a predefined meaning (e.g., one click indicating craving, two clicks indicating a controlled craving).

The primary objective of the platform is to translate raw behavioural signals into clinically meaningful insights through visualisations and summaries, with psychologists as the primary target users. Secondary potential applications include enterprise and wellbeing programmes.

From a technical perspective, the platform combines deterministic metrics derived through data science methods with subjective and interpretative analyses mediated by a Large Language Model (LLM). The main test case used for evaluation is a socially tolerated dependency (coffee consumption), selected to assess feasibility while allowing future generalisation to other disorders.

3.2.3 Current Platform Status

At the time of the interview, the platform was executed locally due to considerations related to operational cost and data privacy. Synthetic data, modelled after real behavioural patterns, was used to avoid exposing real patient information.

The system is currently structured around weekly aggregations, reflecting the most common clinical follow-up cadence, while acknowledging the need to support alternative temporal resolutions (e.g., biweekly). It was noted that several platform features, such as heatmaps and metric aggregations, do not require the use of an LLM, which is instead reserved for subjective interpretation and higher-level clinical insights.

3.2.4 Feature Review and Clinical Feedback

3.2.5 Metrics Dashboard

The psychologist emphasised that averages computed over clinically meaningful windows (weekly and monthly, and daily when appropriate) are significantly more useful than cumulative totals over long periods. Peak indicators (peak day and peak week) were identified as particularly valuable, as they allow joint exploration with the patient of contextual events associated with increased symptom intensity.

Cumulative totals were considered to have limited clinical utility. Conversely, identifying days with the lowest incidence was highlighted as important for understanding protective factors and effective coping strategies. Visualising the use of strategies (as defined in the clicker protocol) was also considered clinically valuable for assessing self-regulation and coping behaviour.

3.2.6 Heatmap Visualisation

Daily heatmaps were regarded as highly valuable for analysing temporal patterns and linking critical days to external stressors or events. As this functionality does not rely on the LLM, it was recommended that it be maintained and enhanced, particularly by improving the inspection of specific days and visualising associated averages and peaks.

3.2.7 Raw Event Logs

Raw event lists were considered of lower priority for routine clinical analysis, with aggregated views preferred. However, access to filtered raw data was deemed useful for investigating specific patient-reported events, provided appropriate temporal filters (e.g., 7- or 14-day windows) are available.

3.2.8 Reports

Weekly reports and key indicators (averages, peaks, strategy usage) were considered appropriate. However, the psychologist noted that repeated information across dashboard and report sections reduces clarity and usability. It was recommended that the section previously titled “Interpretation and Next Steps” be restructured to focus explicitly on “risk hotspots,” avoiding repetition of metrics already presented.

The inclusion of clinical notes written by the psychologist was strongly supported, as this aligns with standard clinical practice and facilitates longitudinal tracking. It was also recommended that protocol descriptions and documentation be removed from the report itself and

instead provided through contextual help mechanisms (e.g., tooltips), as such information is already familiar to clinicians.

3.2.9 Insights by Time of Day

For coffee dependency, time-of-day analysis was considered to have limited relevance, with greater emphasis placed on daily and weekly patterns. However, for certain disorders, such as self-harm, specific hours may be clinically significant due to factors such as isolation or ritualised behaviour. Consequently, time-based analyses should be configurable according to disorder and clinical context.

3.2.10 Model-Generated Recommendations

Model-generated recommendations were viewed as potentially useful in some contexts as support for clinicians, but they should remain optional and never replace professional judgment. Direct exposure of such recommendations to patients was discouraged, as it may be triggering in certain cases. If used, this should be governed by clear policies and safeguards.

3.2.11 Key Decisions

Based on the discussion, several design decisions were agreed upon:

- Prioritise weekly and monthly averages and peak indicators over cumulative totals.
- Maintain and enhance the heatmap, including inspection of peak and lowest-incidence days.
- Restrict raw data access to filtered, case-specific consultations.
- Restructure reports to reduce redundancy and emphasise risk hotspots.
- Integrate clinical notes linked to specific temporal windows.
- Remove protocol documentation from reports and replace it with contextual UI help.
- Make time-of-day insights configurable by disorder.
- Disable model recommendations for patients and retain them as optional clinician support with disclaimers.
- Explicitly state that the platform does not replace the psychologist and that the LLM serves only as interpretative support.

3.2.12 Risks, Limitations, and Future Opportunities

Key risks identified include the operational cost of continuous LLM usage, privacy challenges when transitioning to hosted environments, and the difficulty of supporting a wide range of disorders without increasing system complexity. The clinical validity of LLM-generated insights was also highlighted as requiring ongoing validation.

Future opportunities include the introduction of a patient diary feature, configuration of platform modules by disorder, multilingual support, enterprise wellbeing applications, and a clearer separation between deterministic metric pipelines and LLM-driven interpretative analysis with appropriate clinical governance.

3.2.13 Conclusion

The expert interview provided critical clinical validation and guidance for refining the platform. The feedback reinforced the importance of clinically meaningful aggregations, contextual interpretation, configurability by disorder, and careful governance of LLM-generated insights. These findings directly informed the design decisions implemented in the subsequent development sprint and contributed to assessing the platform’s viability as a clinically oriented data interpretation tool.

3.3 Prompt Engineering

3.3.1 The Concept

Prompt engineering is an emerging discipline that is focused on optimizing interactions with generative AI models. The objective of this optimisation is to maximise the potential and utility of these models. While a prompt serves as the textual interface, communicating a user’s desires through instructions, questions, input data, and examples, prompt engineering transcends the mere construction of these inputs. The term is defined as an iterative process of developing prompts by systematically modifying or changing techniques to achieve a specific goal. In order to engage in this practice, it is first necessary to possess a profound comprehension of the capabilities and limitations of the model, as well as the context within which it operates.

The transition from ”tips and tricks” to a rigorous engineering discipline is evident in the field of prompt engineering, which mirrors traditional software engineering practices, such as version control and regression testing. The process is cyclical and generally comprises three repeated steps:

- Inference: The present study will run the prompt against a dataset.
- Evaluation: The performance of the model’s output must be assessed against the established criteria.
- Modification: The template for the prompt is to be refined on the basis of the evaluation in order to improve results.

3.3.2 Importance of Prompt Engineering

A fundamental objective of prompt engineering is to navigate and mitigate the inherent constraints of Large Language Models (LLMs). The limitations of the model are evident in its transient state, characterised by an absence of persistent memory, its probabilistic nature, which manifests as inconsistency between runs, and its propensity towards hallucination, resulting in content fabrication. The purpose of engineering techniques is to impose structure on stochastic systems. One such technique is ”self-consistency”, which involves prompting the model multiple times and analysing the coherence of the responses to ensure credibility.

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3.4 Context Engineering - A Branch of Prompt Engineering

Context Engineering is an emerging formal discipline that extends beyond traditional prompt engineering to address the increasing complexity of modern Artificial Intelligence (AI) systems,

particularly those based on Large Language Models (LLMs). While prompt engineering primarily focuses on the design of static textual inputs to guide model behavior, Context Engineering adopts a broader and more systematic perspective. It is concerned with the structured optimization of the entire information payload supplied to an LLM at inference time.

Rather than treating context as a single, immutable string of text, Context Engineering conceptualizes it as a dynamic and structured composition of heterogeneous informational components. These components may include task instructions, external knowledge sources, tool definitions, system-level constraints, and persistent or episodic memory. Such elements are selectively sourced, filtered, and orchestrated in order to maximize model performance, robustness, and task alignment.

3.4.1 Formal Definition and Objectives

Formally, Context Engineering transforms the notion of context from a static prompt into a dynamic function. Let the context C be defined as:

$$C = \mathcal{A}(c_1, c_2, \dots, c_n), \quad (1)$$

where $\mathcal{A}$ denotes a context assembly function, and each c_i represents an individual context component. These components may include static instructions as well as dynamic elements such as user interaction history, retrieved external knowledge, or the current system state.

The central objective of Context Engineering can be framed as an optimization problem: identifying the optimal set of context-generating and assembly functions that maximize the quality of the LLM’s output, subject to practical constraints such as the model’s finite context window and computational limitations. This formulation highlights the shift from ad hoc prompt design toward principled, system-level optimization.

3.4.2 Foundational Components of Context Engineering

Context Engineering is structured around three foundational phases, which together form a comprehensive taxonomy of the discipline.

- **Context Retrieval and Generation:** This phase focuses on sourcing and constructing relevant information. It includes traditional prompt engineering for instruction formulation, external knowledge retrieval through mechanisms such as Retrieval-Augmented Generation (RAG), and dynamic context assembly, where multiple information sources are combined into a coherent input representation.
- **Context Processing:** Context processing concerns the transformation and refinement of the assembled information. Key techniques include long-sequence processing for handling extended inputs, self-refinement mechanisms that iteratively improve model outputs, and structured information integration, such as the incorporation of knowledge graphs or multimodal data representations.
- **Context Management:** Context management addresses the efficient organization and maintenance of information over time. This includes the design of memory hierarchies that distinguish between short-term and long-term context, context compression strategies that maximize informational density within limited context windows, and optimization methods that mitigate the inherently stateless nature of LLMs.

3.4.3 System-Level Implementations

The principles of Context Engineering are instantiated in a variety of advanced AI system architectures that enable LLMs to interact effectively with external environments.

Retrieval-Augmented Generation (RAG) systems dynamically inject external knowledge into the inference process and have evolved toward increasingly modular and agentic architectures. Memory systems aim to emulate aspects of human cognition by enabling persistent information retention across interactions. Tool-integrated reasoning frameworks allow LLMs to invoke external functions or tools, such as calculators or web browsers, to augment their reasoning capabilities. Finally, multi-agent systems consist of coordinated collections of agents that communicate and collaborate to solve complex tasks, leveraging shared and distributed context representations.

Together, these implementations demonstrate how Context Engineering serves as a unifying framework for building scalable, adaptive, and intelligent LLM-based systems.

3.5 Prompt Engineering versus Context Engineering

Although Prompt Engineering and Context Engineering share the common objective of improving the performance and reliability of Large Language Models (LLMs), they differ fundamentally in their conceptual scope, formalization, and system-level implications. The progression from prompt engineering to context engineering reflects a broader shift from heuristic text crafting toward principled information orchestration and architectural optimization.

3.5.1 Static Text versus Dynamic System Architecture

Prompt Engineering traditionally treats context (C) as a monolithic and static string of text, typically expressed as:

$$C = \text{prompt}. \quad (2)$$

Within this paradigm, optimization is achieved through iterative refinement of textual elements such as instructions, questions, demonstrations, or examples. The primary challenge lies in exploring the space of possible text formulations—either manually or automatically—to identify phrasings that best elicit the desired model behavior for a given task.

In contrast, Context Engineering reconceptualizes context as a dynamic and structured system. As previously defined, context is expressed as:

$$C = \mathcal{A}(c_1, c_2, \dots, c_n), \quad (3)$$

where $\mathcal{A}$ denotes an assembly function and each c_i represents a distinct informational component, including dynamic elements such as retrieved external knowledge, user interaction history, system state, or tool specifications. Rather than optimizing a static text string, Context Engineering addresses a system-level optimization problem, seeking the optimal configuration of retrieval, selection, and assembly functions that jointly maximize output quality.

3.5.2 Scope of Optimization

The optimization objective in Prompt Engineering is typically framed as maximizing the probability of a desired output Y given a fixed prompt:

$$\max P_\theta(Y \mid \text{prompt}), \quad (4)$$

where P_θ denotes the model parameterized by θ . Techniques such as Chain-of-Thought prompting or Step-Back prompting are employed to encourage reasoning within a single inference window, without altering the broader information pipeline.

Context Engineering expands this objective to encompass the optimization of the entire inference-time information flow. The goal is to maximize expected performance across a distribution of tasks by jointly optimizing multiple subsystems, including:

- **Retrieval**, where external knowledge is sourced (e.g., via Retrieval-Augmented Generation) to maximize relevance or mutual information with the target output;
- **Processing**, which involves handling ultra-long sequences, structured representations, or multimodal inputs that exceed the capacity of standard prompts;
- **Management**, encompassing context compression and memory hierarchies to operate within strict context window and computational constraints.

3.5.3 Statefulness and Memory

Prompt Engineering is predominantly stateless in nature. Although conversational history can be embedded within prompts, each interaction is effectively treated as an isolated instance, requiring all relevant context to be explicitly provided within the current input window.

By contrast, Context Engineering is inherently stateful. It explicitly models persistent memory components (c_{mem}) and dynamic system state (c_{state}) as first-class elements of the context assembly process. This enables the integration of memory architectures inspired by human cognition, such as episodic and semantic memory, allowing LLM-based systems to maintain continuity and coherence across extended interactions.

3.5.4 Scalability and Error Analysis

As task complexity increases, Prompt Engineering typically relies on manual inspection, empirical trial-and-error, and heuristic refinement. Its scalability is constrained by context window limitations and the increasing brittleness of long, densely packed prompts.

Context Engineering addresses scalability through modular system design. By decomposing context generation into distinct functional stages—such as retrieval, filtering, formatting, and compression—individual components can be independently evaluated, optimized, and debugged. This modularity enables systematic error analysis, for example by assessing retrieval quality separately from generation performance, and supports the construction of robust, large-scale LLM-based applications.

4 Motivation for a Context Engineering Approach in This Thesis

Given the objectives and constraints of the present thesis, a Context Engineering approach is substantially more appropriate than traditional prompt engineering. The core requirement of the proposed system is the handling of longitudinal mental health data that evolves on a weekly basis and where historical information directly influences future interpretations. Such data is inherently dynamic and stateful, rendering a static, prompt-centric interaction paradigm insufficient.

Prompt engineering assumes that all relevant information can be provided explicitly within a single inference window. However, in the context of clinical or psychological monitoring, each interaction is conditioned on prior sessions, long-term patient characteristics, and evolving patterns of behavior. Treating each session as an isolated text input not only fails to capture this temporal dependency but also introduces risks related to context overflow, noise accumulation, and loss of clinically relevant structure. Context Engineering, by contrast, explicitly addresses information logistics: how data is stored, retrieved, compressed, and reasoned over time.

4.1 Context Management through Structured Memory Systems

A central requirement of this project is the ability to track patient progress over extended periods. This necessitates a structured memory hierarchy rather than reliance on a single, ever-

growing context window.

Weekly therapy sessions are naturally modeled as episodic memory units, where each session constitutes a distinct episode. Feeding raw session transcripts directly into the model is neither scalable nor reliable, as context windows are finite and unfiltered data quickly becomes noisy. Instead, Context Compression techniques are employed to distill each week’s raw input into concise, clinically meaningful summaries (e.g., changes in symptoms, notable stressors, or improvements). These summaries preserve salient information while minimizing redundancy.

In parallel, a semantic memory component maintains a persistent clinical profile representing the patient’s baseline state. This includes relatively stable information such as diagnoses, medication, long-term therapeutic goals, and known risk factors. By separating episodic updates from long-term state, the system ensures that the model consistently operates with an accurate understanding of the patient’s clinical background.

To further prevent context drift, selective forgetting mechanisms are incorporated. Retrieval is biased toward recent episodes, reflecting recency effects, while still allowing access to historically significant events based on importance. This mirrors cognitive principles of human memory and ensures that outdated or irrelevant information does not degrade model performance.

4.2 Dynamic Context Retrieval and Assembly

For the system to provide meaningful clinical support, context must be dynamically assembled rather than statically defined. Formally, the context is constructed as:

$$C = \mathcal{A}(c_{\text{week}}, c_{\text{history}}, c_{\text{knowledge}}), \quad (5)$$

where c_{week} represents the current session summary, c_{history} denotes retrieved historical episodes, and $c_{\text{knowledge}}$ encodes relevant clinical definitions and frameworks.

Retrieval-Augmented Generation plays a critical role in this process. When a psychologist queries whether a specific trigger or symptom has occurred previously, the system does not rely on implicit model memory. Instead, it retrieves relevant past episodes from a structured storage mechanism, such as a vector database, ensuring accurate and verifiable recall.

Moreover, mental health data is inherently relational. Relationships between entities—such as family members, stressors, and emotional responses—often span multiple sessions. Incorporating graph-based representations enables multi-hop reasoning over these relationships, allowing the model to identify higher-level patterns (e.g., the recurrence of work-related stress following family conflict) that are difficult to capture through linear text retrieval alone.

4.3 Context Processing for Clinical Reasoning

Beyond retrieval and storage, the system must actively reason over the assembled context to generate useful insights. Simple summarization is insufficient in a clinical setting, where the goal is to support professional judgment rather than replace it.

Chain-of-Thought prompting is employed to enforce structured analytical reasoning. Instead of directly requesting conclusions, the model is guided through intermediate steps such as identifying current symptoms, comparing them to previous sessions, detecting discrepancies or trends, and considering potential interventions. This explicit reasoning structure improves transparency and reduces the likelihood of unsupported or generic outputs.

To ensure clinical validity, domain-specific rubrics and definitions are injected into the context. Without such grounding, LLMs tend to default to generic advice derived from broad internet data. By incorporating established clinical frameworks and diagnostic criteria, the model’s reasoning is constrained to professionally accepted standards.

Finally, a self-refinement stage is introduced in which the model evaluates its own outputs for consistency, safety, and alignment with the patient’s risk profile. This reflective step acts as

an internal safeguard before insights are presented to the psychologist, reinforcing reliability and ethical responsibility.

4.4 End-to-End Workflow Summary

The resulting system follows a structured and interpretable workflow:

1. Weekly patient data is received.
2. Raw input is compressed into a structured episodic summary.
3. Relevant long-term state and historical episodes are retrieved.
4. Context is dynamically assembled with injected clinical knowledge.
5. The model performs step-by-step reasoning over the assembled context.
6. A structured briefing is generated for the psychologist.

This workflow illustrates how Context Engineering provides a coherent and scalable framework for longitudinal mental health analysis. By explicitly modeling memory, retrieval, and reasoning, it aligns naturally with the requirements of this thesis and offers a principled alternative to traditional prompt-based approaches.

4.5 Context Engineering in Our Platform

Our platform uses context engineering to make LLM outputs predictable, grounded, and safe across three surfaces: dashboard, report, and chat. On the server (`app/src/server.py`), click-stream data are transformed into structured aggregates and narratives (`simulate_patient_activity`, `build_state_narrative`), then assembled into surface-specific prompts:

- **Dashboard:** `request_llm_dashboard` sends metrics + a schema hint and requires JSON-only responses; `merge_dashboard_with_llm` validates and falls back to deterministic data when needed.
- **Report:** `request_llm_report` runs separately, with rules to avoid repeating dashboard insights; `validate_llm_report_payload` enforces structure, safety, and deduplication; `reportSource` indicates whether the LLM output was accepted or fell back.
- **Chat:** `build_chat_system_prompt` feeds a compact snapshot of metrics and context, keeping responses short, grounded, and non-diagnostic.

Dynamic context assembly (`assemble_context_components`) injects state narratives, structured snapshots, and placeholders for vetted resources (RAG-ready). Prompts include explicit reasoning and safety guardrails (e.g., no diagnoses/treatments), and `safe_json_loads` plus validators harden against malformed or fenced JSON. Together, these layers convert raw button presses into stable, clinically safe outputs tuned to each UI surface.

4.6 Llama 3 as Standard Baseline

The selection of the underlying Large Language Model (LLM) was a critical design decision, as it directly influences the feasibility, reliability, and clinical relevance of the proposed platform. In this project, the Llama3 model was selected as the baseline LLM due to a combination of practical, methodological, and research-driven considerations.

A primary motivation for choosing Llama3 is its lightweight nature and ability to run locally. This characteristic significantly reduces dependency on external, cloud-based services, which often introduce constraints related to cost, latency, data privacy, and long-term availability. Given

the sensitive and potentially clinical nature of the data handled by the platform, local execution aligns more closely with privacy-aware and clinically responsible design principles. Moreover, local deployment enables greater control over the system pipeline, facilitating reproducibility and experimentation without reliance on third-party APIs or online infrastructure.

Another key factor influencing this decision is that Llama3 is a pre-trained and well-established model. The objective of this thesis is not to design or train a Large Language Model from scratch, but rather to investigate the viability and behaviour of LLMs when integrated into a clinically oriented platform. Developing a model from the ground up would shift the focus of the project toward low-level model engineering, detracting from the core research aim of evaluating interaction quality, prompt design, and clinical applicability. By leveraging a pre-built model, development effort can be directed toward system integration, prompt engineering, and evaluation, which are more closely aligned with the research questions of this work.

Despite these advantages, the use of Llama3 also presents certain limitations. As a relatively compact model compared to large, cloud-hosted alternatives, it may exhibit reduced performance in complex reasoning tasks, nuanced language understanding, or domain-specific clinical interpretation. These limitations necessitate careful prompt design and output evaluation, and they motivate the deeper analysis of model behaviour conducted in this thesis.

Llama3 was selected not as an optimal or definitive solution, but as an appropriate and justified baseline model that supports the exploratory and evaluative goals of this thesis. Its use enables a focused investigation into the integration and viability of LLMs within a clinically informed platform, while maintaining flexibility for future model substitution if required.

Final Sprint

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17. februar 2026

1 Planning and Objectives

Sprint 6: Final Development and Testing

Duration: January 7th, 2026 – February 1st, 2026

Sprint Objective: The primary objective of this sprint is to attain a final MVP solution for the project outlined in this thesis. In order to achieve this objective, the platform is being developed and modified based on the results and feedback received from the investors. This process is being overseen by a professional psychologist, whose expertise is guiding the enhancement of the platform to ensure that it better reflects a more clinical and professional perspective. This approach will enable the team to concentrate on refining the product and implementing the feedback received during the interview process.

Subsequent to the development phase, the present sprint will comprise a concluding testing phase. This will entail the re-execution and re-working of the testing framework, in addition to the conducting of a manual quality assessment of the LLM output. The objective of this final testing phase is to obtain an assessment of the platform and the LLM output itself from a more direct perspective.

Key Activities & Deliverables:

- Last Feature Development
 - **Activity:** A review of the feedback received from the interview conducted with a professional psychologist. This will inform the implementation of any changes deemed necessary or significant within the scope of this project and sprint.
 - **Deliverable:** A final study on the possible minimum viable product of the proposed solution for this project.
- Test Sessions
 - **Activity:** Repeat the conducted one to one sessions for the main developed features, similarly to the previous sprint while still aiming to keep the core testing framework to later compare how the user feedback deviated.
 - **Deliverable:** Another set of user responses focused on the presented features and the usability of the platform and how viable it is.
- LLM Quality Assessment
 - **Activity:** A final and thorough evaluation of the LLM's portrayal of information and its viability within the context of the study.
 - **Deliverable:** A report on the quality of the LLM's output.

Rationale: The sixth sprint constitutes the final consolidation phase of the project, with the objective of bringing the platform to a stable and defensible Minimum Viable Product (MVP). At this stage, the emphasis shifts from exploratory development to refinement, validation, and final assessment, ensuring that the solution presented in this thesis is coherent, usable, and clinically informed.

The primary rationale for this sprint is the integration of feedback obtained from investors and, most importantly, from a professional psychologist. Under the oversight of clinical experts, the platform is continually refined to align more closely with professional standards, authentic clinical workflows, and appropriate interpretative boundaries. This process ensures that the final implementation moves beyond a research prototype toward a solution that is credible within a clinical-adjacent context.

It is evident that Sprint 6 encompasses a concluding testing phase, a necessity that is substantiated by the imperative for comparative evaluation. The project employs a systematic approach by re-adopting the established testing framework and conducting one-to-one sessions, facilitating a direct comparison with earlier sprints. This enables a systematic assessment of changes in usability, perceived value and platform viability.

A comprehensive manual quality assessment of the LLM output is finally conducted. Given the pivotal function of the LLM in the generation of interpretations and insights, this summative evaluation is imperative to critically assess its reliability, limitations, and suitability within the scope of the study.

In essence, Sprint 6 functions as a synthesis sprint, integrating final development, expert oversight, and structured evaluation to substantiate the conclusions of the thesis and ascertain the viability of the proposed platform as a final MVP.

2 Development

2.1 Last Feature Development

This project followed an iterative, clinician-informed development approach. After presenting the platform to a clinical psychologist and collecting structured feedback, we translated the findings into concrete product and system changes. The objective of this phase was not to validate clinical outcomes, but to align the platform’s data flows, UX structure, and model usage with real-world clinical workflows.

2.1.1 Rebalancing of Core Metrics and Summaries

Feedback emphasized that totals over long periods are less clinically actionable than averages, peaks, and low-incidence days. In response, the aggregation layer was extended to compute daily, weekly, and monthly averages; explicit peak and low trigger days; peak trigger weeks; and weekly log totals. These metrics are now treated as primary indicators throughout the system. The dashboard and report payloads were revised so that averages, peak/low markers, and weekly log volume are surfaced first, while cumulative totals remain available but de-emphasized.

2.1.2 Separation of Deterministic Metrics vs. LLM-Mediated Insights

To reduce unnecessary model use and keep deterministic logic transparent, we strengthened the distinction between statistical outputs (handled by the data pipeline) and narrative/interpretive outputs (handled by the LLM). The deterministic pipeline now computes averages, peak/low days, trigger windows, and weekly log volume before any model call. LLM prompts were adjusted to use these metrics exactly (no invented values), and validation was tightened to enforce structured JSON outputs and safety constraints.

2.1.3 LLM Context Completion and Multi-Feed Grounding

Follow-up errors were traced to incomplete context exposure in conversational prompts. To mitigate this, the conversational context assembly was extended to deliver the full dataset and dashboard payload in multiple supplemental chunks. The system prompt now explicitly instructs the model to treat these chunks as authoritative. This multi-feed strategy preserves safety constraints while ensuring the model can answer questions that require granular or longitudinal references (e.g., peak log week).

2.1.4 Report Restructuring Toward Clinical Hotspots

The report was redesigned to reduce redundancy and focus on clinically meaningful risk hotspots. The headline metrics now prioritize success rate and most-used protocol alongside daily average activity and the toughest day. Total logged events and peak craving windows were moved into Interpretation & Next Steps, where they provide contextual support rather than headline emphasis. Repeated content across dashboard and report was minimized. Additionally, protocol definitions were removed from the report body (to avoid redundancy for clinicians) and are now presented via contextual documentation in the UI rather than repeated inside narrative sections.

2.1.5 Clinical Notes Integration

The psychologist highlighted the importance of clinician annotations for contextualizing events and tracking evolution across sessions. We introduced clinical note support as a first-class object, with storage, indexing by week/date, and API endpoints for retrieval and creation. These notes are now integrated into dashboard/report payloads, enabling the UI to display clinician-entered context alongside weekly summaries and hotspots.

2.1.6 Raw Event Log Access and Filters

The raw click stream was retained but repositioned as a low-priority, filtered view. We maintained short-window filters (7/14/30 days) and removed any notion of exposing continuous streams in the interface. This preserves targeted lookup when a patient references a specific incident without pulling attention away from aggregated clinical summaries.

2.1.7 Hour-of-Day Insights as Configurable

Feedback stressed that hourly granularity is clinically relevant only for some disorders (e.g., self-harm) and less so for mild dependencies like coffee. We introduced configuration flags in dataset definitions to enable or disable hourly insights per case. This enables disorder-specific tailoring while keeping the baseline experience simple.

2.1.8 Recommendation Governance and Audience Control

To reflect the concern that LLM recommendations must not replace clinical judgment, we implemented a recommendation policy flag. Model-generated suggestions are now explicitly marked as optional and clinician-only, and hidden from patient-facing contexts. This policy is delivered in the API payload to ensure downstream UI behavior remains aligned with the intended governance model.

2.1.9 Dashboard Layout and Insight Density

The dashboard was reorganized to place key analyst insights in the left column and data insights in the middle column, improving visual balance and reducing empty space. The LLM prompt and schema were updated to request more detailed bullet points (4–6 key insights, 4–5 data

insights), and fallback content was expanded accordingly. This ensures that the insight cards remain populated even when the model is unavailable and reduces the appearance of sparse containers.

2.1.10 Stability Improvements in Sync Behavior

During platform initialization, the dashboard sometimes displayed a preliminary insight before quickly updating. This behavior was mitigated by adding a request-tracking mechanism so only the latest sync response can update state. This results in a more stable and predictable initial render without transient content flashes.

2.1.11 Summary of Development Outcomes (Non-Evaluative)

These changes collectively shift the platform toward clinically meaningful aggregation (averages, hotspots, and weekly log volume), reduce redundant or non-actionable content, and formalize clinician control through notes and recommendation governance. The system now enforces a cleaner separation between deterministic metrics and LLM-derived narrative outputs, the LLM receives complete structured context across multiple feeds, and the UI structure reflects the clinical priorities identified in the feedback.

2.2 LLM Quality Assessment

The development of the manual LLM assessment framework focused on delivering a rigorous and reproducible evaluation workflow that mirrors the platform’s production data pipeline and user-facing outputs. The objective was to replace ad-hoc checks with a structured, end-to-end checklist aligned with the actual dashboard, report, and conversational surfaces of the system. This ensured that evaluation targets not only numeric outputs but the complete set of metrics, summaries, and narrative elements presented by the platform, without introducing parallel logic that could diverge from application behavior.

The assessment script was re-engineered to import and reuse the server-side functions responsible for generating platform outputs. Instead of using custom prompts or synthetic data, it now loads processed One-Button Tracker (OBT) data via the dataset configuration, constructs the bundle and prompt block exactly as the application does, and issues the same LLM requests used to generate dashboard and report JSON. The resulting outputs are merged with fallback metrics using the platform’s merge logic, and validated against the same JSON schema checks used in production. This design ensures that evaluation inputs and outputs remain consistent with the system under assessment.

To move beyond limited example questions, the assessment workflow was extended to generate a comprehensive evaluation set across all platform features. The script now builds verification items for headline metrics (averages, totals, peaks and lows), trigger timelines, cycle progress distributions, timeline events, weekly highlights, and calendar heatmaps. Each evaluation item stores the LLM value, the expected value derived from processed data, and a structured evidence snapshot for traceability. The same approach is applied to the report view: key metrics, weekly breakdowns, and protocol metadata are validated against processed aggregates, while narrative insights and action plans are captured as qualitative checks linked to evidence for manual review.

The conversational assistant output was integrated into the same evaluation pipeline. The script reuses the platform’s chat system prompt and context assembly, including dashboard snapshots and chunked context payloads, to ensure that chat queries are evaluated under identical constraints. A default set of targeted questions was defined to cover study duration, totals, peaks, averages, and protocol distributions, with optional custom question sets supported via JSON input. This brings the conversational layer into the same structured assessment process as the dashboard and report outputs.

The manual review workflow was also extended to support the richer evaluation structure. The reviewer interface now surfaces the evaluation area (dashboard, report, history, chat), the specific feature and field, the expected value, the LLM value, and optional evidence. Rating labels remain simple (correct, partial, incorrect, skip) to preserve consistency while improving traceability and efficiency during review. A reporting mode was retained for quick inspection without rating.

Overall, the development effort transformed a lightweight question-answer script into a full-fidelity assessment harness tightly coupled to the production pipeline. The resulting framework provides a structured and repeatable evaluation dataset, grounded in the same data transformations, prompts, and outputs used by the platform itself.

2.3 Testing Framework

Reusing the same testing framework used on sprint 4, an experiment was similarly conducted trying to assess the current quality and viability of the platform and respective output containing the following tasks.

Task List:

- Task 1: Review the key and data insights
- Task 2: Ascertain which week presented the most significant challenges for Alex.
- Task 3: Ascertain the day that proved most challenging for Alex.
- Task 4: Investigate the list of cravings and logs.
- Task 5: Analyse the dashboard and reports metrics.
- Task 6: Interact with the LLM for more information.
- Task 7: Please provide suggestions for the next sessions with Alex.
- Task 8: Add you own note, in the role of psychologist, to week 5.

3 Results

3.1 LLM Quality Assessment

The results were derived from the evaluation log in `manual_llm_eval.jsonl`, which contains 184 assessment items spanning dashboard, report, history, and chat outputs. The distribution across platform surfaces was as follows: dashboard (134 items), report (28 items), history (5 items), and chat (17 items). Each item is assigned a check type, thereby distinguishing between structured, automatically verifiable fields and qualitative or freeform responses.

The evaluation set was dominated by structured checks. A total of 36 items were designated as exact checks and 118 as object checks, encompassing headline metrics, timelines, weekly summaries, and heatmap entries. It was established that all 154 structured checks were in alignment with the processed-data expectations delineated within the evaluation log, thereby confirming 100% equality between `llm_value` and the expected values. This encompassed all structured dashboard items (17 exact, 111 objects), all structured report items (14 exact, 7 objects), and all history summary checks (5 exact). These results indicate full agreement between the produced structured outputs and the processed metrics for the evaluated dataset.

The remaining 30 items were comprised of qualitative and freeform elements. Thirteen qualitative checks were conducted, in addition to seventeen freeform checks facilitated via the chat function. These items capture narrative insights, action suggestions, and conversational answers that require manual judgment rather than direct equality comparison. A manual review of these

freeform elements was conducted, and it was observed that the model provides highly accurate feedback across all features in very tight and controlled conditions. This indicates the model’s true reliability, as the expected answers provided by the LLM corresponded to the processed data, even in the case of more narrative-type answers, which were previously deemed unreliable.

3.2 Last Test Session

After conducting one last test session, the following feedback was received. The cycle progress visualization could be enhanced through refinement of the colour scheme, as the current palette hinders clear distinction between cycles. The enhancement of colour contrast has been demonstrated to engender improved visual clarity and user interpretation (Jones, 2019).

The OBT pattern heatmap was also identified as potentially confusing. It has been demonstrated that issues arise in relation to the interaction design when the user hovers over tabs. This is due to the fact that the displayed information does not always fit well within the graph boundaries. This has the potential to hinder comprehension and negatively affect the user experience. From a scalability perspective, the current design may not be optimally suited to longer study durations. For instance, if the study were to be extended to 20 weeks, the continuous presentation of all data in a single view would likely become excessively protracted. The introduction of pagination, such as the division of data into multiple pages (e.g., weeks 1–10 and weeks 11–20), has the potential to enhance both usability and navigability.

With regard to the clinical summary section, a significant amount of unused space was identified at the upper portion of the page. It is evident that this layout is sub-optimal in terms of efficiency. In order to address this issue, it is recommended that measures be taken to utilise the available screen space in a more effective manner and to enhance the density of the information presented.

It is evident that the placement of the date range selection and sorting controls in the session history view is not consistent. It is recommended that the layout be adjusted by positioning the sorting functionality in closer proximity to the date range controls. This adjustment would result in a more coherent and intuitive interface design.

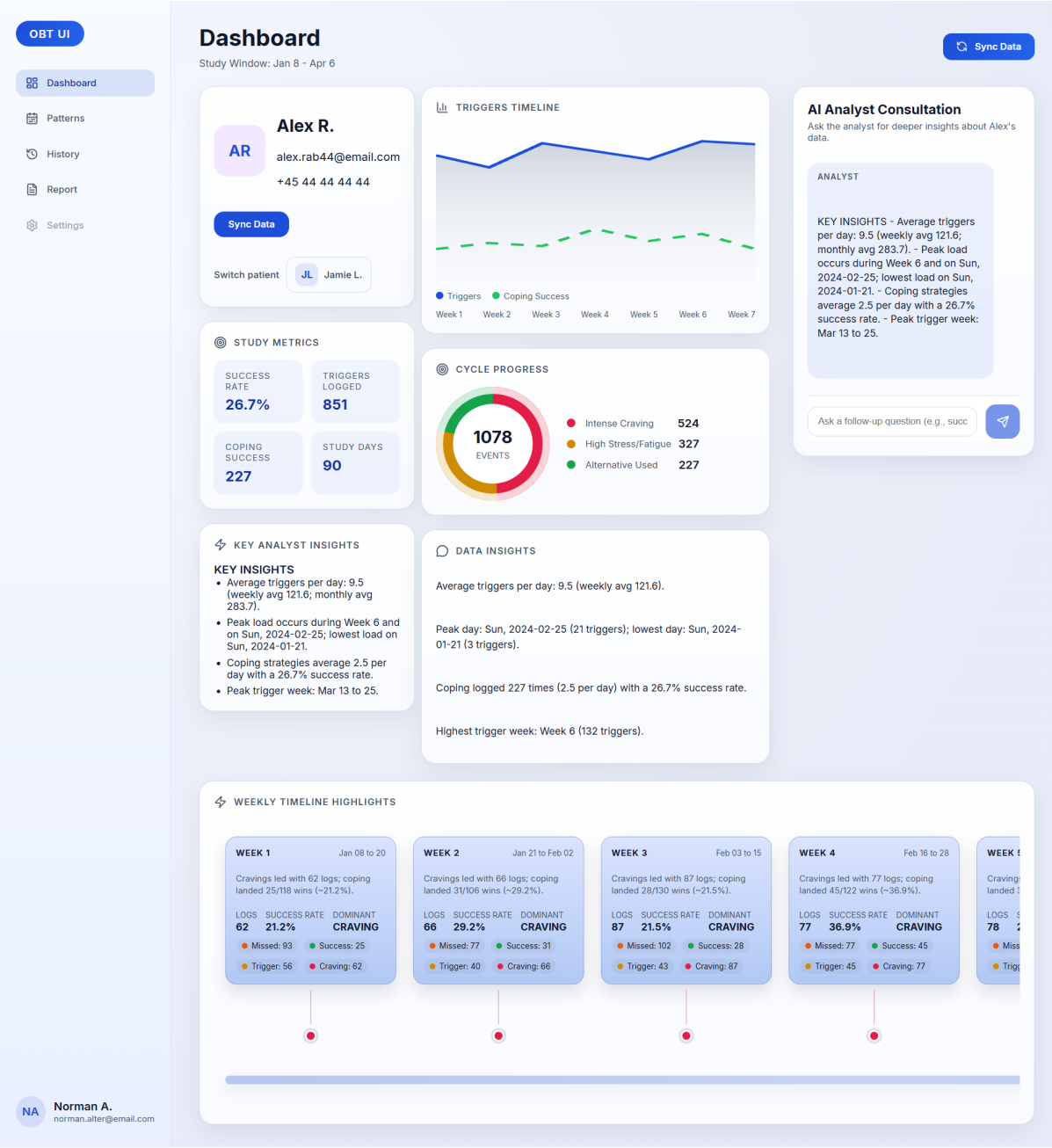
3.3 MVP Solution

The iteration, driven by feedback, resulted in quantifiable enhancements in analytical coverage, report emphasis, and conversational reliability. Firstly, it is important to note that the deterministic aggregation layer has been developed to now compute peak log week and weekly log totals in addition to trigger-based peaks. This development has the capacity to resolve questions that hitherto could not be answered, for example, questions concerning the number of logs for a given period. Furthermore, it enables insights into log volume to be provided alongside metrics focused on triggers. Secondly, the report highlights were rebalanced to emphasise daily average activity and the toughest day (peak trigger load), while maintaining the success rate and most-used protocol as the primary indicators. The total logged events and the peak craving window have been relocated to the "Interpretation & Next Steps" section, thereby enhancing the clinical significance of the headline panel without discarding the contextual totals.

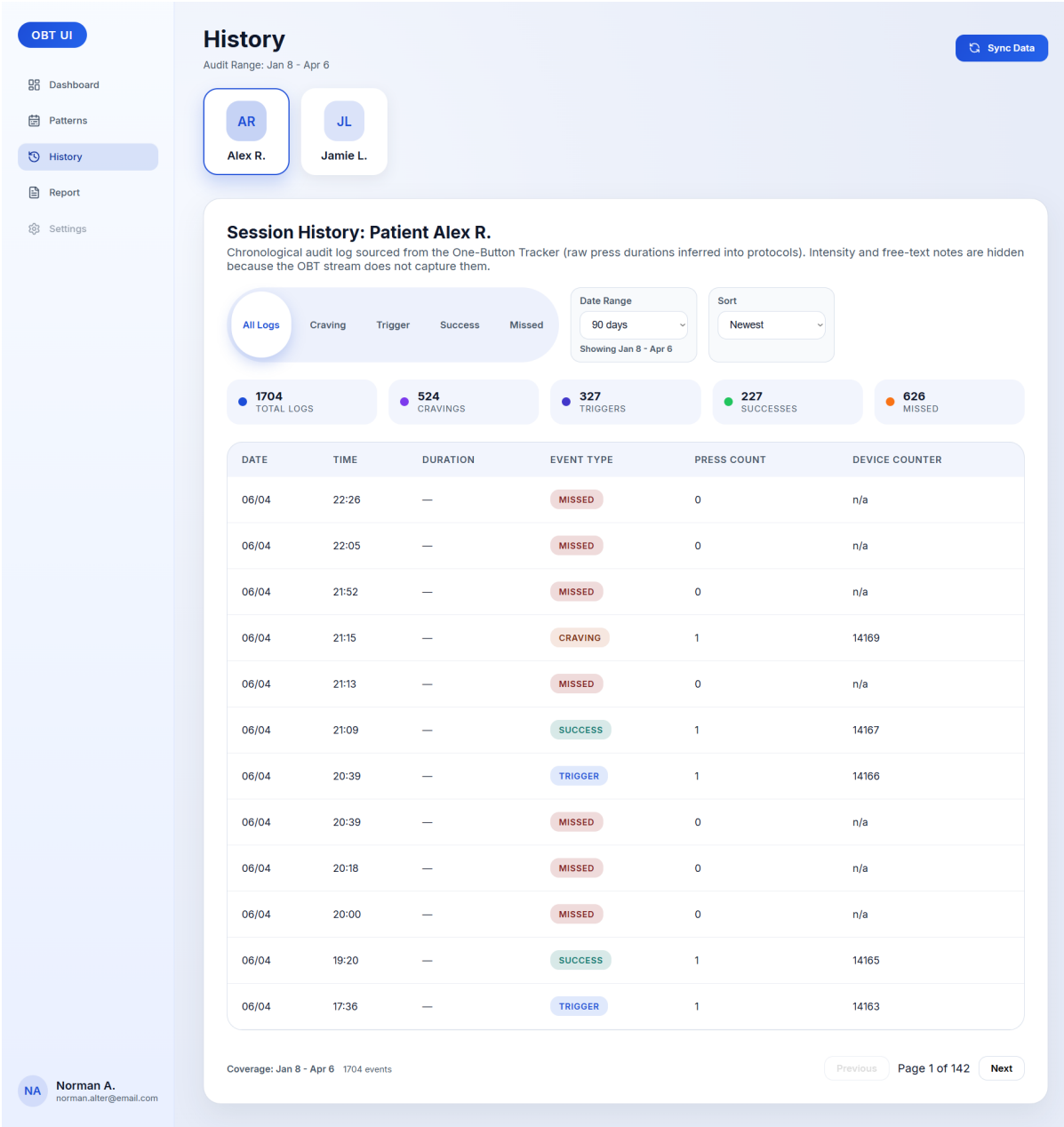
Conversational accuracy was further enhanced by the provision of the full dataset and dashboard payload in supplemental context chunks, explicitly designated as authoritative in the system prompt. This multi-feed grounding reduced model uncertainty in follow-up questions that require longitudinal references. It is evident that the aforementioned changes serve to enhance the accessibility of clinically meaningful metrics, thereby reducing ambiguity in analytical responses and aligning report emphasis with actionable patterns, such as the typical daily load and the highest-strain days.

In addition to the alterations that were implemented in response to the feedback received at the conclusion of Sprint 5, the final session conducted within this sprint revealed flaws that

required resolution in order to demonstrate the platform as a more comprehensive solution. Consequently, the final days of development were dedicated to implementing these modifications and ensuring that the platform approached a proper minimum viable product.



Figur 1: Dashboard



Patterns

Heatmap coverage: Jan 8 - Apr 6

Sync Data

Pattern Intelligence

Clinical heatmap for cravings and coping

Align interventions with the densest stress windows and quickly see where coping is holding versus slipping.

Clinical-grade readout

STUDY SPAN

90 days • 13 week window

0 idle days

COPING COVERAGE

26.7%

1704 total events captured

PEAK DAY

Sun, 2024-02-25 • 42 logs

Quietest: Tue, 2024-02-20 • 6 logs

AVERAGE LOAD

18.9

Logs per day

OBT PATTERN HEATMAP

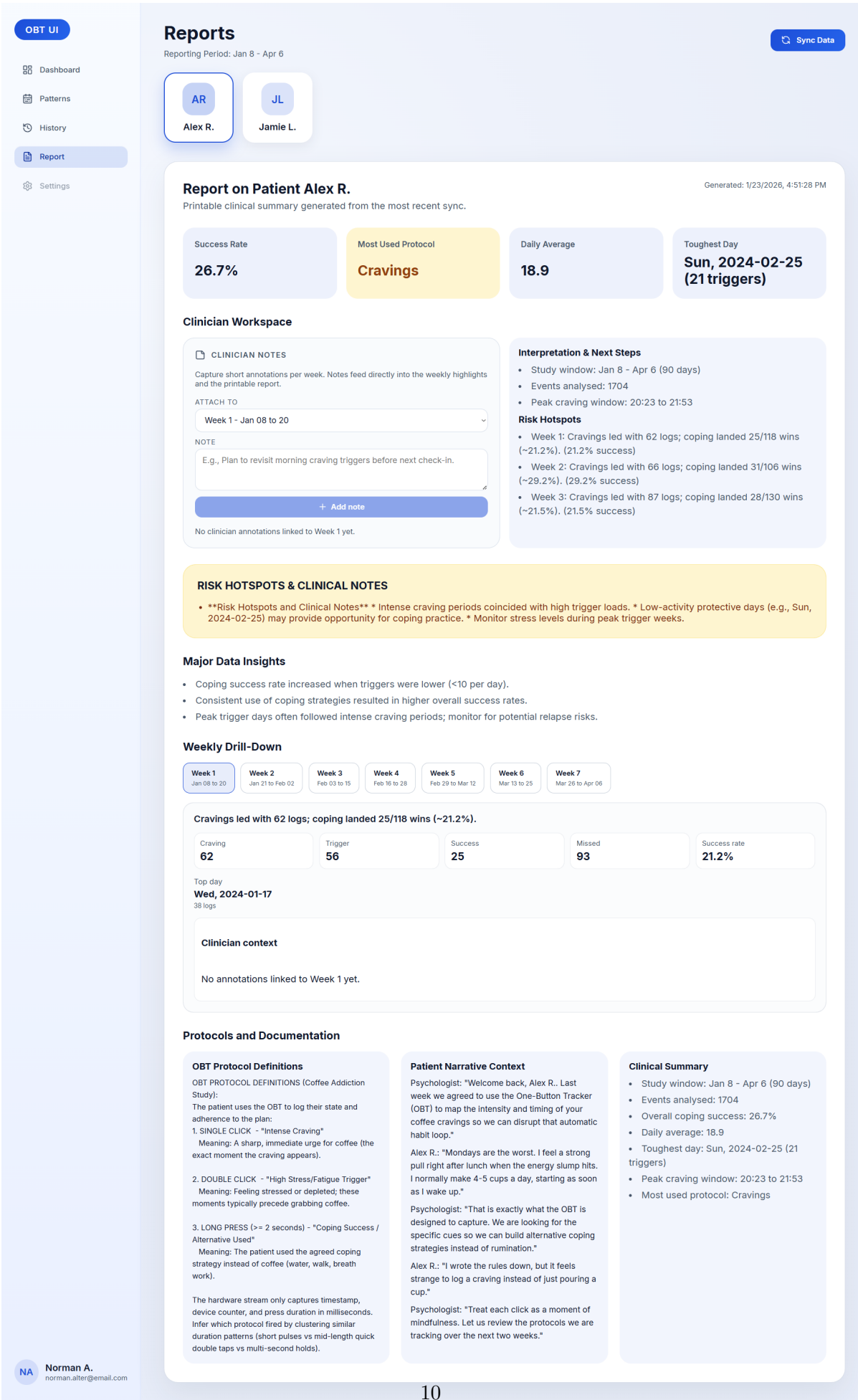
Weeks 1-10 of 13

Previous

Next

■ Craving (single press) ■ Trigger (double press) ■ Success (long hold)





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